30.8.2025

CS 310

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Bits/ Alphabet

Encoding

00111 01111 00100 01001 10011 ...

GODISNOWHERE

"stand on the shoulders of giants"

Natural Language

AMBIGUITY

Formal → we define the Syntax, semantics

3 models

Finite State Automata — FSA

Push Down " — PDA

Turing machine — TM

Computer **Science** & Engineering

What

abacus

slide rule

Casio FX

mobile phone

⋮

Turing test

$3x^2 - 5x + 7 = 0$

Hilbert's 10th problem

$\rightarrow 7x^3y^2 - 9x^2y^3 + 13xyz + \dots = 0$

undecidable

Is it

→ divisible by 7?

→ Give a solution (diagram) that any 10-year old can use to answer in "real-time"

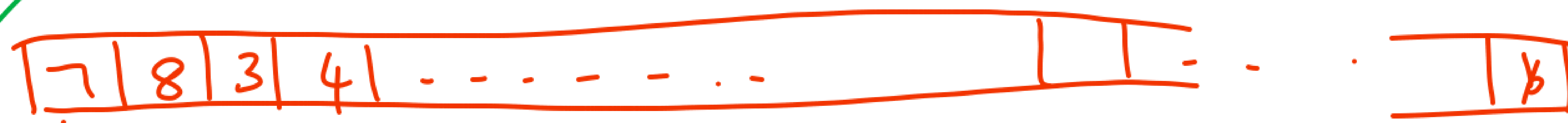
FSA

PDA

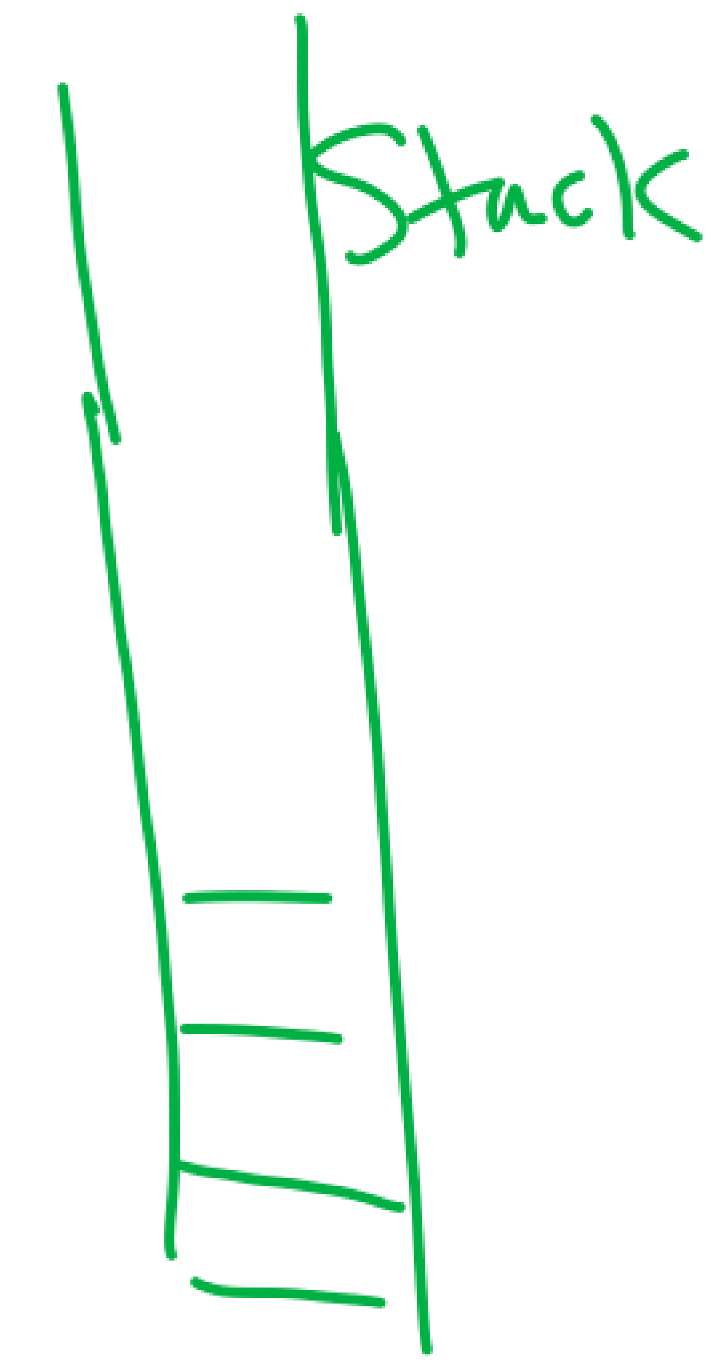
Input Tape

Is the string of the form $7^n 9^n$?

$n \geq 1$



Is it a palindrome?



→ non-determinism
(guess)

Turing machine

turingmachinesimulator.io

~~Input~~

tape

Read/write

5 | 7 | 8 | 3 | . . . _ _ _ _ _ | b

- ① replace a digit by some symbol.
- ② move 1-step L or R

q^k

q

Is n prime?

Machines	Language
FSA	Regular Expression
PDA	Context free Grammar
TM	Type 0 grammar

Chomsky hierarchy

$(0+00)1^*$

$S \to aSb$

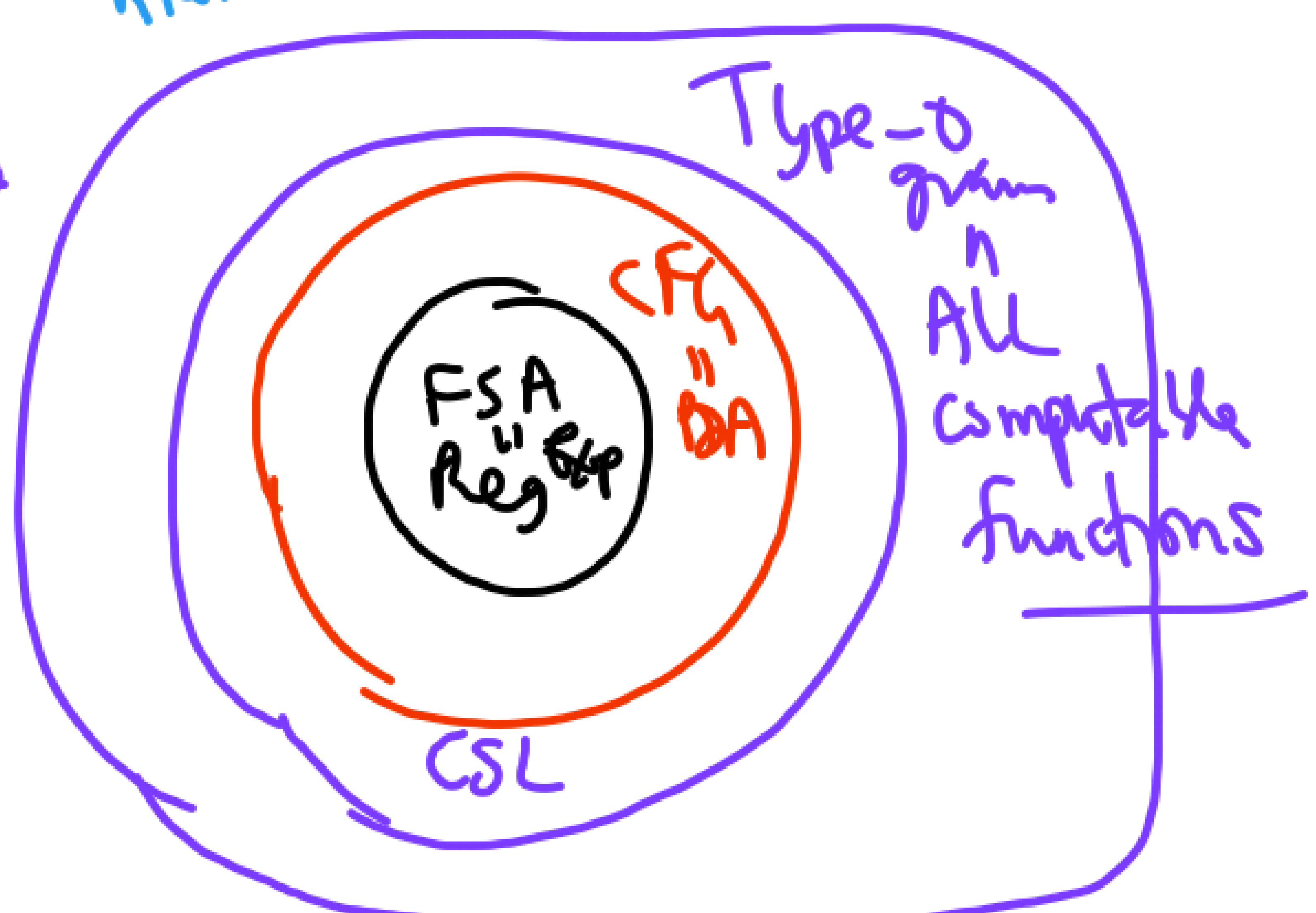
$S \to ab$

$B \to cb$

$a^n b^n c^n$

lhs can have any string

$aSC \to Sbc$



8.8.2025

Agenda

1. Languages
2. Enumeration/Encoding
3. Define (How to?)
4. Grammars
5. Properties — Decision, closure

Alphabet

Σ (symbols)

0, 1, 2
a, b, c

Strings
Words

Concatenation ·

a·b·c

Σ^k

$\Sigma^3 \rightarrow$ 3 letters

$$\Sigma^* = \bigcup_{k=0}^{\infty} \Sigma^k \rightarrow \text{Sentence}$$

all possible strings

length

0

ϵ λ

- Moodle post
- Assign 1

LLM

Satan-Cantor

Enumeration

$\Sigma = \{a, b, c\}$

Length

$\langle 1 \rangle$

$\langle 1, 2 \rangle$

$\langle 3 \rangle$

$\langle 3, 1, 5 \rangle$

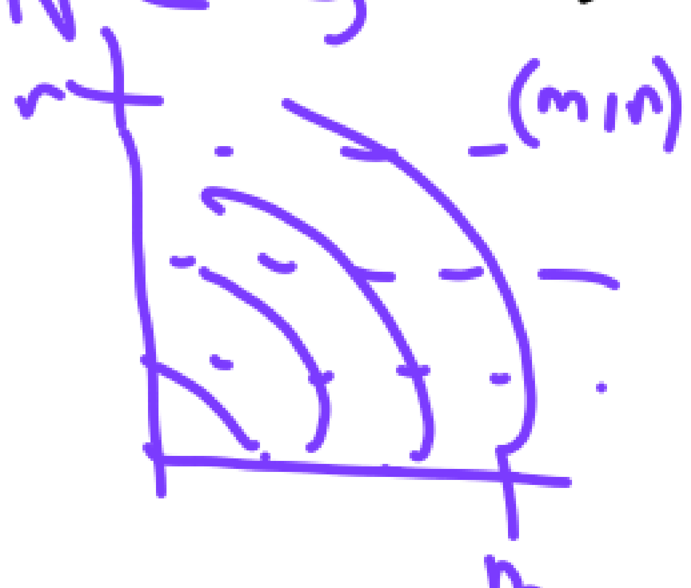
$\mathbb{N}$		Σ^*		
1	—	ϵ	—	0
2	—	a		
3	—	b	}	1
4	—	c		
5	—		}	2

Bijection

$$\mathbb{N} \leftrightarrow \Sigma^*$$

Pairs

$\mathbb{N} \times \mathbb{N} \leftrightarrow \mathbb{N}$



$\mathbb{N} \times \mathbb{N} \times \mathbb{N} \leftrightarrow \mathbb{N}$

Dovetailing

$\mathbb{N} \leftrightarrow \mathbb{N}^k$ ← unknown (finite)

$\Sigma \quad \Sigma^*$

Define Language

$$L \subseteq \Sigma^*$$

Can we enumerate all Languages?

$$\Sigma = \{a\}$$

$N = \{1, \ldots\}$ $\quad 2^N$

$S_d = \{a, a^2, a^4$

	a	a^2	a^3	…	a^{100}	…
1 — S_1	0	1	1	0	…	
2 — S_2	1	0	1	0	0	-
3 — S_3	1	1	1			
4 — S_4	1	0	1	0		
⋮						

diagonal

$$S_d = \{ a^i \mid a^i \notin S_i \}$$

Terminals — Non Terminals

Grammars (to define languages)

$$G = \langle \Sigma, V, S, P \rangle$$

Σ: alphabet — a, b, c — 0, 1

V: Variables — $V = \{ S, A, B, P, R, L \}$ — ~~$x \leftarrow x+1$~~

AA

BB

S: Starting Variable

P: Set of Productions / Rules

Productions

$lhs \rightarrow rhs$

$\alpha \rightarrow \beta$

any string $(\Sigma \cup V)^*$

1) Type 0 (unrestricted) $\alpha, \beta \in (\Sigma \cup V)^*$

Context

2) CSL — Sensitive — any — $+ \; |\alpha| \geq |\beta|$

3)

$(\Sigma \cup V)^*$ — aA — aBcA — cBBAd — AAcBx

$$\alpha \longrightarrow \beta$$

1. Type-0 (Unrestricted) $\alpha, \beta \in (\Sigma \cup V)^*$

2. Context-Sensitive (CSG) lhs at least as long as rhs $|\alpha| \geq |\beta|$

3. Context-Free Grammar (C.F.G) lhs $\in V$ (simple variable)

Reg. Expr

4. Right Linear Left $\rightarrow$ $\alpha \in V$, $\beta \rightarrow$ has at most one variable.

$S \rightarrow aa$

linear: $S \rightarrow aBc$

$S \rightarrow ccBa$

left-linear: $S \rightarrow Acc$

right-linear: $S \rightarrow abC$

Derivations

$Xb \to bbY$

$L(G) = \{ w \in \Sigma^* \mid S \Rightarrow^* w \}$

$bbYa \to aa$

S

$\Downarrow$

$\Downarrow$

$\Downarrow$

"1-step replacement"

$abc\,\widetilde{Xb}\,aYD$

$\Downarrow$

$abc\,\underline{bbY\,a}\,YD$

$\Downarrow$

Equal # of same a's and b's

$\Sigma = \{a, b\}$

$L = \{ w \in \Sigma^* \mid n_a(w) = n_b(w) \}$

$L_1 = \{ a^n b^n \mid n \geq 1 \}$

$a^n b^{2n}$

$a^{10} 2^{20}$

Write a CFG G $\quad L(G) = L_1$

Technique

Induction or Recursion

Base

Build-up

$S \rightarrow ab$

$S \rightarrow aSb$

1. $S \rightarrow abb$
2. $S \rightarrow aSbb$
3. $\cancel{S \rightarrow bb}$

All good strings

$S \Rightarrow aSbb \Rightarrow aaSbbbb$

No BAD string

$$L = \{ a^n b^n c^n \mid n \geq 1 \}$$

Type 0

$$L = \{ a^{i^2} \mid i \geq 1 \}$$

13.8.25

0. Languages

$L \subseteq \Sigma^*$

$\Sigma = \{0, 1\}$

Generate-and-test

- Complete
- Termination

Soundness

$L = \{ w \in (0+1)^* \mid w \text{ is a prime} \}$

$L = \{ \langle w_1, w_2, w_3 \rangle \mid w_i \in (0+1)^* \text{ and } w_1, w_2, w_3 \text{ are the roots of } 7x^3 - 3y^2z + \dots = 0 \}$

1. Grammars

$\langle \Sigma, V, S, \underline{\underline{P}} \rangle$

2. Machines

- FSA
- PDA (stack)
- TM

$L = \{a^n b^n c^n \mid n \geq 1\}$

G

Σ extra

$L(G) = \{w \mid w \in \Sigma^*, S \overset{*}{\Rightarrow} w\}$

Generate template

$S \to S_1 \#$ R

$S_1 \to ABCS_1$

$S_1 \to \epsilon$

× loops

$AB \to BA$

$BA \to AB$

Swap rules

$CA \to AC$

$CB \to BC$

$C\# \to \epsilon$

$Cc \to cc$

$BA \to AB$

$Bc \to bc$

$Bb \to bb$

$Ab \to ab$

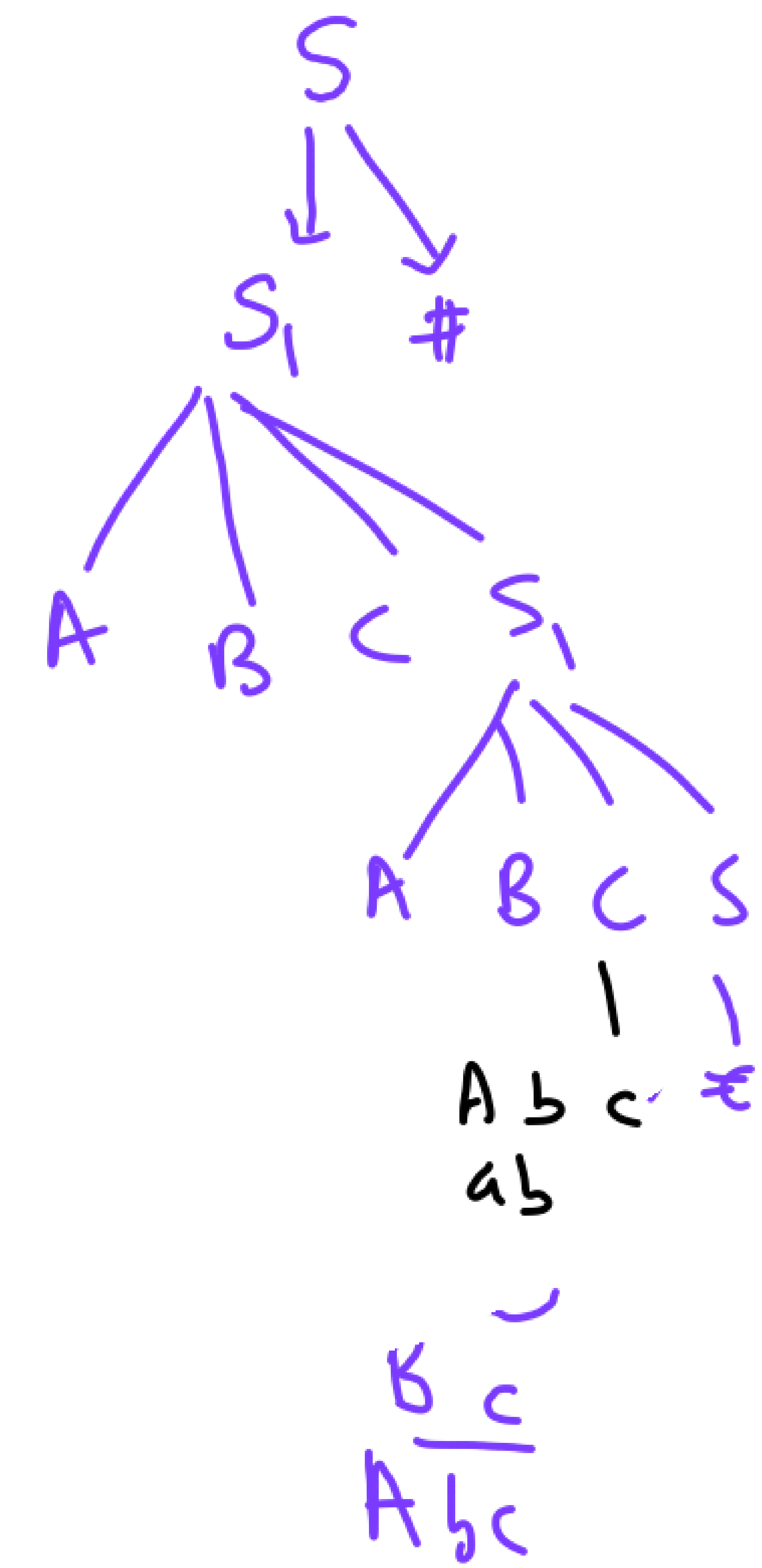
$Aa \to aa$

$Ca \to aC$

$Cb \to bc$

$Ba \to aB$

Derivation-tree



$$L = \{a^n b^n c^n \mid n \geq 1\}$$

Base case 1. $S \to abc$

Longer 2. $S \to aAbc$

go fwd 3. $Ab \to bA$

4. $Ac \to Bbcc$

go back 5. $bB \to Bb$

6. $aB \to aa$

7. $aB \to aa\underline{\underline{A}}$

S
2 $\downarrow$
$a\underline{A}bc$
3 $\downarrow$
$ab\underline{A}c$
4 $\downarrow$
$ab\underline{B}bcc$
5 $\downarrow$
$\underline{aB}bbcc$
7 $\downarrow$
$aaAbbcc$
$\downarrow$

ambiguity

2 different derivation trees.

non-isomorphic

~~Type-0~~

CFG

$L=\{w\in(a+b)^* \mid n_a(w)=n_b(w)\}$

Kolmogorov Complexity

1

$S \to \epsilon$

$S \to aSb$

$S \to bSa$

$\boxed{S \to SS}$

$\underbrace{a}_{S} \quad \underbrace{a}_{S}$

2

$S \to \epsilon$

$S \to a\underline{B}$

$S \to b\underline{A}$

$B \to bS$

$B \to a\underline{B}\,\underline{B}$

$A \to aS$

$A \to bAA$

~~$a^n b^n c^n$~~

Closure Properties

$L = \{ a^i b^j \mid i > j \}$

$S \to S_1 / S_2$

Concatenation

$\boxed{S \to S_1 C}$

$L = \{ a^i b^j c^k \mid i = j \text{ or } j = k \}$

$S_1 \to \epsilon$
$S_1 \to a S_1 b$

$\underline{a^i b^i} c^k$

$C \to \epsilon$
$C \to c C$

L_1 L_2

$\cup$ $a^i \underline{b^j c^j}$

$S \to S_1$
$S \to S_2$

Union

Left-linear

Right-linear

Regular Grammars

Regular Expression

FSA

Det

Non-Det

$$L = \{ w \in (a+b)^* \mid n_a(w) \text{ is even AND } n_b(w) \text{ is odd} \}$$

$S \to aA$

$S \to bB$

$A \to aS$

$A \to bC$

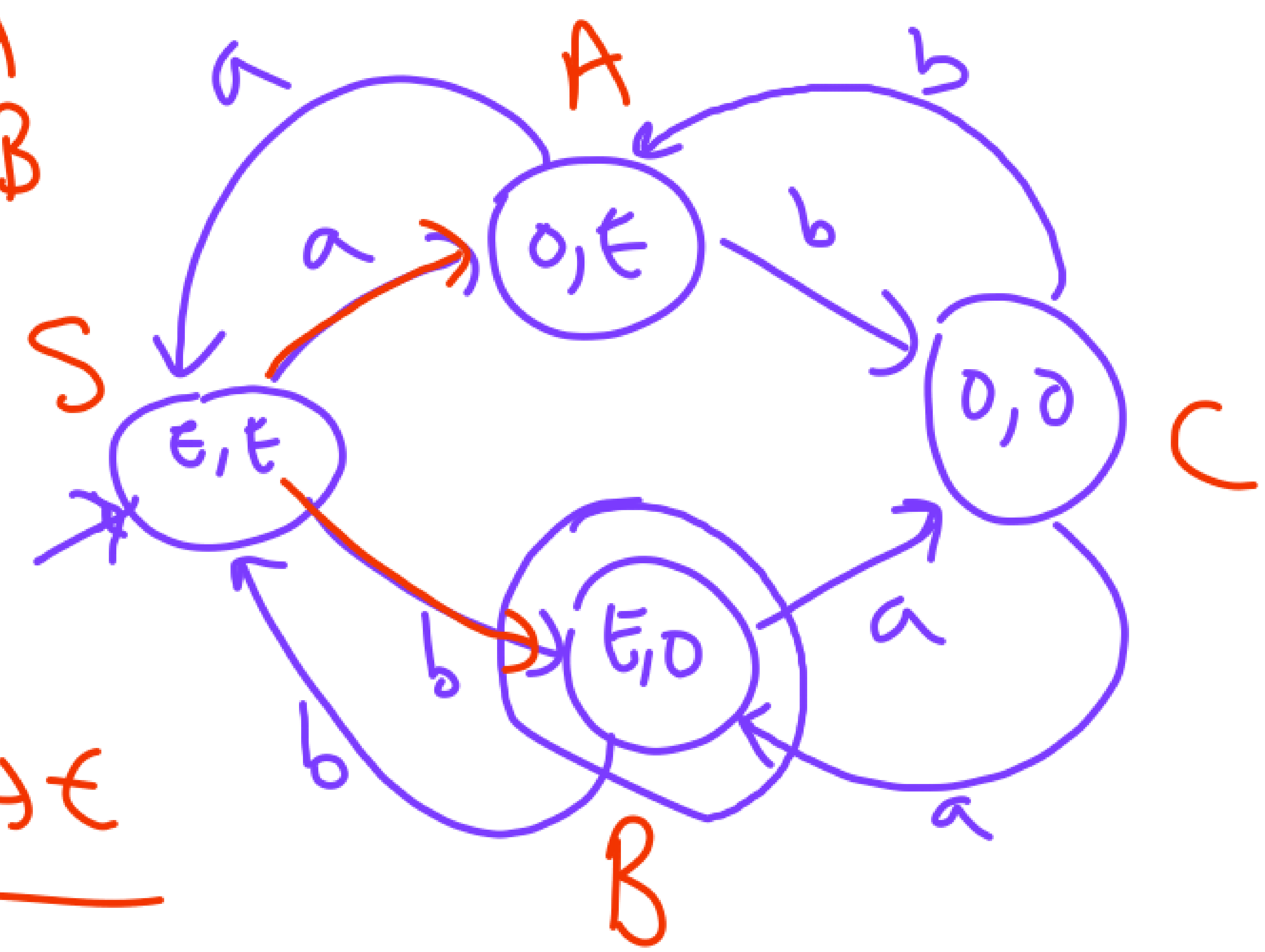
$B \to bS$

$B \to aC$

$C \to bA$

$C \to aB$

$B \to \epsilon$



20/8/25

CS310

- Alphabet, Words, Lang
  Σ, w, $L \subseteq \Sigma^*$
- Grammars $\langle \Sigma, V, S, P \rangle$

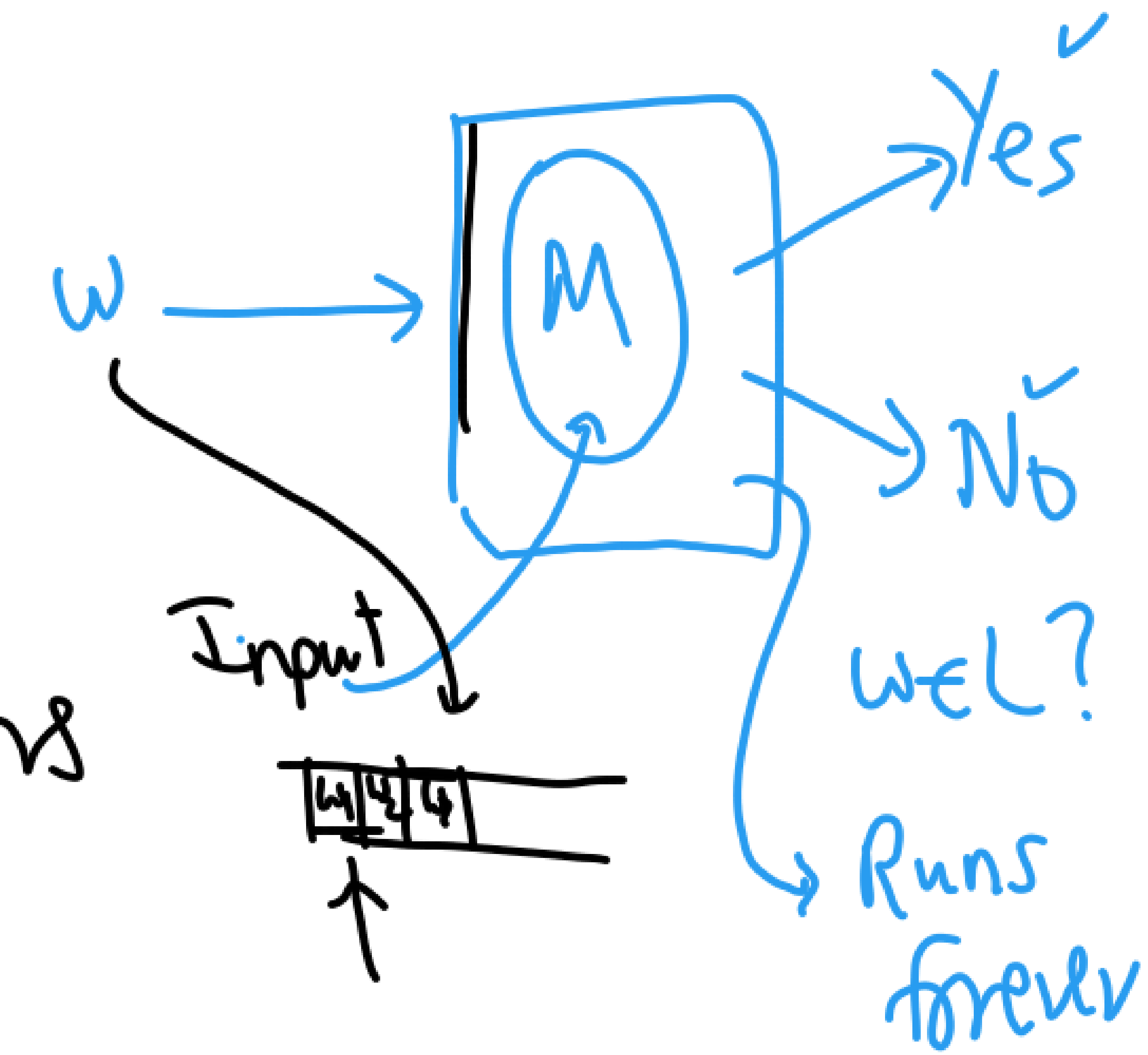
Regular	CFG	Type 0
$\equiv$	$\equiv$	$\equiv$
FSA	PDA	TM

- Machines
  - State
  - Transitions
  - RUN on an Input

Language Acceptors

(1) · Non-Determinism

(2) ϵ - moves



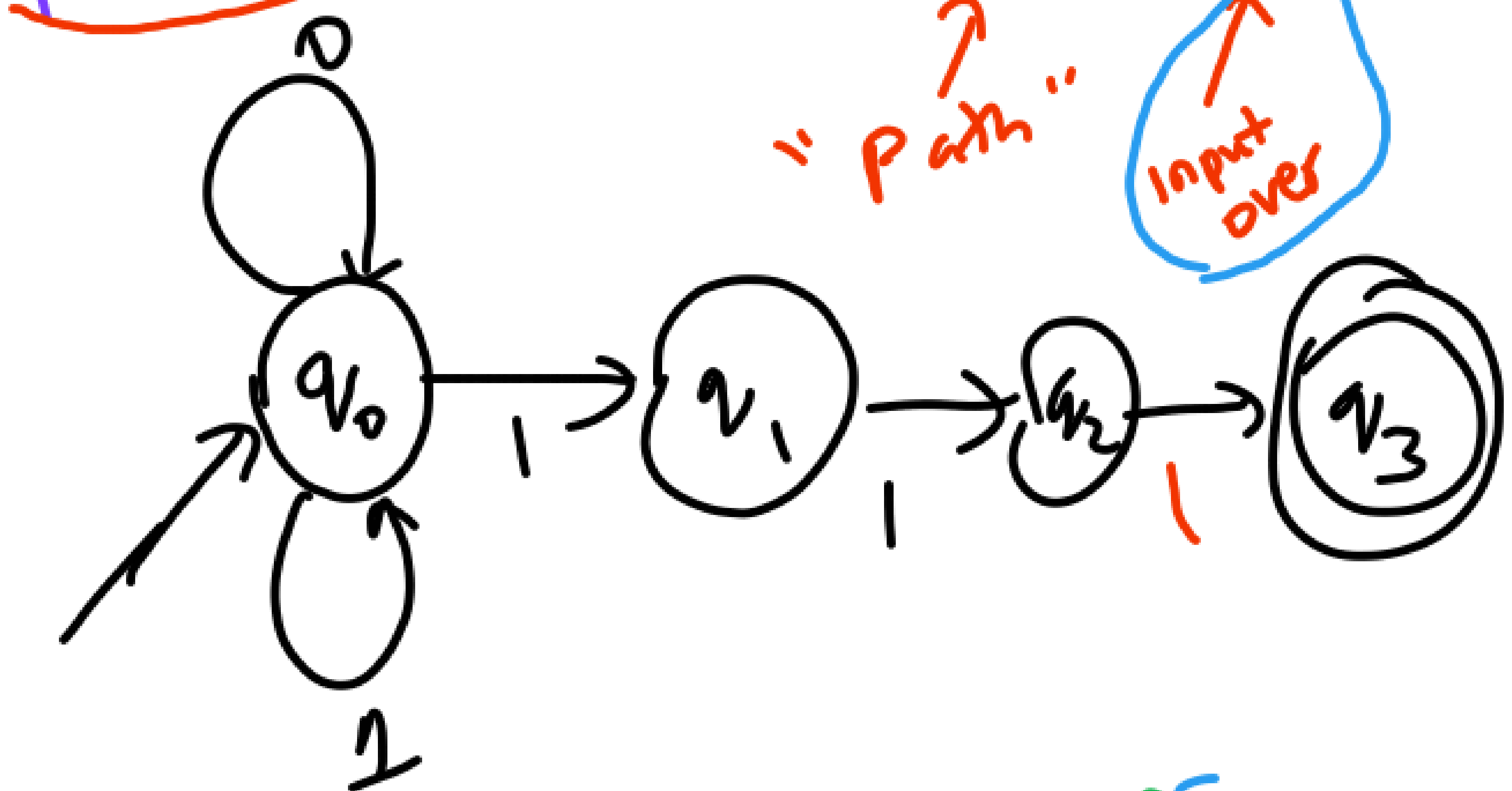
Don't know non-det → try all choices

FSA

M Accepts w → $q_0 w \vdash^* q_f \in$ "path" input over

$q_f \in F$ → double circle

$L = \{ w \in (0+1)^* \mid w \text{ ends with } 3\ 1\text{s} \}$



ID Instantaneous Description → q w_{rest}

Current state | remaining input

Run → Halts → input over / no more possible

start

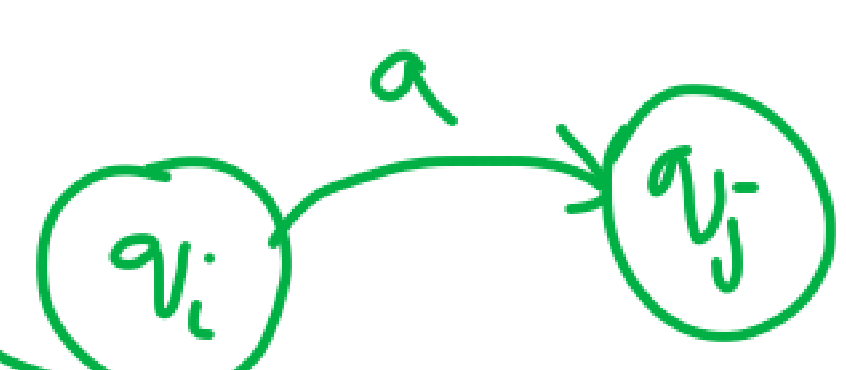
$q_0 w \vdash^* \cdots$

$\underbrace{\vdash^*}_{\text{0 or more steps}}$

1-step

1-step

$q_i a w \vdash q_j w$



One of the runs

$q_0 11011$

$\vdash q_0 1011$

$\vdash q_1 011$

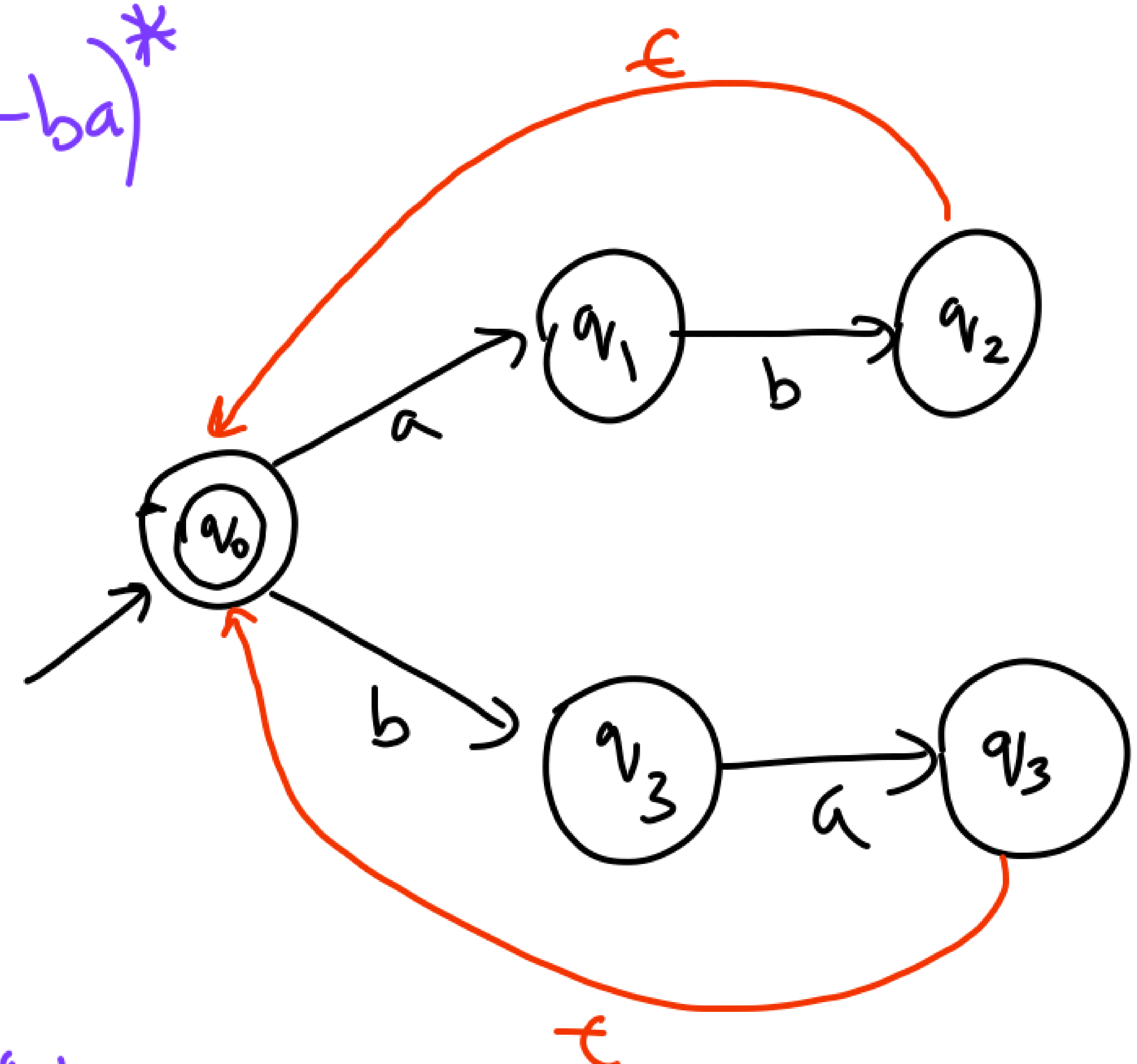
$q_0 10110111$ → acceptance run?

ε-transitions

$L = (ab+ba)^*$

DFA

NFA $\cdot$ - no ϵ



$M = \langle \Sigma, Q, \delta, F \rangle$

- Σ: labels $\Sigma \cup \{\epsilon\}$
- Q: States $q_0 \dots q_n$
- δ: transitions "arrows" from $q_i \xrightarrow{a} q_j$, ϵ
- F: final $F \subseteq Q$

Accepting Run

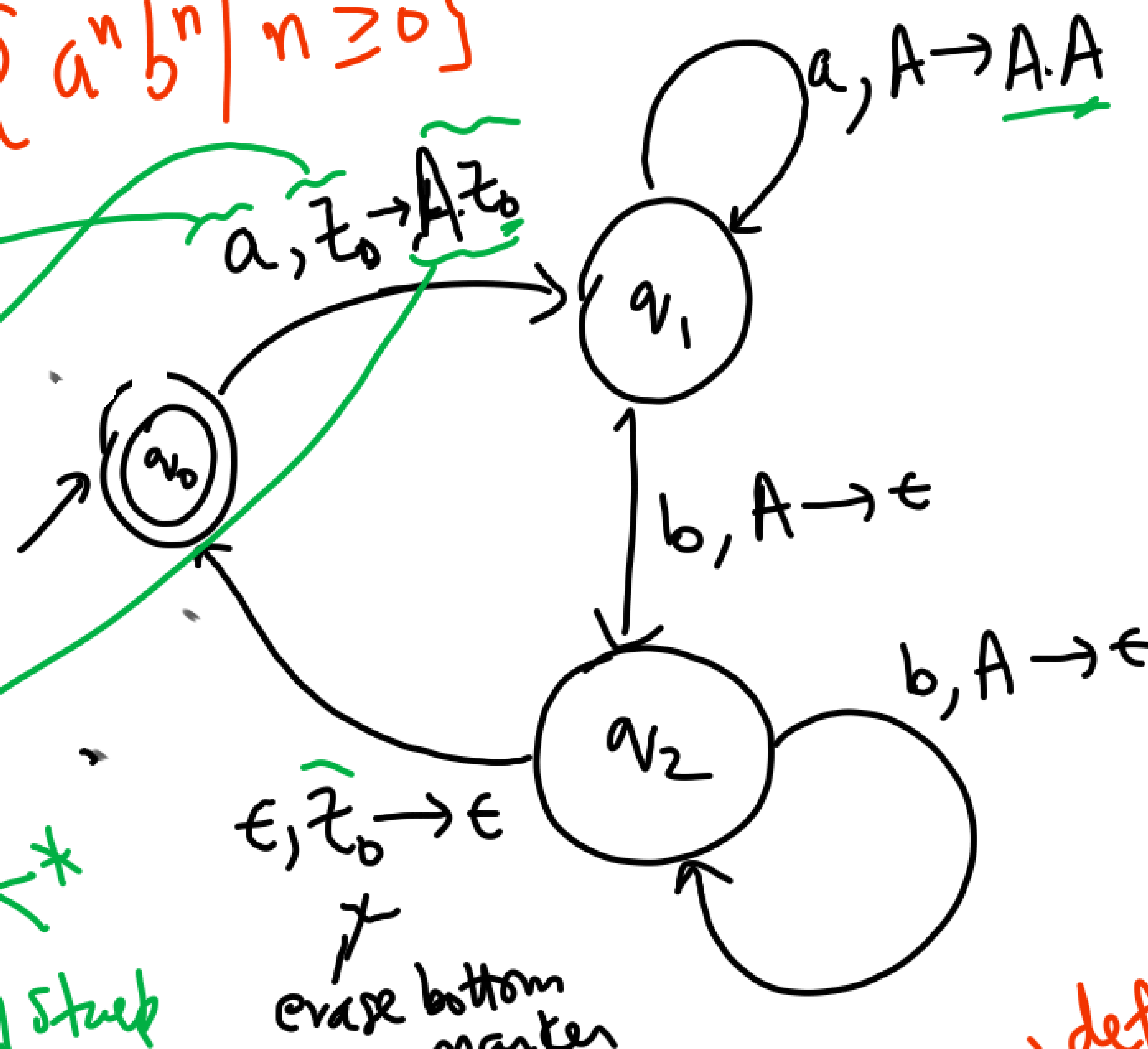
$q_0\,abba \vdash q_1\,bba \vdash \underline{q_2}\,ba \vdash \underbrace{\underline{q_0}\,ba}_{\epsilon\text{-tran}} \vdash q_3\,a \vdash q_3\,\epsilon \vdash \underbrace{q_0\,\epsilon}_{\text{stop}}$

$q_2 \xrightarrow{\epsilon} q_0$

Push Down Automata

Stack

$L = \{a^n b^n \mid n \geq 0\}$



input symbol

top-of-stack

Push string in Γ^* on top of stack

Γ → stack symbols

$\{A, B, C\}$

$\Sigma \subset \Gamma - \{Z_0\}$

$\{a, b\}$

bottom marker

Sample

$a^3 b^3$

a a a b b b ϵ

q_0 ..... $\in \Gamma^*$

accept

defined $a, B \to \alpha$

$q_i \to q_j$

ID $(q_0, w, Z_0) \vdash$

Triple: state, input, stack

$(q_i, aw, B\gamma) \vdash (q_j, w, \alpha\gamma)$

$\gamma \in \Gamma^*$

$(q_0, w, Z_0) \vdash^* (q_f, \epsilon, \alpha)$

anything

input fully consumed

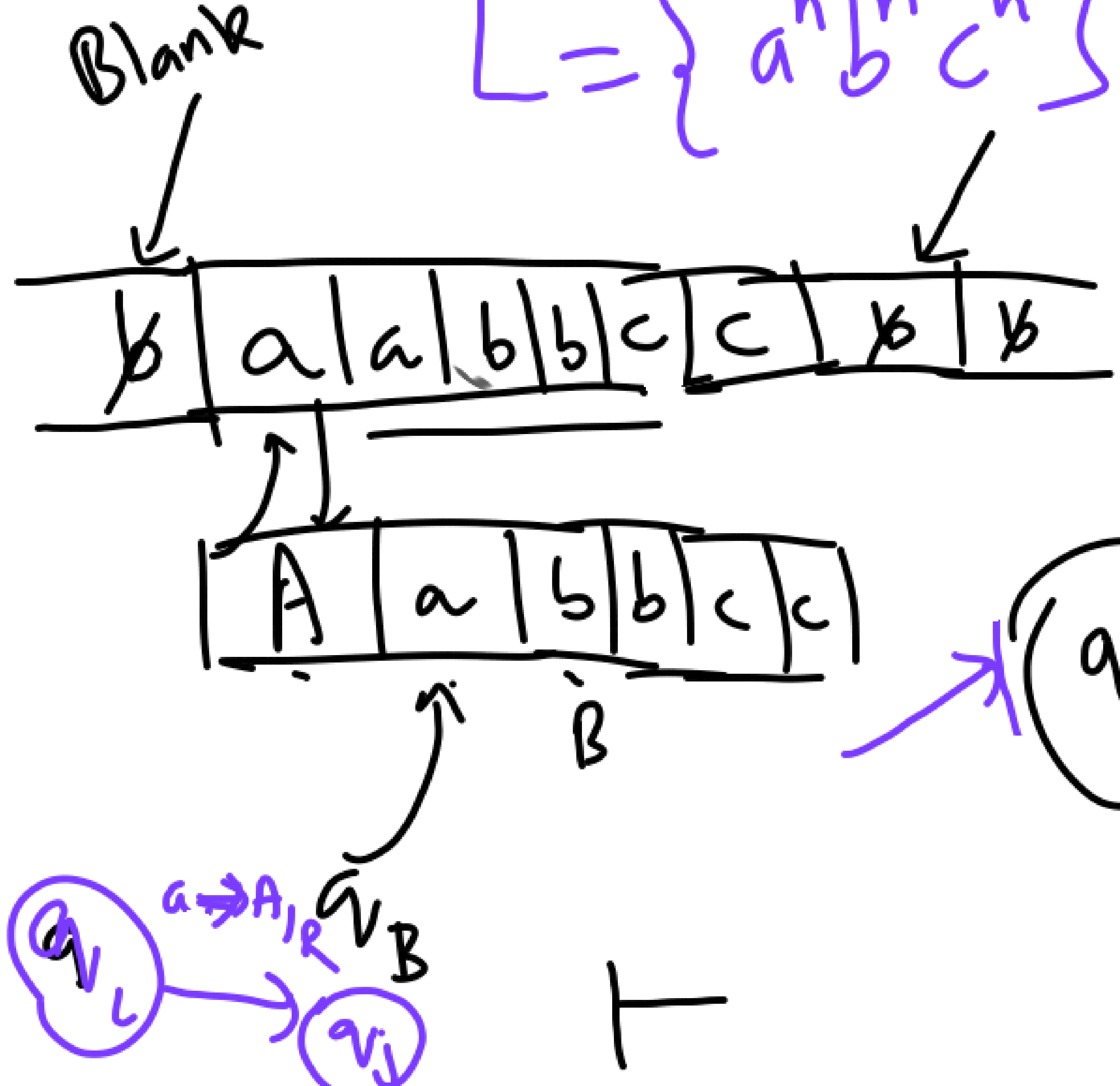
Accept by final state

after all input is consumed

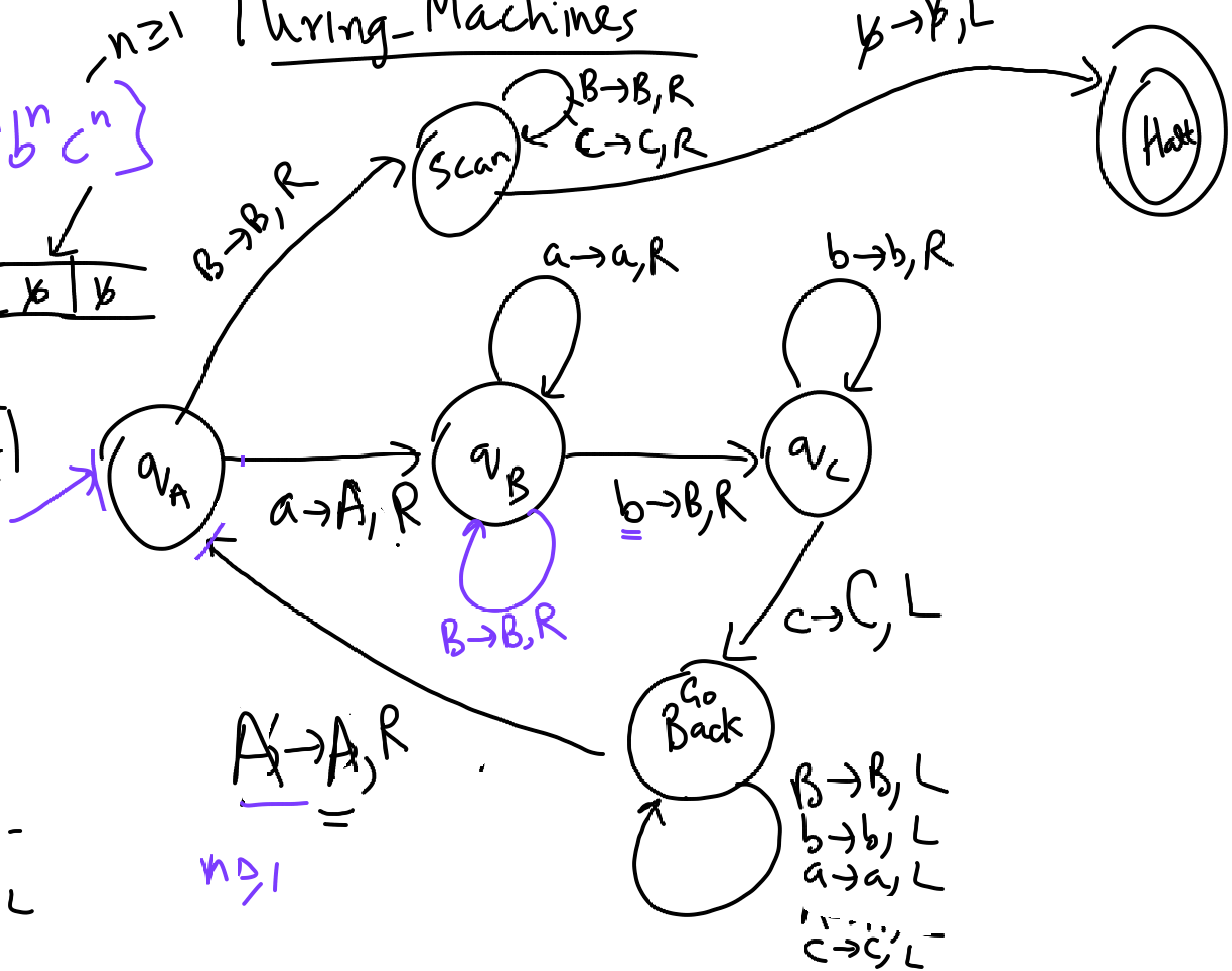
turingmachine.io

Turing Machines

$L = \{a^n b^n c^n\}$ $n \geq 1$



$\underline{w_L}\, q_i\, \underline{a}\, \underline{w_L} \vdash w_L A q_j w_L$



22.8.2025

FSA

M

$\langle \Sigma, Q, \delta, q_0, F \rangle$

δ: transitions

q_0: start

$\to q_0$

$q_i \xrightarrow{a \in \Sigma \cup \{\epsilon\}} q_j$

Systematic

NFA+ϵ $\to$ $\delta : Q \times \Sigma \cup \{\epsilon\} \to 2^Q$

any subset incl. $\emptyset$

$\Updownarrow$

No ϵ-transitions

NFA

$\Updownarrow$

function $Q \times \Sigma \to Q$

DFA $\to$ exactly one next state

PDA

$\Sigma \subseteq \Gamma - \{Z_0\}$

$\langle Q, \Sigma, \Gamma, \delta, q_0, Z_0, F \rangle$

Γ: stack symbols

Z_0: bottom of stack

$q_i \xrightarrow{a,\, t \to \alpha} q_j$

a: input tape symbol, $\Sigma \cup \{\epsilon\}$, empty

t: top of stack, $t \in \Gamma$

α: replacement string, $\alpha \in \Gamma^*$

$\delta : Q \times \Sigma \cup \{\epsilon\} \times \Gamma \to 2^{Q \times \Gamma^*}$

can replace only top symbol

at most one next state

no choice $\Rightarrow$ deterministic

no arrow for a, t is ok

TM

Input tape

$\not b$	a	---

$\Sigma \subseteq \Gamma - \{\not b\}$

$\langle Q, \Sigma, \Gamma, \delta, q_0, \not b, F \rangle$

Γ: tape symbols

$\not b$: blank

$q_i \xrightarrow{a \to t, L} q_j$

t: "change 1-symbol"

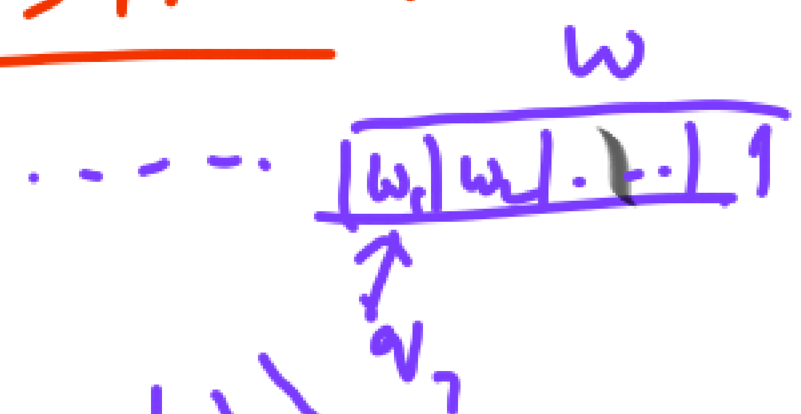
L (or R): move 1-step

$\delta : Q \times \Gamma \to 2^{Q \times \Gamma \times \{L, R\}}$

crash if no move possible

$\emptyset$ = no move possible

FSA M

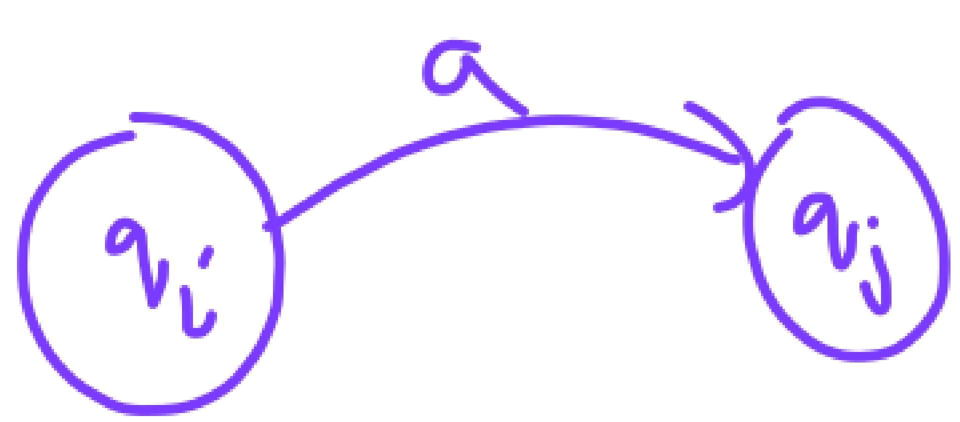


ID $\langle q_1, w \rangle$

Move $\vdash$

$ID_1 \vdash ID_2$

① $\langle q_i, aw \rangle \vdash \langle q_j, w \rangle$



② $\langle q_i, aw \rangle \vdash \langle q_j, aw \rangle$

③ $\langle q_i, \epsilon \rangle \to \langle q_j, \epsilon \rangle$ $q_i \xrightarrow{\epsilon} q_j$

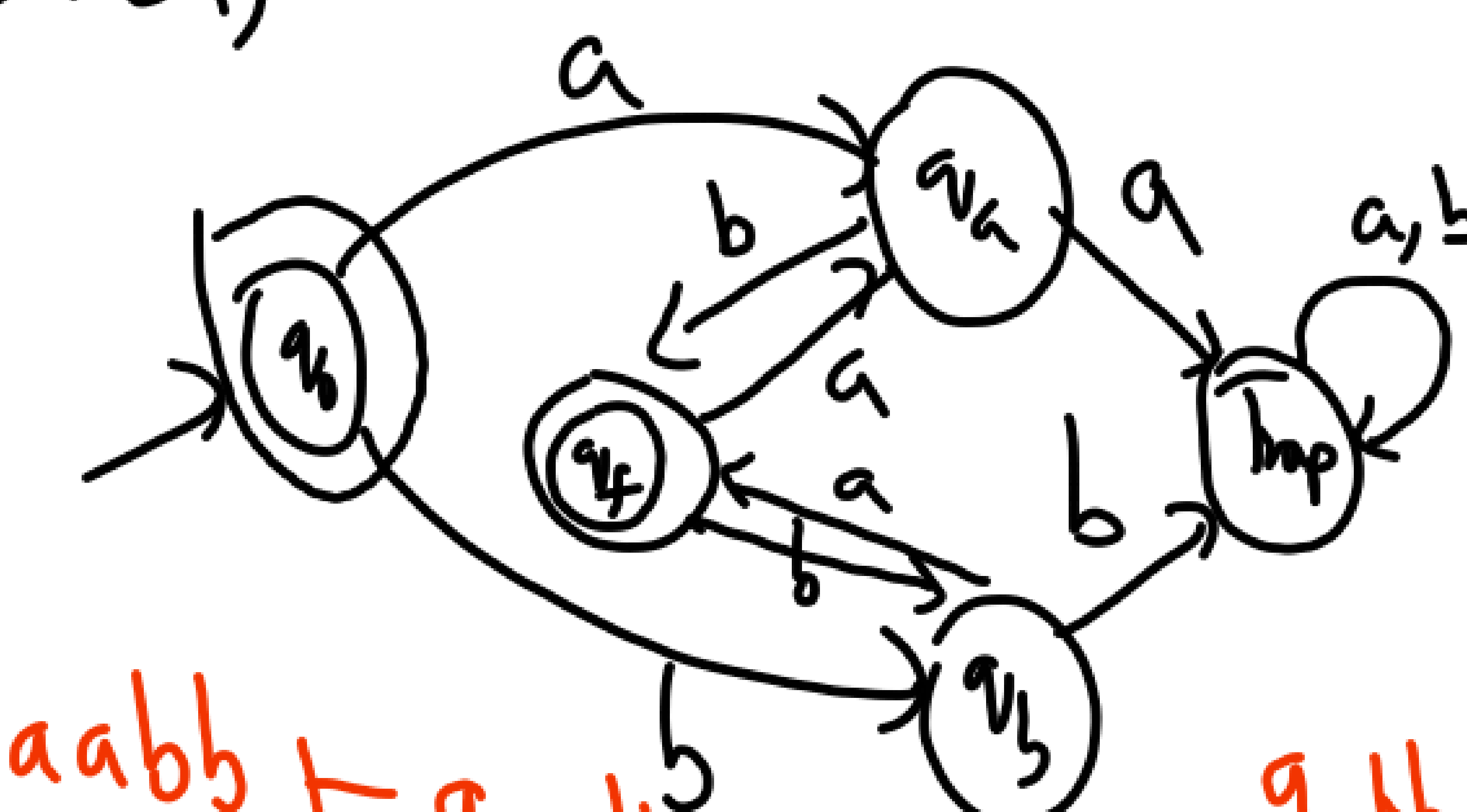
Run is $\vdash^*$ any number of steps

$L(M) = \{ w \mid M \text{ accepts } w \}$ try all for non-det M

there is a run $\langle q_0, w \rangle \vdash^* \langle q_f, \epsilon \rangle$

$q_f \in F$

$(ab+ba)^*$



$q_0 baabb \vdash q_b aabb \vdash q_f abb \vdash q_a bb \to q_f b \vdash q_b$

~~PDA~~ M

ID $\langle q, w, \alpha \rangle$, $w \in \Sigma^*$, $\alpha \in \Gamma^*$

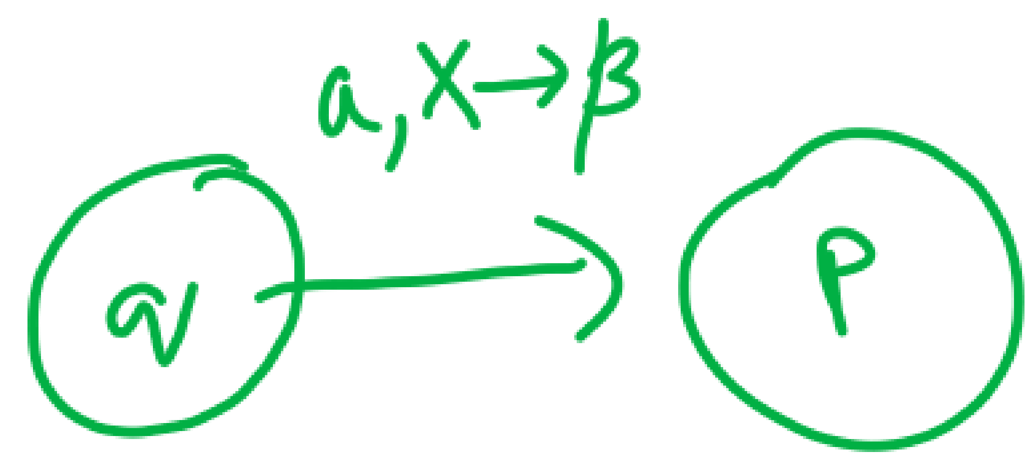
$L(M) = \{ w \mid M \text{ accepts } w \}$ — by final state

$\langle q_0, w, Z_0 \rangle \vdash^* \langle q_f, \epsilon, \alpha \rangle$

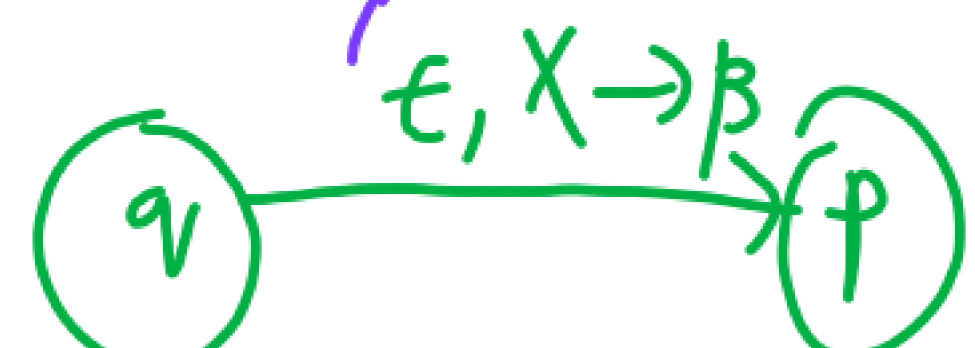
q_f in F; ϵ: empty (all input processed); α: anything on stack

a: input symbol; X: top of stack

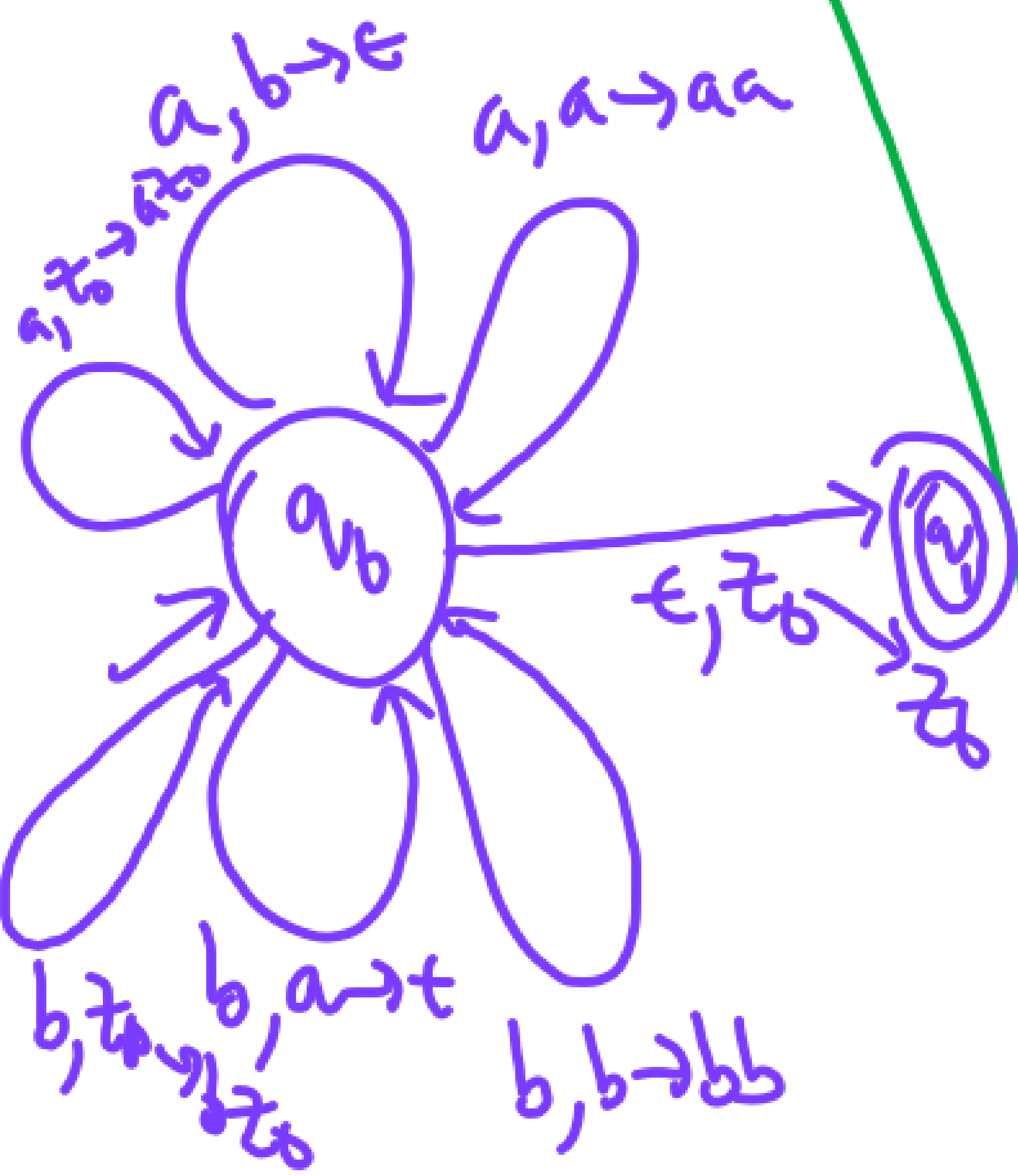
(1) $\langle q, aw, X\alpha \rangle \vdash \langle p, w, \beta\alpha \rangle$



(2) $\langle q, aw, X\alpha \rangle \vdash \langle p, aw, \beta\alpha \rangle$



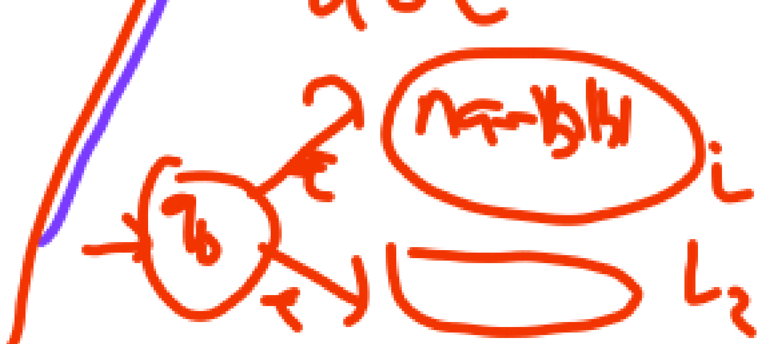
equal # a's b's



$\{ a^i b^j c^k \mid i = j \text{ or } j = k \}$

$L_1 \cup L_2$

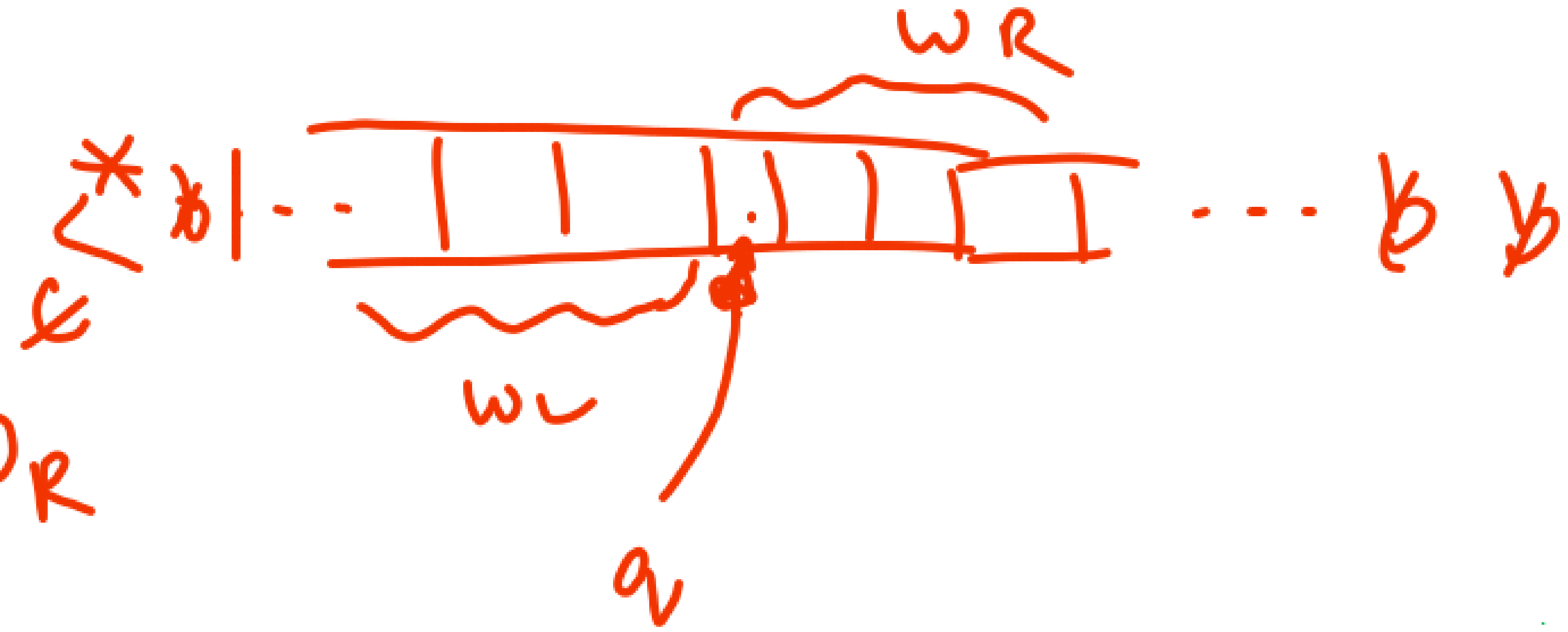
$a^i b^i c^k$ $\quad$ $a^i b^j c^j$



TM

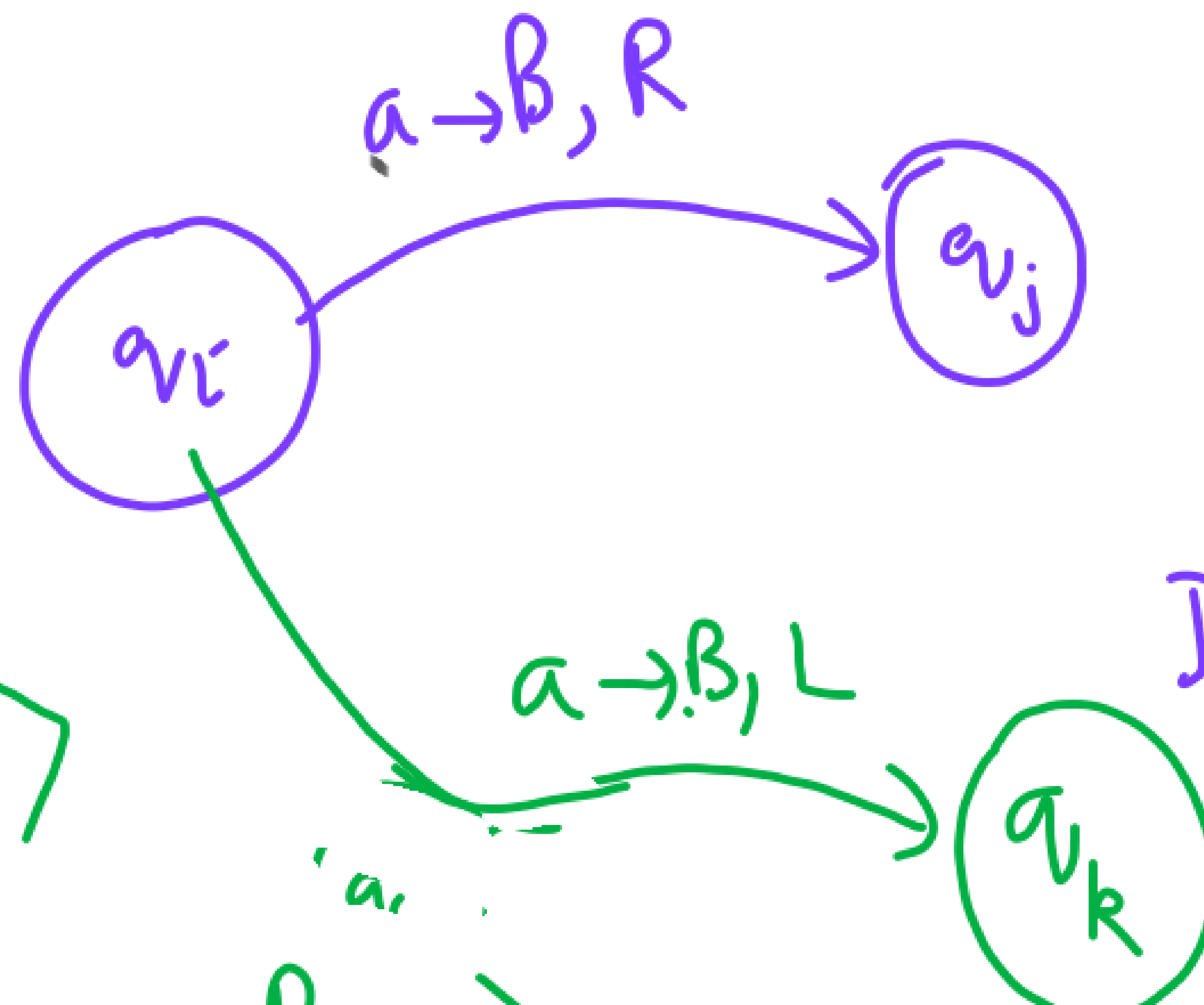
ID $w_L \; q \; w_R$

$\vdash^*$



Move $\vdash$

$$w_L \, q_i \, a \, w_R \vdash w_L \, B \, q_j \, w_R$$



$w_L | c | a | w_R$

q_i

last symbol of left

$$\langle w_L \, c \, q_i a w_R \rangle$$

$$\vdash \langle w_L \, q_k c B \, w_R \rangle$$

$w_L | c | B | w_R$

q_k

accepts

$$q_0 \, w \vdash^* \underline{x_1} \, q_f \, \underline{x_2}$$

any string

any string

Final state

(Halt)

(no move possible)

turing machine .jb	
	a,b
ww	
w#w given	

Do without #

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CS 310

① FSA

Closure

Pumping Lemma

② PDA → accept by
- empty stack
- final state

DPDA vs NPDA

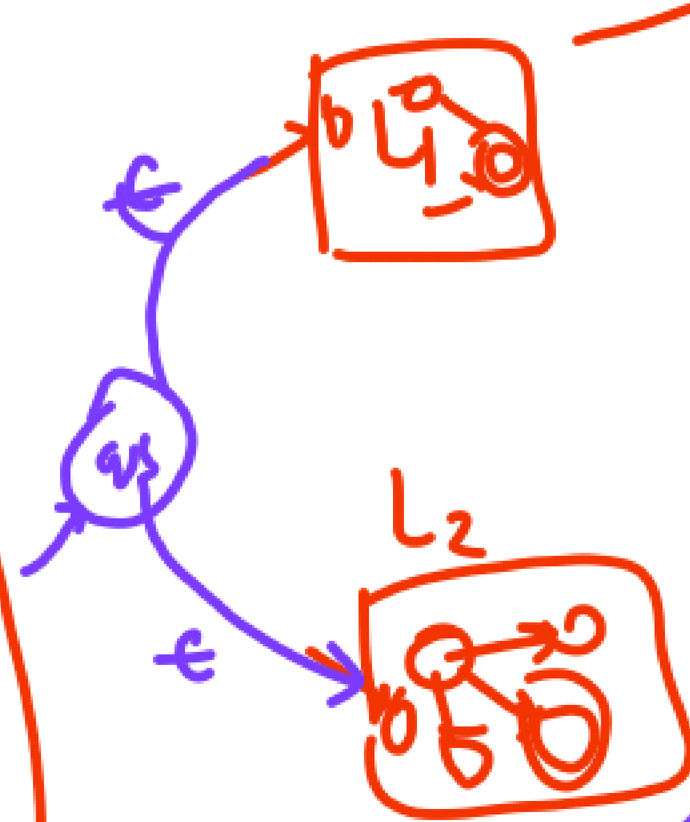
Minimize

Product Automata

DFA
M_1 — 6 states

DFA
M_2 — 7 states

Given L_1 L_2 regular



- ✓ Complement $\overline{L_1}$ — DFA+FLIP L_1
- ✓ Union $L_1 \cup L_2$
- ✓ Intersection $L_1 \cap L_2$
- ** ✓ Kleene Closure L_1^*
- ✓ Concat. $L_1 L_2$

- Difference $L_1 - L_2$ — Symm. Diff
- Reverse L_1^R
- Half
- Odd
- Even
- ⋮
- Alternate

Pumping Lemma

$L = a^n b^n \mid n \geq 0$ is not regular

Proof by Contradiction

You

A

Adversary

B $\xrightarrow{1}$ L is regular

~~I then~~

2 Please build DFA & tell me how many states

348

3 Split $a^N b^N$ — N-states

loop

$u v w$

repeats state

$a^{400} b^{400}$

$|uv| \leq N$

Then $u v^i w \in L$ for $i \geq 0$

$q_0 \xrightarrow{a} q_7 \xrightarrow{a} q_{13}$ $q_i \longrightarrow q_i$

k - a's were used

23

PDA

$L = \{ a^n b^m a^n \mid n, m \ge 0 \}$

final state

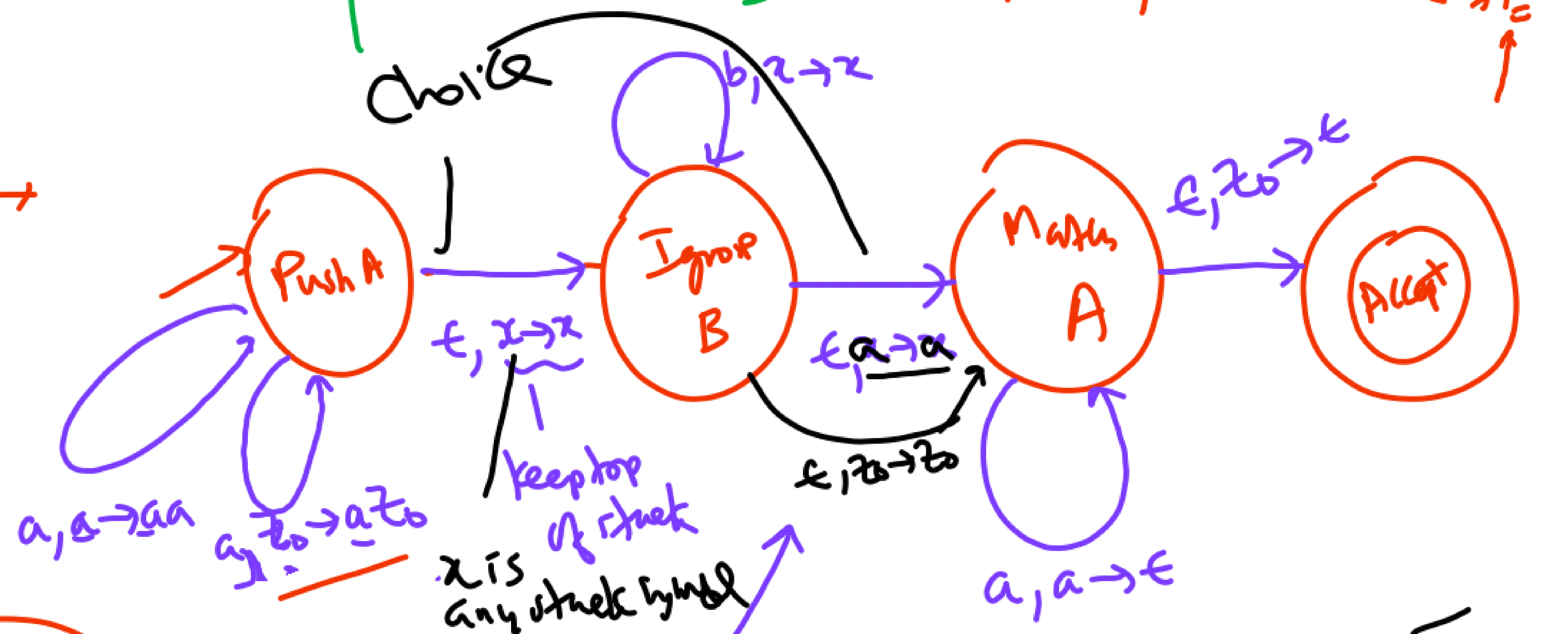
$(q_0, w, Z_0) \vdash \cdots \vdash (q_f, \epsilon, \alpha)$

1 $S \to T$

2 $S \to aSa$

$T \to bT$

$T \to \epsilon$



NPDA → Non-deterministic

(b-states)

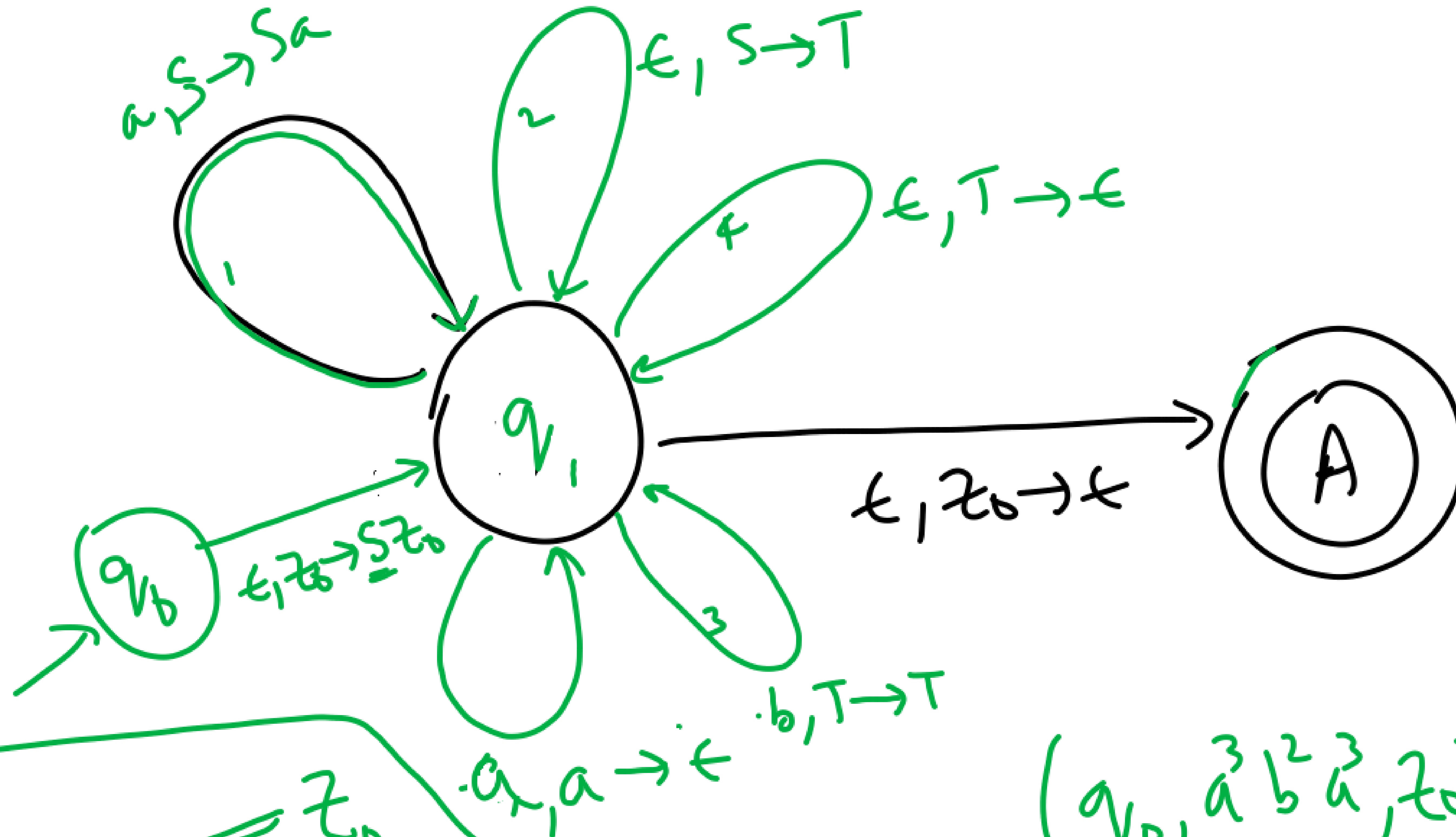
atmost one choice

PDA Deterministic

exactly 1 choice

$(q_0, a^3 b^2 a^3, Z_0) \vdash (q_0, a^2 b^2 a^3, aZ_0)$

$\vdash \quad \vdash \quad \vdash (q_0, b^2 a^3, a^3 Z_0) \vdash (q_1, b a^3, a^3 Z_0)$



Σ alphabet

Z_0

1. $S \to aSa$
2. $S \to T$
3. $T \to bT$
4. $T \to \epsilon$

$(q_0,\ \omega,\ Z_0)$

bottom of stack

$(q_0, a^3 b^2 a^3, Z_0)$

$\vdash$

$\vdash$

$\vdash (A, \epsilon, \epsilon)$

30.8.2025

Bits / Alphabet

CS 310

→ G. Sivakumar

siva@iitb.ac.in

00111 01111 00100 01001 10011 ...

Encoding

G O D I S N O W H E R E

"stand on the shoulders of giants"

Natural Language → AMBIGUITY

Formal → we define the syntax, semantics

3 models

Finite State Automata FSA

Push Down " PDA

Turing machine TM

Computer Science & Engineering

What

abacus

Slide rule

Casio FX

mobile phone

⋮

Turing test

$3x^2 - 5x + 7 = 0$

Hilbert's 10th problem

$\to 7x^3y^2 - 9x^2y^3 + 13xyz + \ldots = 0$

undecidable

→ Is it divisible by 7?

→ Give a solution (diagram) that any 10-year old can use to answer in "real-time"

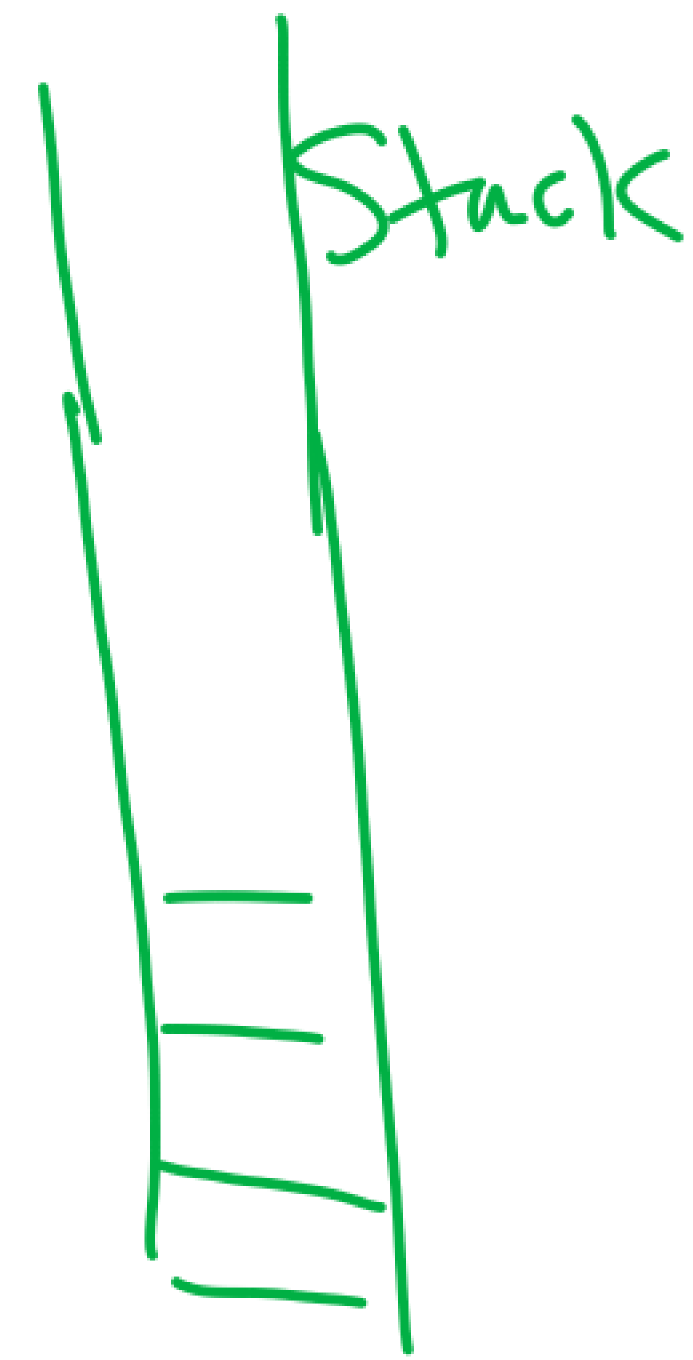
FSA

PDA

Is the string of the form $7^n 9^n$?

$n \geq 1$

Input Tape



Is it a palindrome?

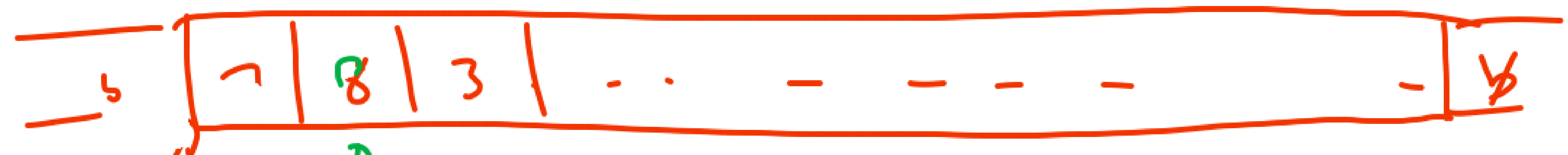
$\longrightarrow$ non-determinism (guess)

Turing machine

turingmachinesimulator.io

Input tape

Read/write



① replace a digit by some symbol.

② move 1-step L or R   q   k   q

Is n prime?

Machines	Language
FSA	Regular Expression
PDA	Context free Grammar
TM	Type 0 grammar

Chomsky hierarchy

$(0+00)1^*$

$S \to aSb$

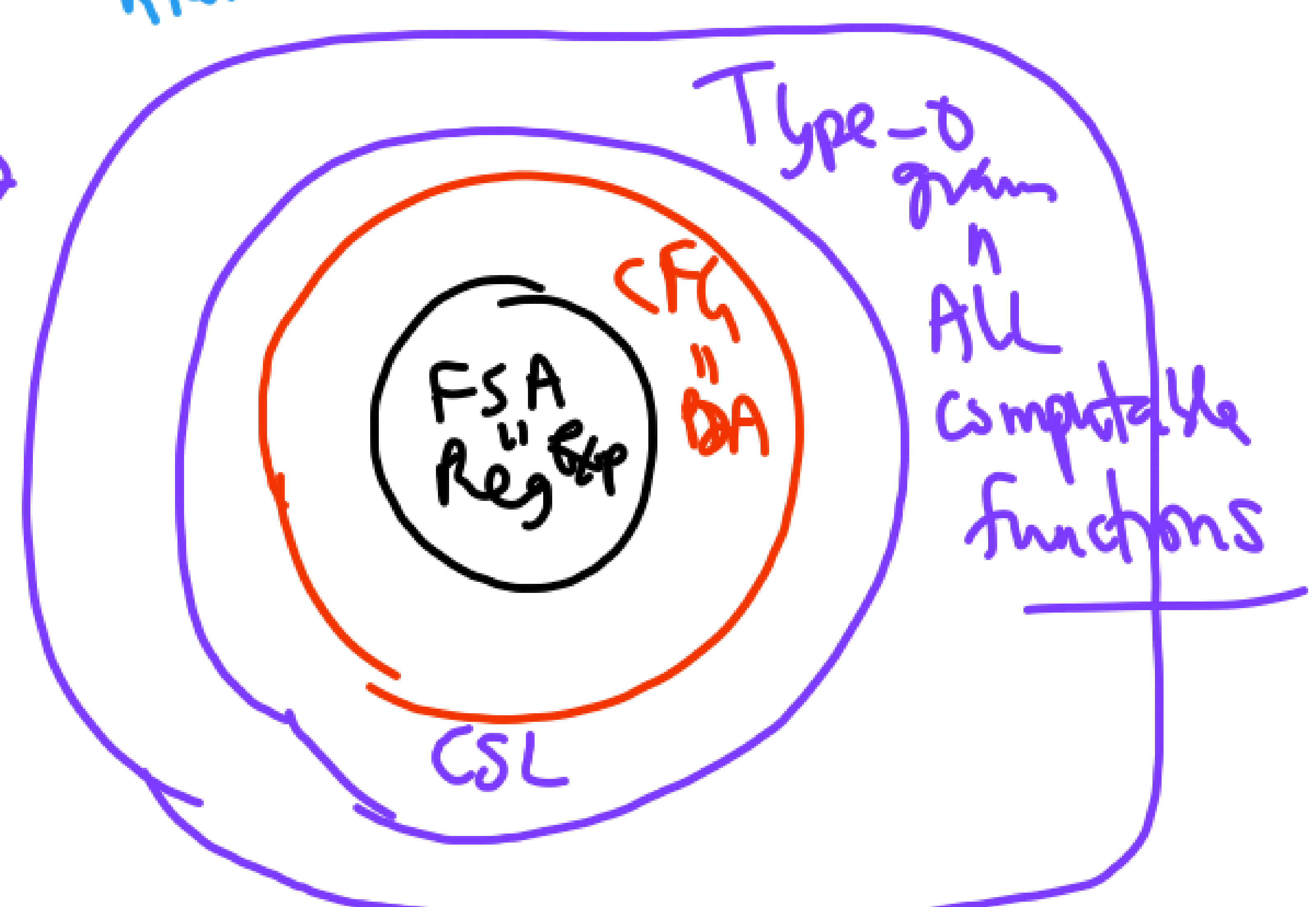
$S \to ab$

$B \to cb$

$a^n b^n c^n$

lhs can have any string

$aSC \to Sbc$



12.9.2025

CS 310

Languages — Machines

→ Closure Properties

→ Determinism

Canonical (Normal) Forms

① (NFA+ε) → (NFA) → DFA → RegExp → (NFA+ε)

10 mins

10 mins

② CFG → CHOMSKY, Greibach

50 mins

③ PUMPING Lemma

MIDSEM

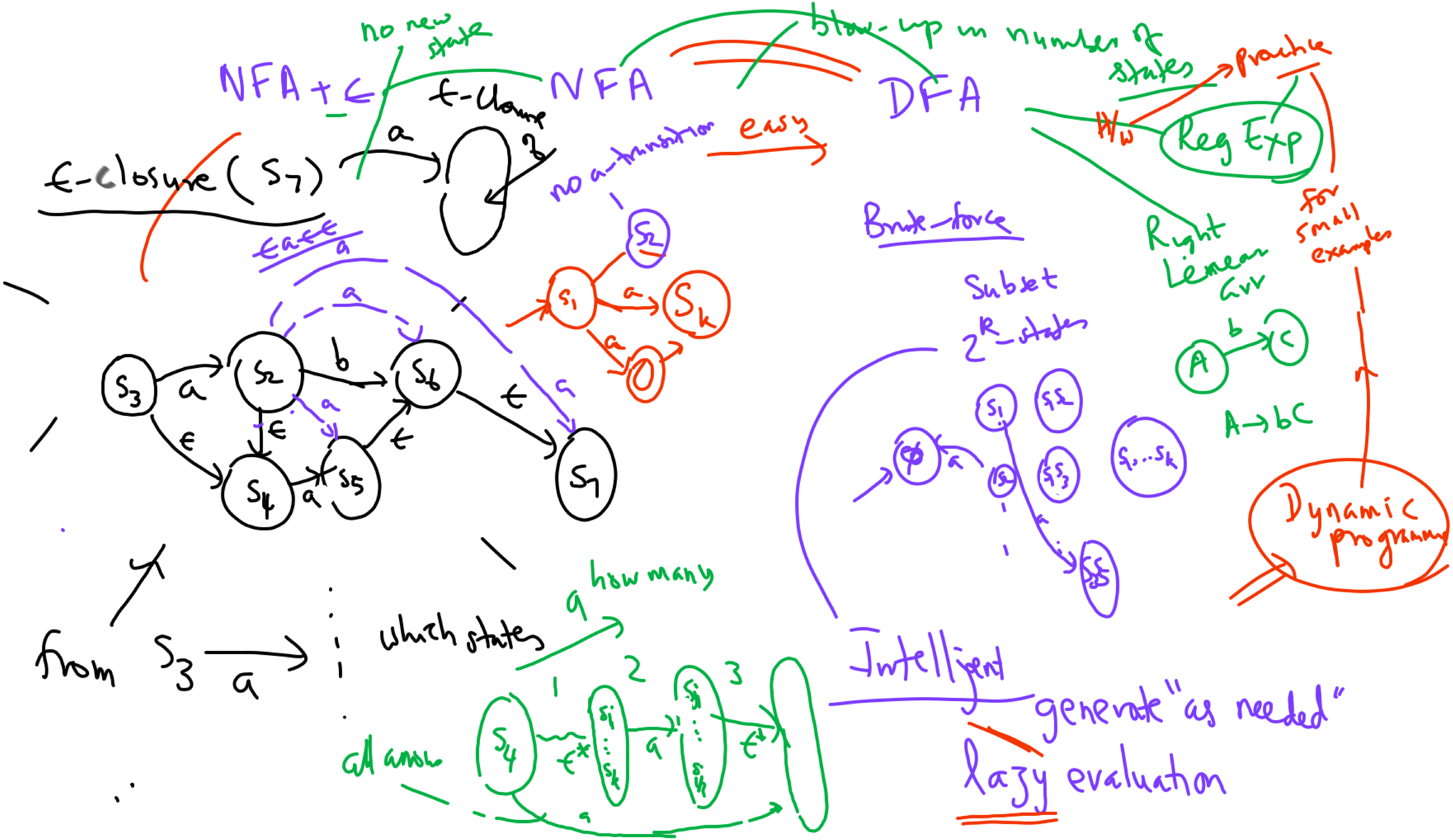
→ Sat 20/9

~~LA 001, 002, 207~~

Ensemble

Federated

Moodle



NFA + ε
no new state
ε-closure
NFA
blow-up in number of states
DFA
practice
easy
no a-transition
ε-closure(S7)
Reg Exp
Right Linear Grr
for small examples
Brute-force
Subset
2^k states
A → bC
Dynamic programming
from S3 —a→ which states
how many
all arrows
Intelligent
generate "as needed"
lazy evaluation

CFG Normal forms (Σ, V, S, P)

$S \to ABCD$

$S \to AX_1$

$X_1 \to BX_2$

$X_2 \to CD$

Chomsky

for any $A \in V$

$c \in \Sigma$

type 1 (T1) $A \to c$

$B, D \in V$

type 2 or (T2) $A \to BD$

only 2 types of rules

G

$S \to aSb$

$S \to ab$

$X_i \; X_j$

$X_2 \to b$

$X_1 \to a$

$\cancel{S \to X_1 S X_2}$

$S \to X_1 S_2$

$S_2 \to S X_2$

$S \to X_1 X_2$

Sheila Greibach

only 1 type of rule

$A \to c\alpha$ — α: anything

$c \in \Sigma$ — one alphabet

$\alpha \in (V \cup \Sigma)^*$

G

non-chomsky rules

③ ε-rule

$D \to \epsilon$

④ unit rule

$A \to B$

Useless Symbols

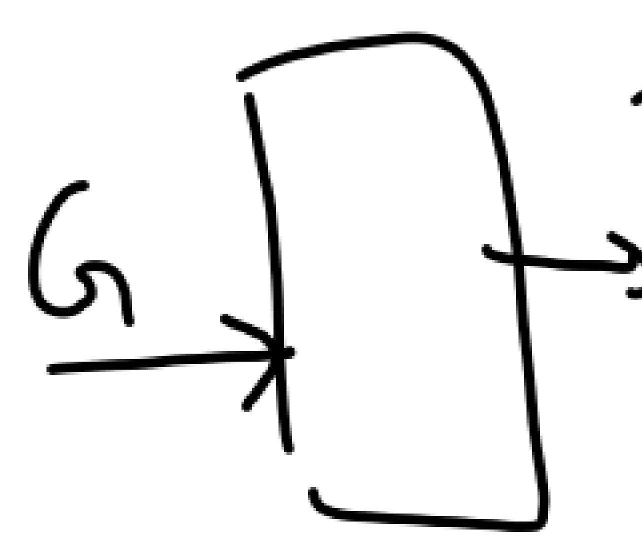
Removal by adding "extra" rules

Useful iff both productive & reachable

① productive: $A \xrightarrow{*} w \in \Sigma^*$

② reachable: $S \xrightarrow{*} \alpha_L \, A \, \alpha_R$

Iterative Fix-Point Method



find all productive symbols

1-step

① $P_0 = \{$ all symbols $D \to \alpha \in \Sigma^* \}$

② $P_{i+1} = P_i \cup \{$ all symbols whose rules have RHS $\in P_i$ all symbols $\}$

③ until $P_{i+1} = P_i$

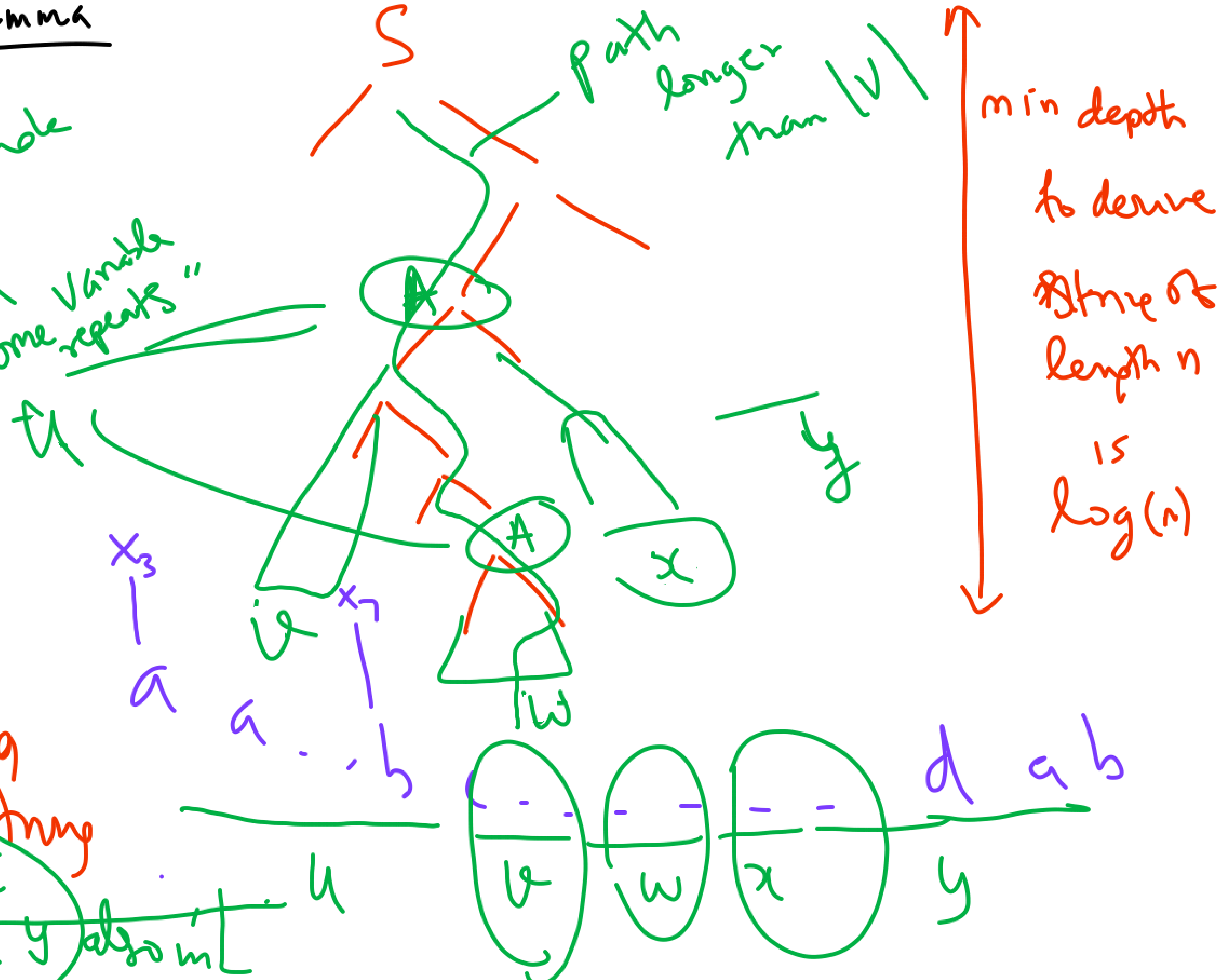
Pumping Lemma

Chomsky Normal Form

Derivation tree

Pigeon hole

"Some variable repeats"



long string

$uv^i w x^i y$

algorithm

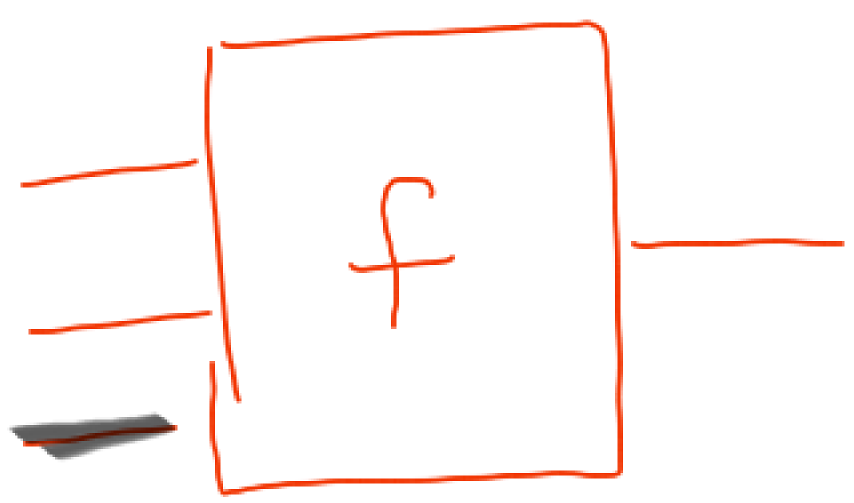
22.9.25

"Computer"

$N=\{1,2,\dots\}$

Inputs

Output



function

$N \to N$

$N^k \to N$

Language $\subseteq \Sigma^*$

$\{0,1\}$

Reg
CFG

Pumping Lemma

there is a language that a TM cannot ACCEPT

① define

I.D.

TM

$\overset{\text{tape}}{\alpha}\ q_k\ \overset{\text{right}}{\beta}$

$M = \langle Q, \Sigma, \Gamma, \delta, q_1, b\!\!/, F \rangle$

transitions

$Q \times \Gamma \to Q \times \Gamma \times D$

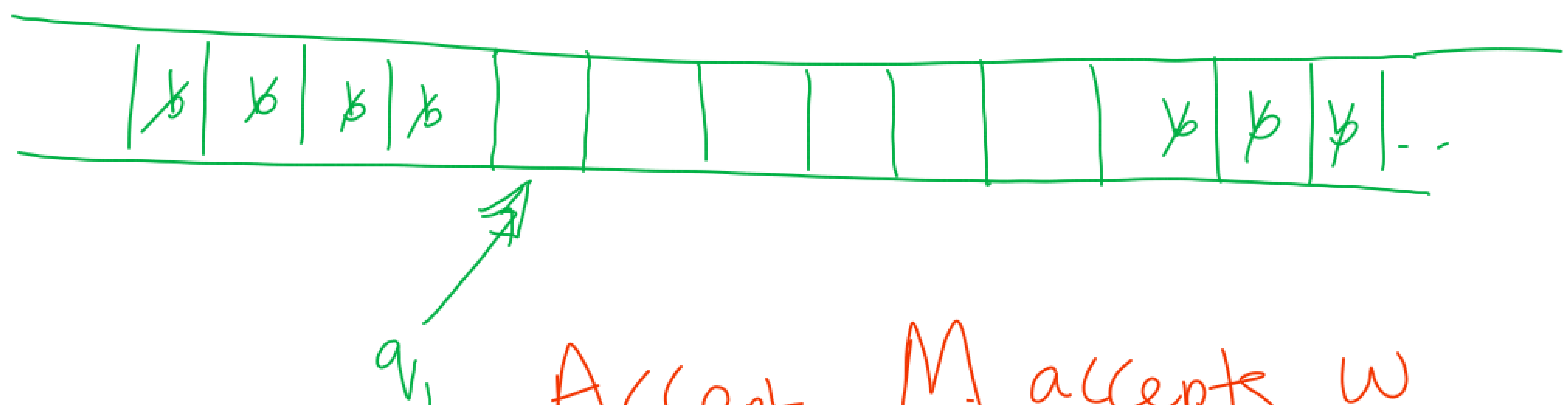
tape symbols

direction L, R

$q_1, w \vdash \dots \vdash \;\vdash$

Start

w



State transitions

Accept

M accepts w

① Final state

② machine Halts

$F = \phi$ (no final states)

$q_1 w \vdash \dots \vdash \alpha q_k \beta$

no further move

$q_0 w \vdash \dots \vdash \alpha q_k \beta$

$q_k \in F$ q_k

TM $\rightarrow$ Unique Bitstring

Encode a TM as a bitstring

$\Sigma = \{0, 1\}$

$a_1 \; a_2 \cdots a_k$

$\Gamma = \{X, Y \ldots\}$

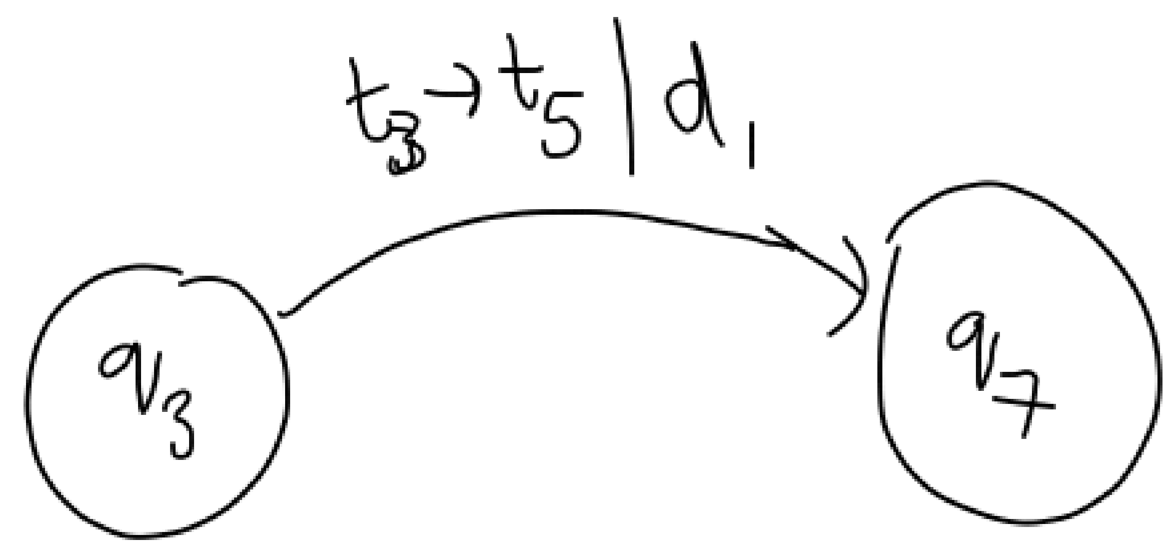
$t_1 \; t_2 \ldots$

$D = \{L, R\}$

$d_1 \; d_2$

$\|\| \quad \| \quad \underbrace{\;000\;}_{3}1\underbrace{\;0000000\;}_{7}1\underbrace{\;000\;}_{3}1\underbrace{\;00000\;}_{5}1\underbrace{\;0\;}_{1} \quad \|$

Single move



$\{1, 2, \ldots\}$

Bitstring $\longleftrightarrow \mathbb{N} \longleftrightarrow \mathbb{N}^k$

Unique prime factorization

ϵ — 1 →

0 — 10 → 2

1 — 11 → 3

00 → 100 → 4

01 → 101 → 5

⋮

10 → 2

9 → 1001 → 9

01001

11001

101001

41 → 101001 → 41

11001

Binary form of Decimal

1	ϵ	1
2	0	2
3	1	3
4	00	4
5	01	5
6	10	6
7	11	
8	000	
9	001	
10	010	
11	011	
	100	

leading 1

Decide $w \in L$? Yes / No

Languages

Recursive = halts on all inputs / in accept state if $w \in L$

& halts in non-accept state if $w \notin L$

L is Recursive Enumerable: if there is a TM M such that M accepts L

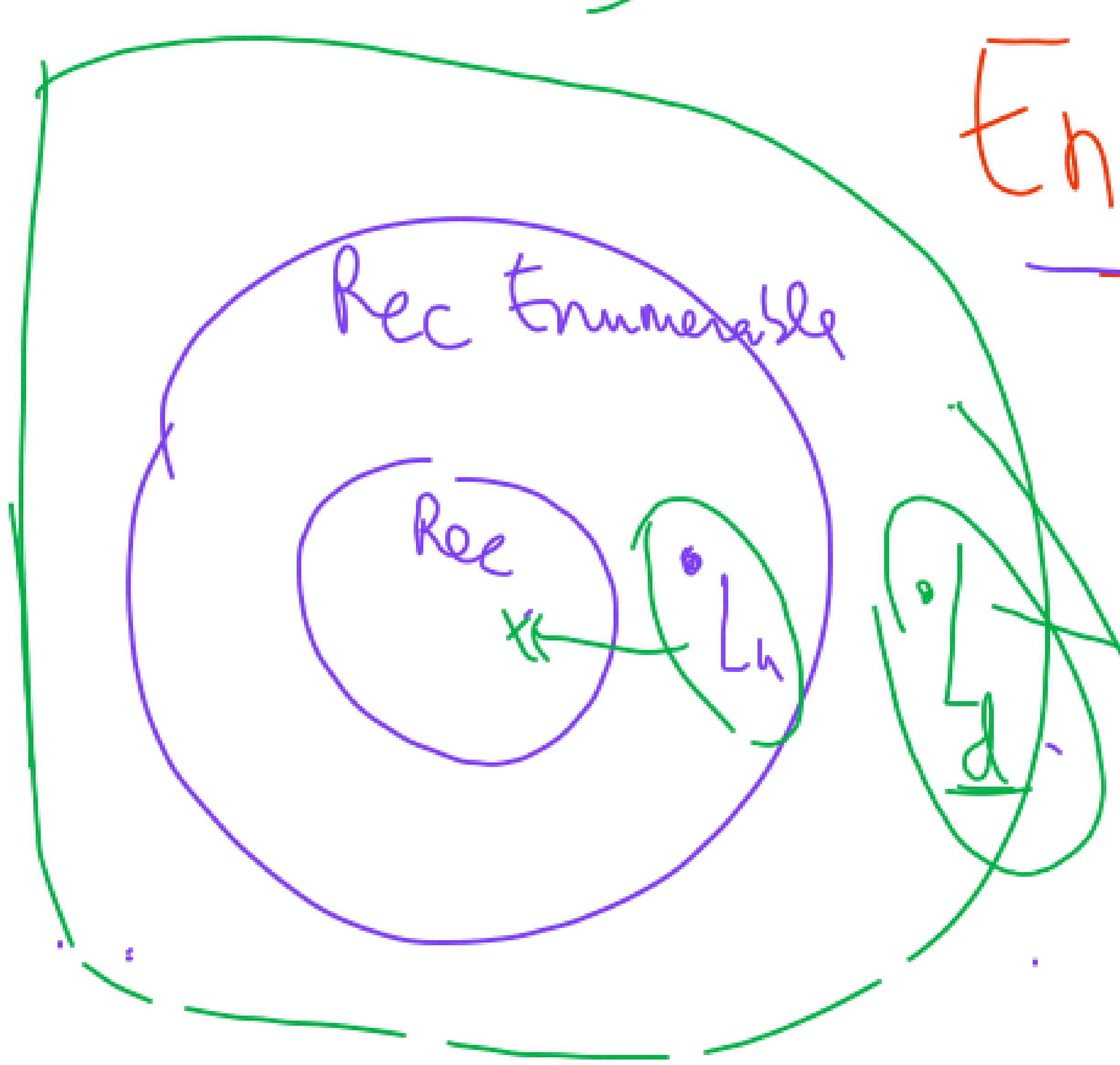
prime?

Programming

$7x^3y^2 + \dots \stackrel{?}{=} 0$

Hilbert's 10th prob

Semi-decidable



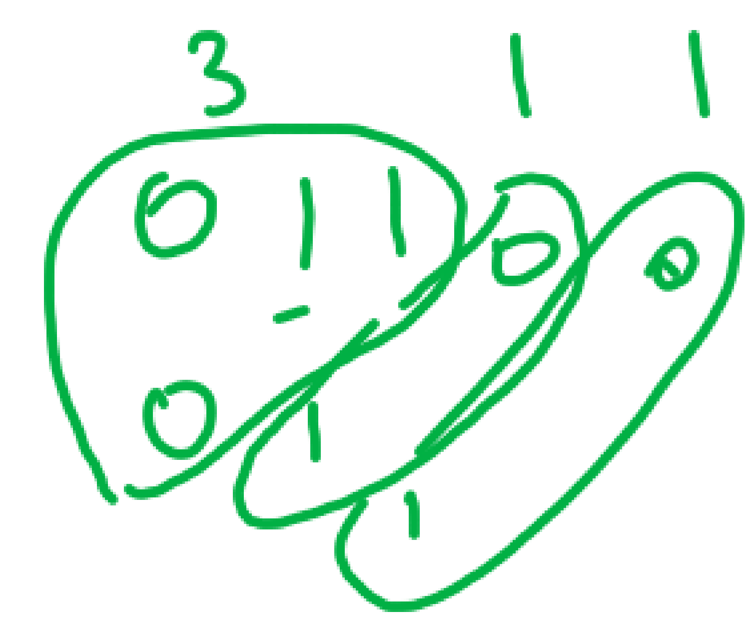
"no TM can be built"

Undecidable

1.10.2025

(Emily)

· **Post** Correspondence Problem



· Assignment 4 on moodle

x_1	x_2	x_3

Sequence → $X = \{ 0 , 1 , 011 \}$

$\updownarrow$

$Y = \{ 1 , 011 , 0 \}$

y_1	y_2	y_3

Corresponding elements

Smallest solution is length 75.

· Quiz 2 (oct 15) Wed

90% Stanford studies

Solution: $\overset{3}{i_1} , \overset{1}{i_2} \dots , i_m$

$\cancel{2}$

$*$

sequence

identical

$$\begin{array}{cccc} x_{i_1} & x_{i_2} & \dots & x_{i_m} \\ y_{i_1} & y_{i_2} & & y_{i_m} \end{array}$$

Future

1. TM ≡ Type 0
2. Rec. Fn. Theory $f(N): N \to N$
3. λ-Calculus
4. Term Rewriting
5. NP-completeness

TMs

Variants

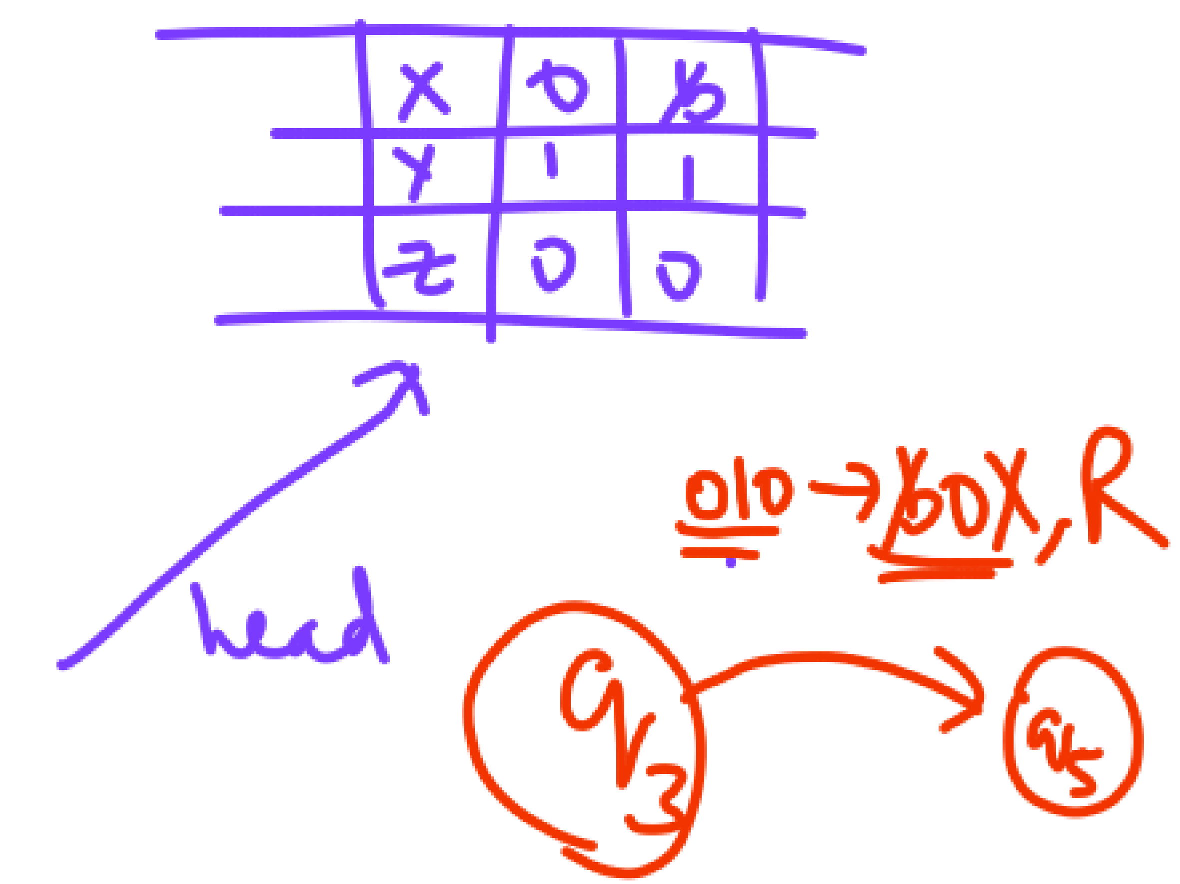
tricks

Assembly lang ↔

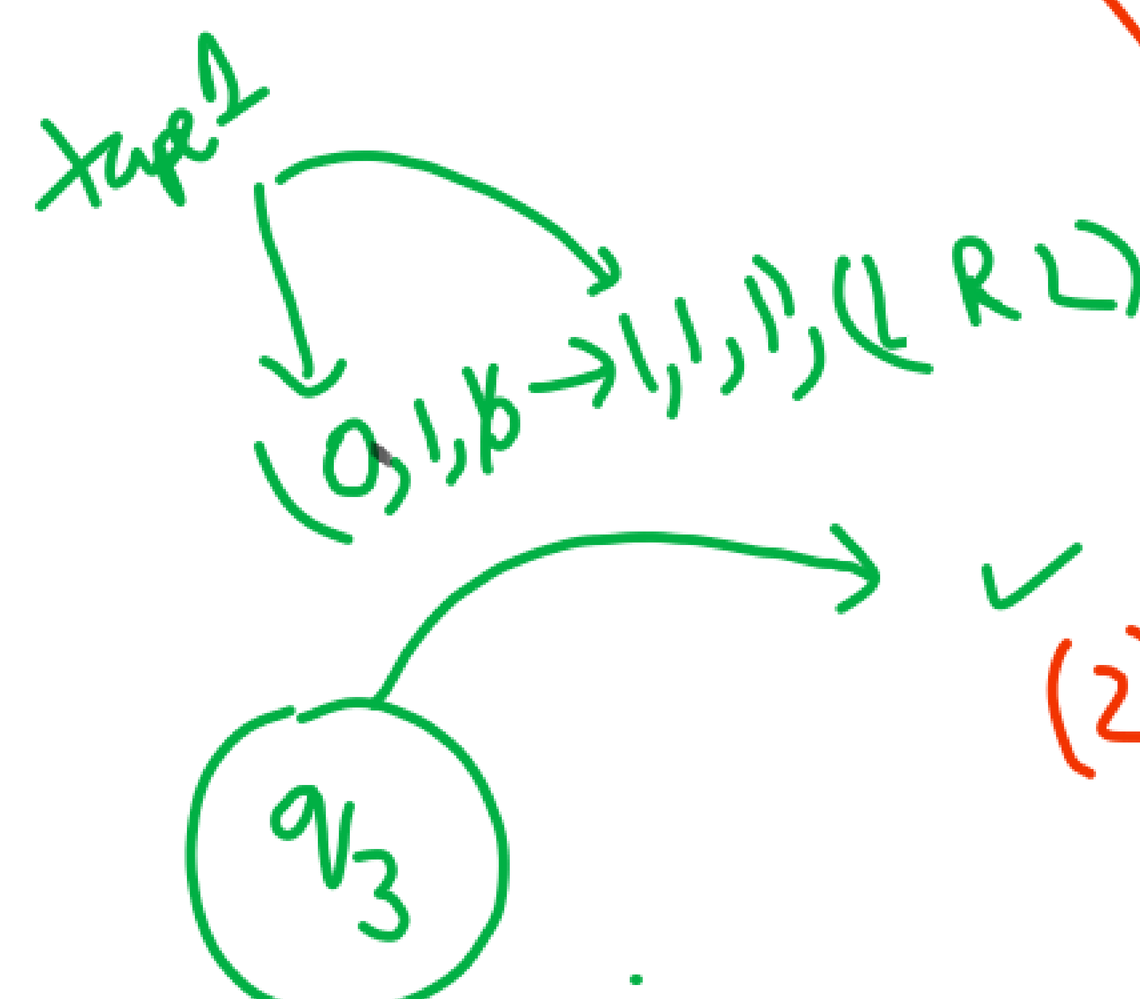
$\langle Q, \Sigma, \Gamma, \delta, q_0, b\!\!\!/, F \rangle$

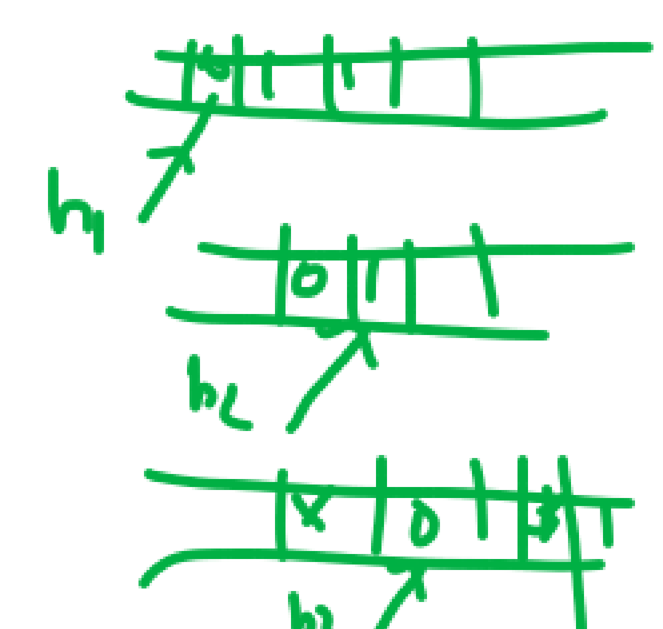
Variants

(1). Multi-track



(2) Multi-tape





(3) Non-deterministic

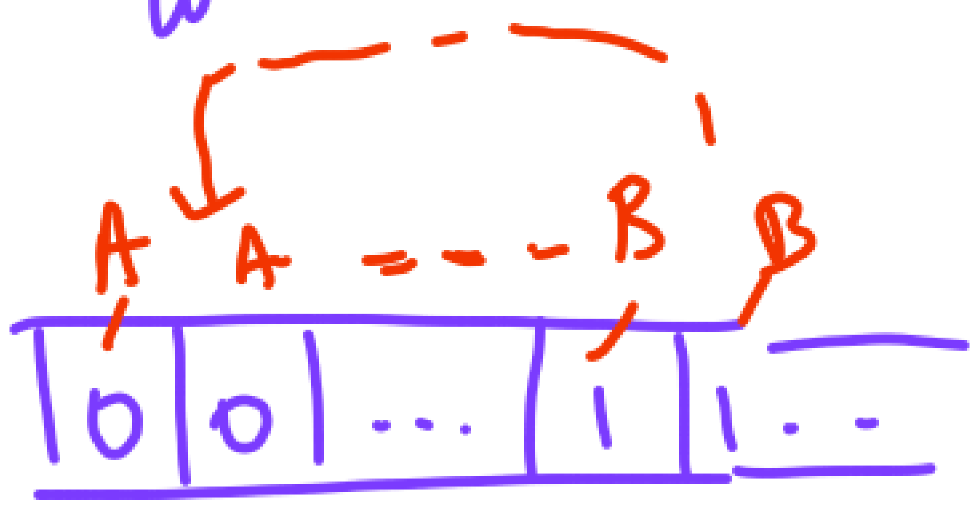
(4) 1-way tape

turingmachine.10

$L = \{0^n 1^n \mid n \geq 1\}$

✓ ~~2-tape~~

H/w

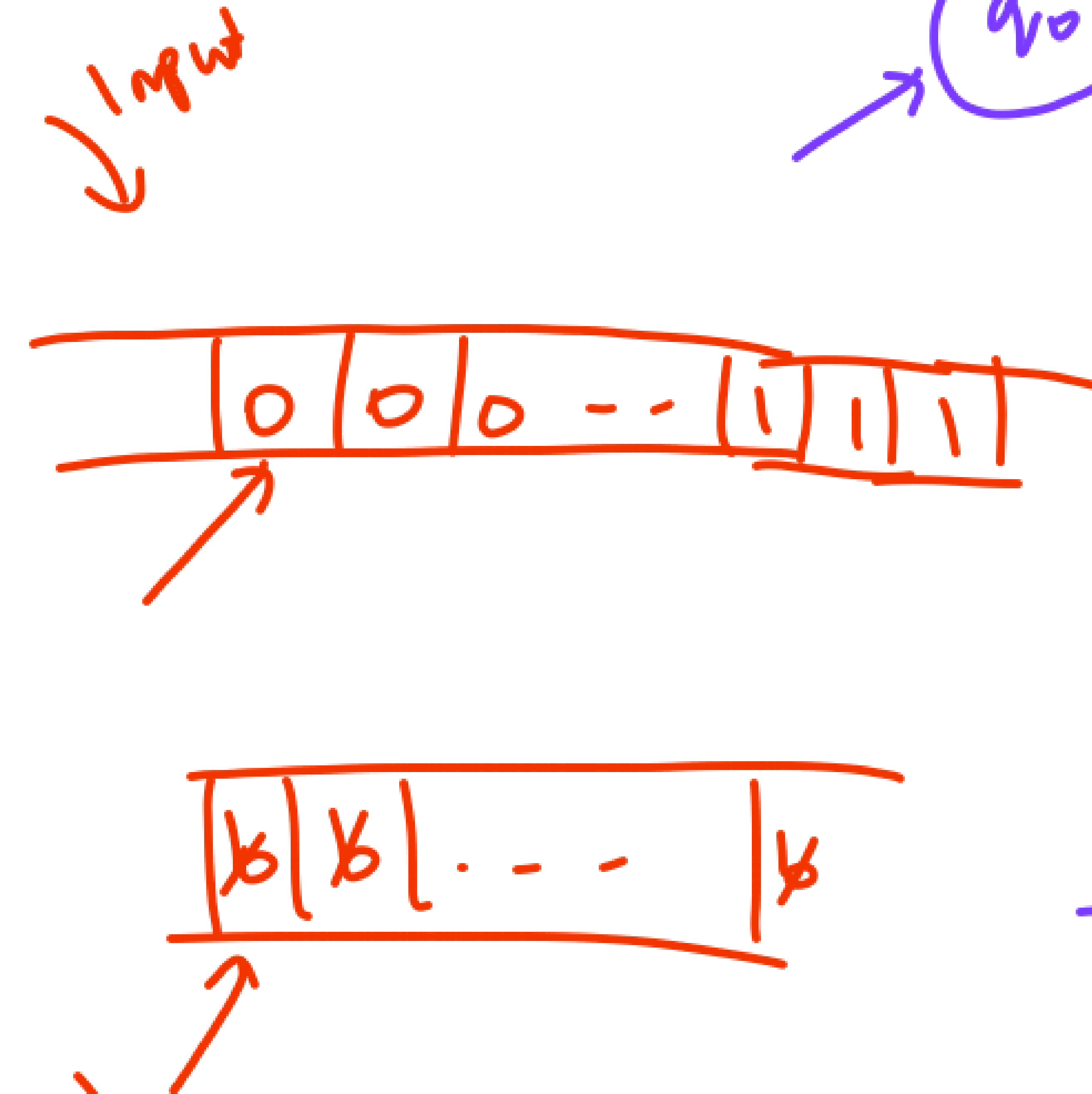


$q_0 w \vdash^{1} A\, 0 \ldots \vdash^{2} \ldots$

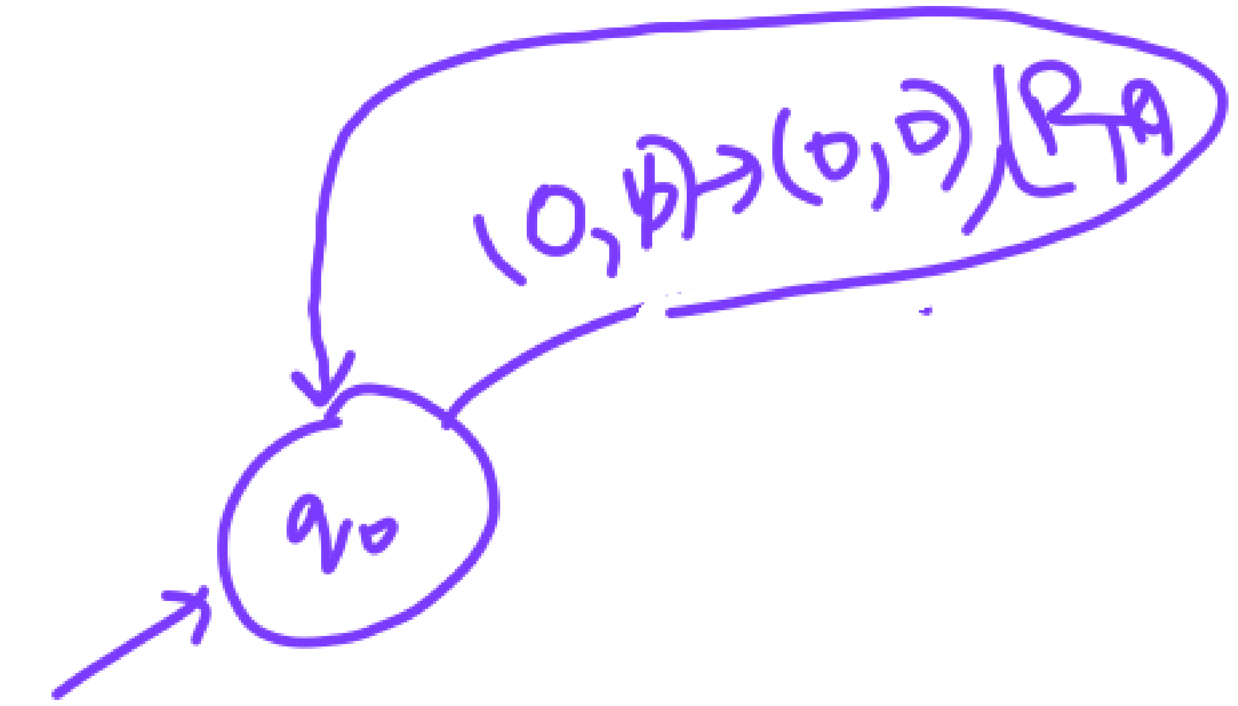
q_1

$2n$

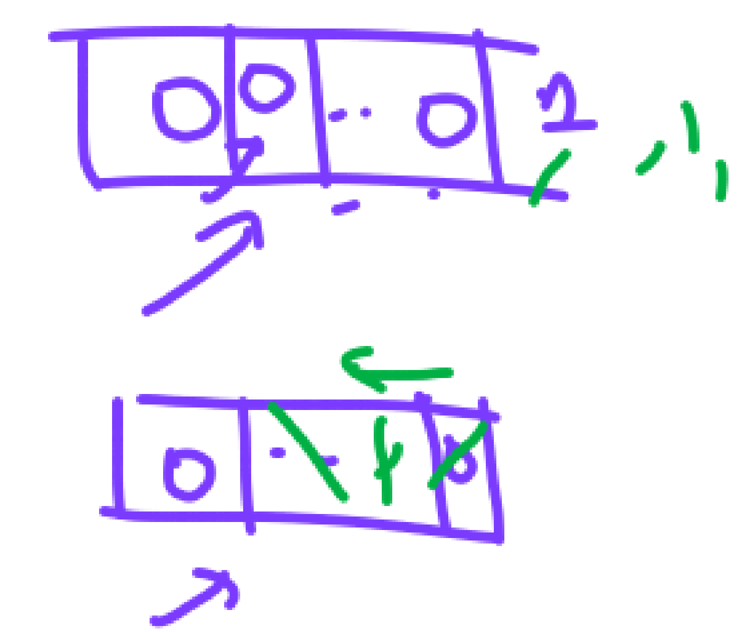
$O(n^2) \leftarrow O(n)$



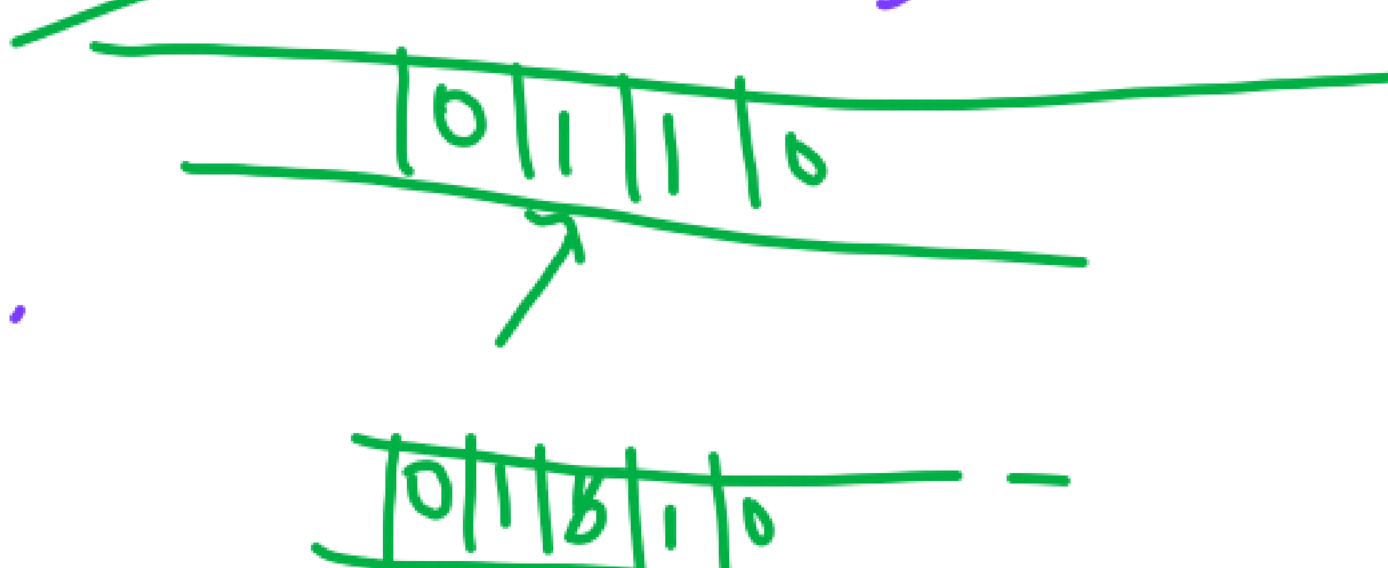
easier

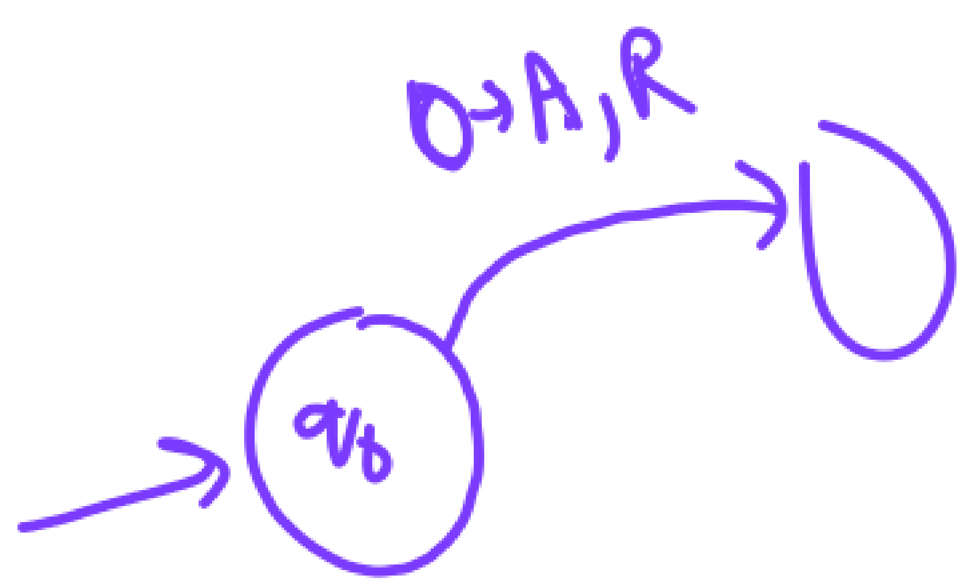


Idea



shift-right





No extra-power for any of the variants

proofs (very technical)

Can convert systematically to equivalent

Simple-tape deterministic TM

can mimic all variants

j-th tape symbol

l-th

m-th dir
0 → L
1 → R

$a_j \to a_l \mid d_m$

$q_i \to q_k$

H/W

DFA → Reg

Dyn. Prog

Model of acceptance

Encoding ~~Machine begins~~

① Accept state

$q_0 w \vdash \cdots \vdash \alpha q_f \beta$

$q_f \in F$

② Halt $q_0 w \vdash \cdots \vdash \alpha q_i \beta$ → no moves possible

more

$0^i 1 0^j 1 0^l 1 0^m 1 0^k 11$

move 2

11

move 3

111

Languages

$L_d \notin$ Rec. Enumerable

$L_u \in$ Rec. Enumerable

$L_u \notin$ Recursive

Rec. Enumerable → L is r.e. $\Longleftrightarrow$ there is a TM M such that $L(M) = L$

M accepts all $w \in L$

Recursive → L is rec. $\Longleftrightarrow$ $\exists$ TM M that halts on all w

and is in accept state if $w \in L$

in reject state if $w \notin L$

L

Closure

Union	Rev
Inters	Difference
Concat	Complement

3.10.2025

① Problem Reduction

PCP

undecidable

Example

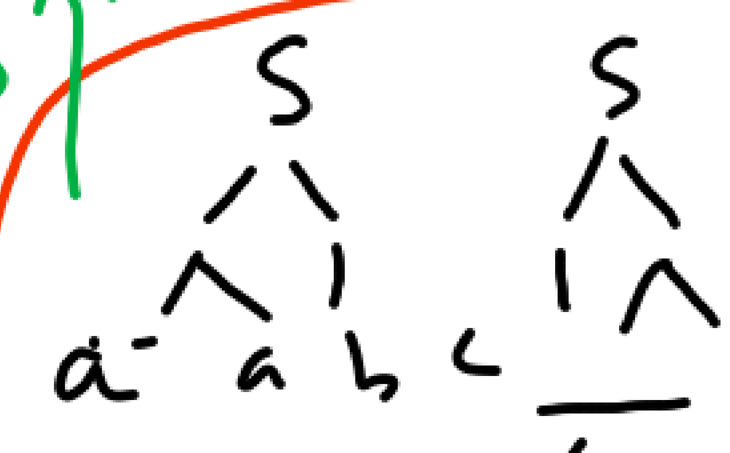
	1	2	3
A → X	a	baa	bba
B → y	ab	aab	ba

Soln

$12^{*}3$

13: a bba / ab ba (1 3)

also undec



CFG

Is G ambiguous?

same string with 2 "different" trees

S →
⋮

PCP —"convert"— → Gr. Ambiguity

"cost"

A/w

PCP has a soln ⟺ Grammar is ambiguous

$S \to A \mid B$ (equivalent for decidability)

$A \to aA1 \mid a1 \mid baaA2 \mid baa2 \mid bbaA3 \mid bba3$

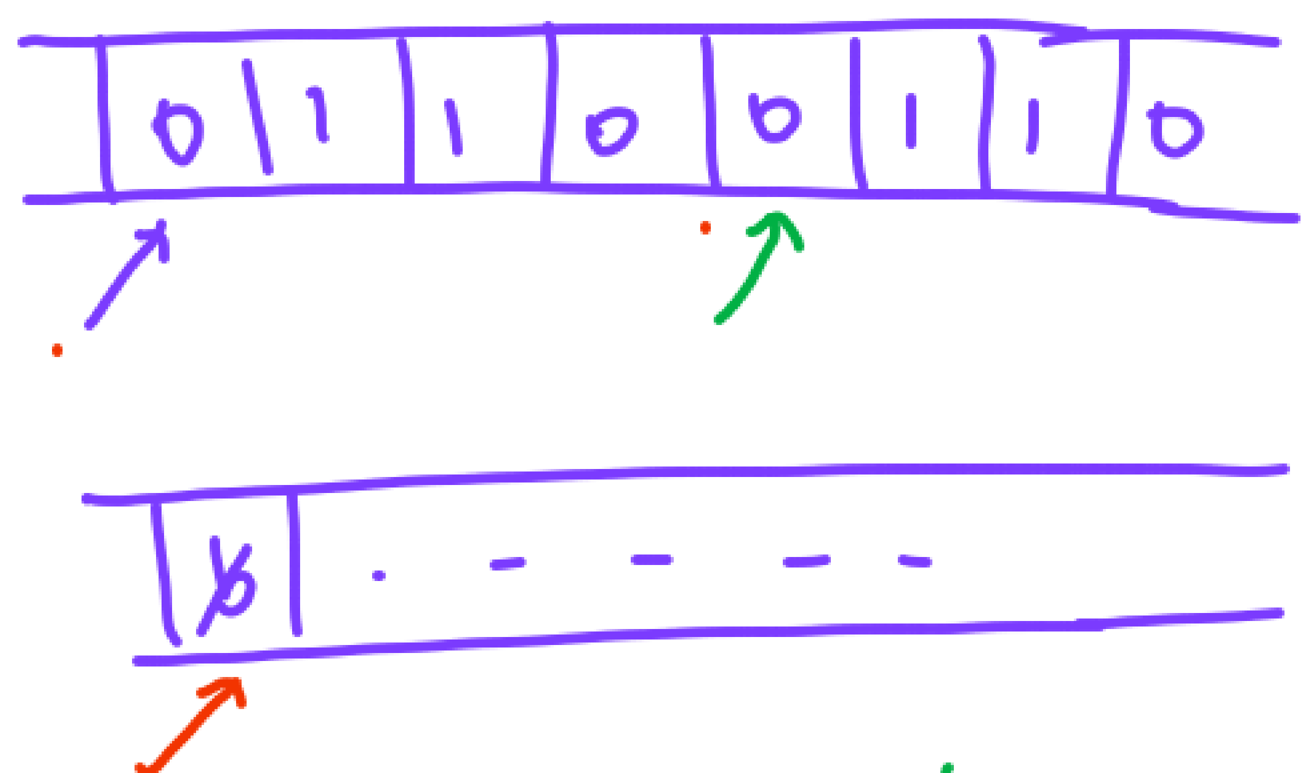
$B \to abB1 \mid ab1 \mid aabB2 \mid aab2 \mid baB3 \mid ba3$

2-tape

Non-det TM

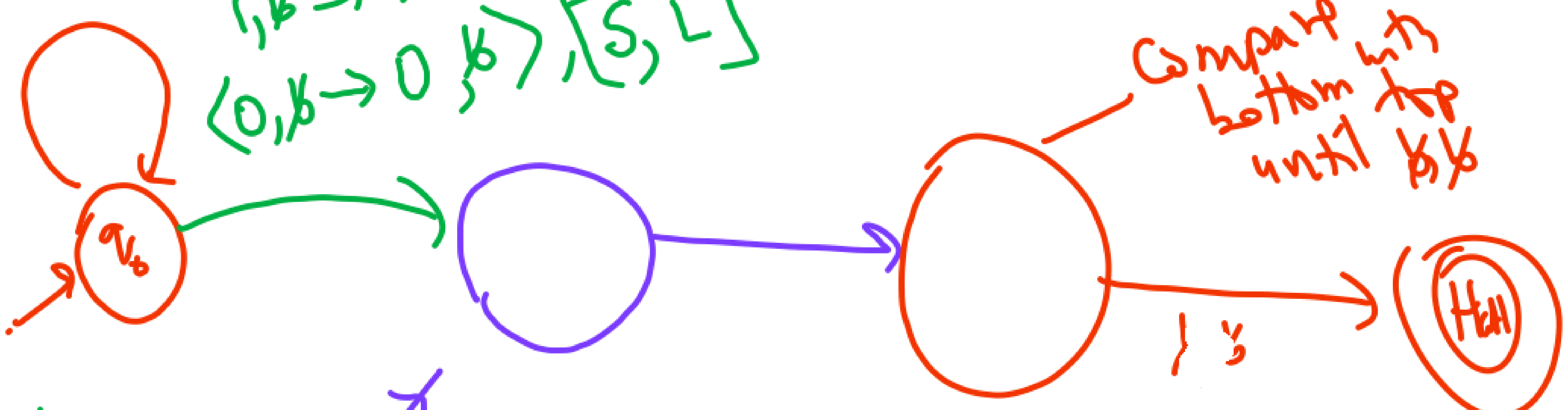
TM-Extensions

for $L = \{ww \mid w \in (0+1)^*\}$



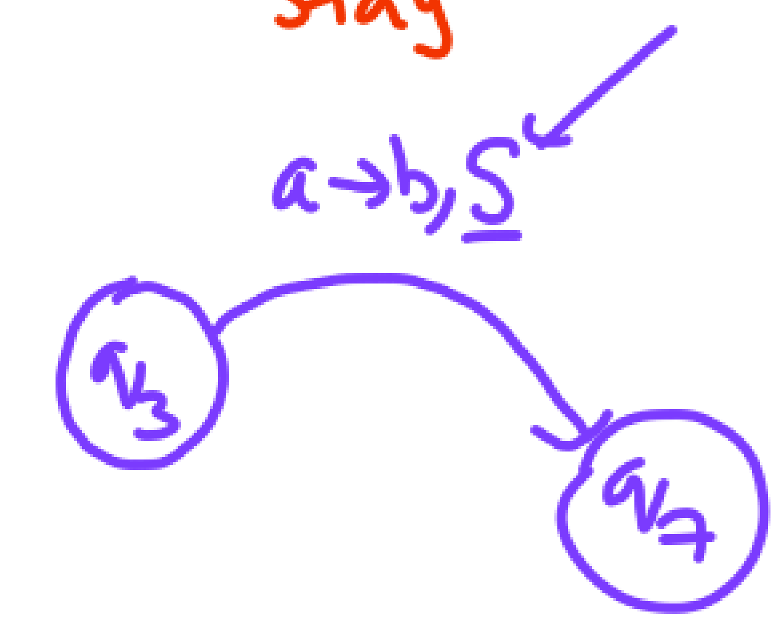
0 1 1 0

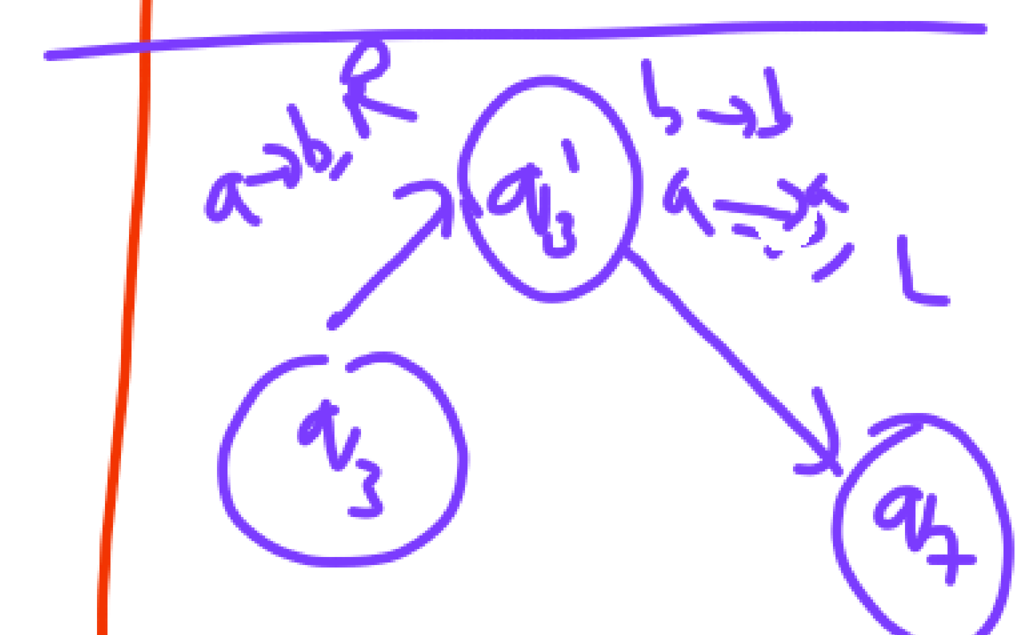
$\langle 1,b \to 1,b \rangle$, [R,R]
$\langle 0,b \to 0,0 \rangle$, [R,R]



Keep going left on bottom tap until b

Extension
Stay
L, R → 2-tape
→ non-det
to
L, R, S
stay





Decidability

L is decidable if

$w \to \boxed{M}$ → Yes if $w \in L$, No if $w \notin L$

and M always halts.

All languages $L \subset \Sigma^*$

Rec. Enumerable = Semi-Decidable

Rec. = Decidable

L_u L_d

Universal

Diagonalized

L is r.e (semi-decidable) if there is a TM, M

$w \to \boxed{M}$ → Yes if $w \in L$ & halt

No & halt

Run forever

$w \notin L$

L_d fails here also

Is G ambiguous?

Does PCP have a soln

Hilbert's 10th problem

$7x^3y^2 - 11xy^2z - \cdots = 0?$

not recursive

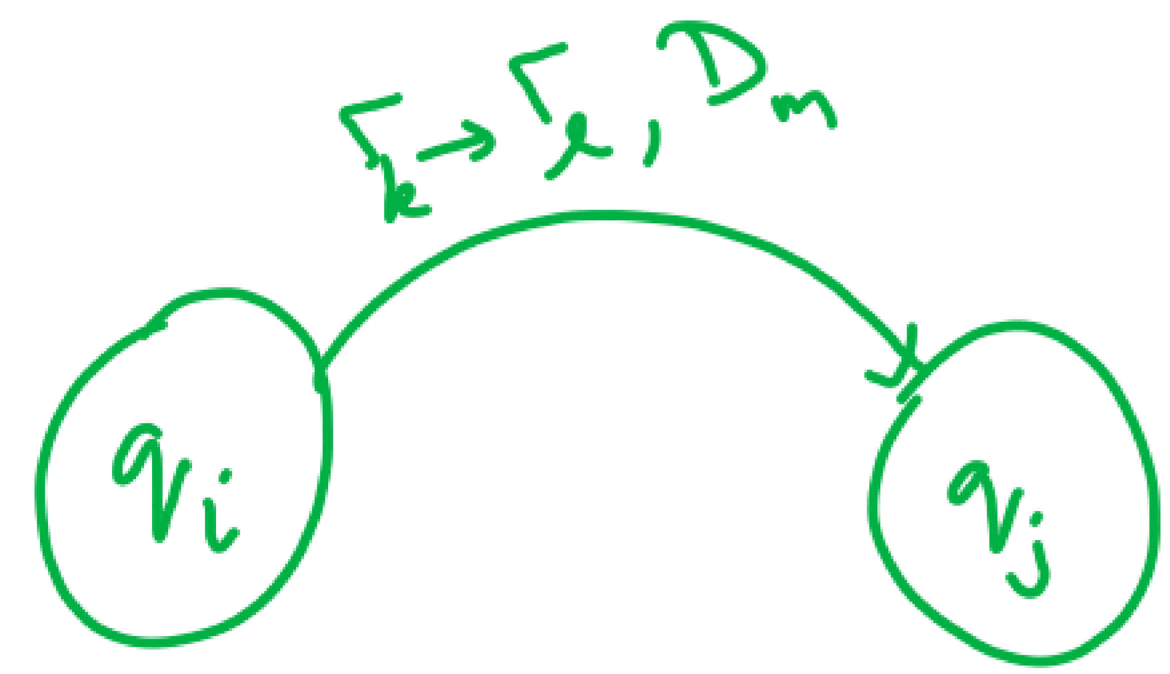
(1) TMs are countable (encoding)

(2) there is a language L_d no TM can accept.

(3) there is a language L_u that a TM can enumerate but cannot "decide"

1-move

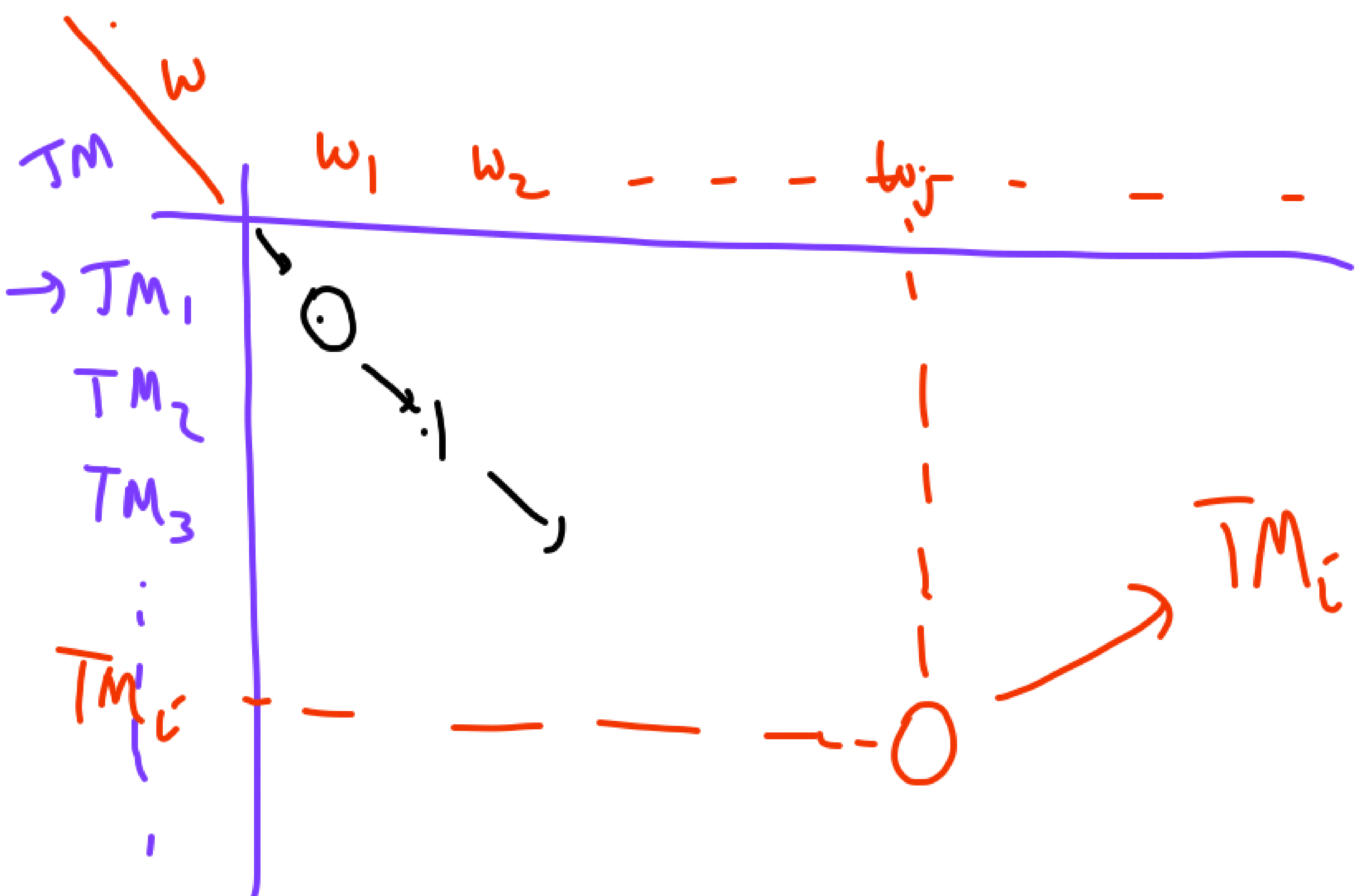
$0^i 1 0^j 1 0^k 1 0^l 1 0^m$



$\underbrace{\text{move}_1}$ $\underbrace{11}_{\text{2 one's}}$ move_2 11 . . . 11 move_n

$111 \underbrace{\qquad\qquad\qquad}_{} 111$

M



$L_d = \{ w \mid w = w_i$ and TM_i does not accept $w_i \}$

cannot be accepted by any TM

TM_i does not accept w_j

Proof by contradiction — if T_k accepts L_d then T_k gives wrong answer on w_k

L_u is a language for which we can build a TM M_u — Universal TM

but M_u will not always halt

$\langle M, w \rangle \rightarrow M_u$

- Yes & halt If M accepts w
- No & halt or Run forever.

Input

M w

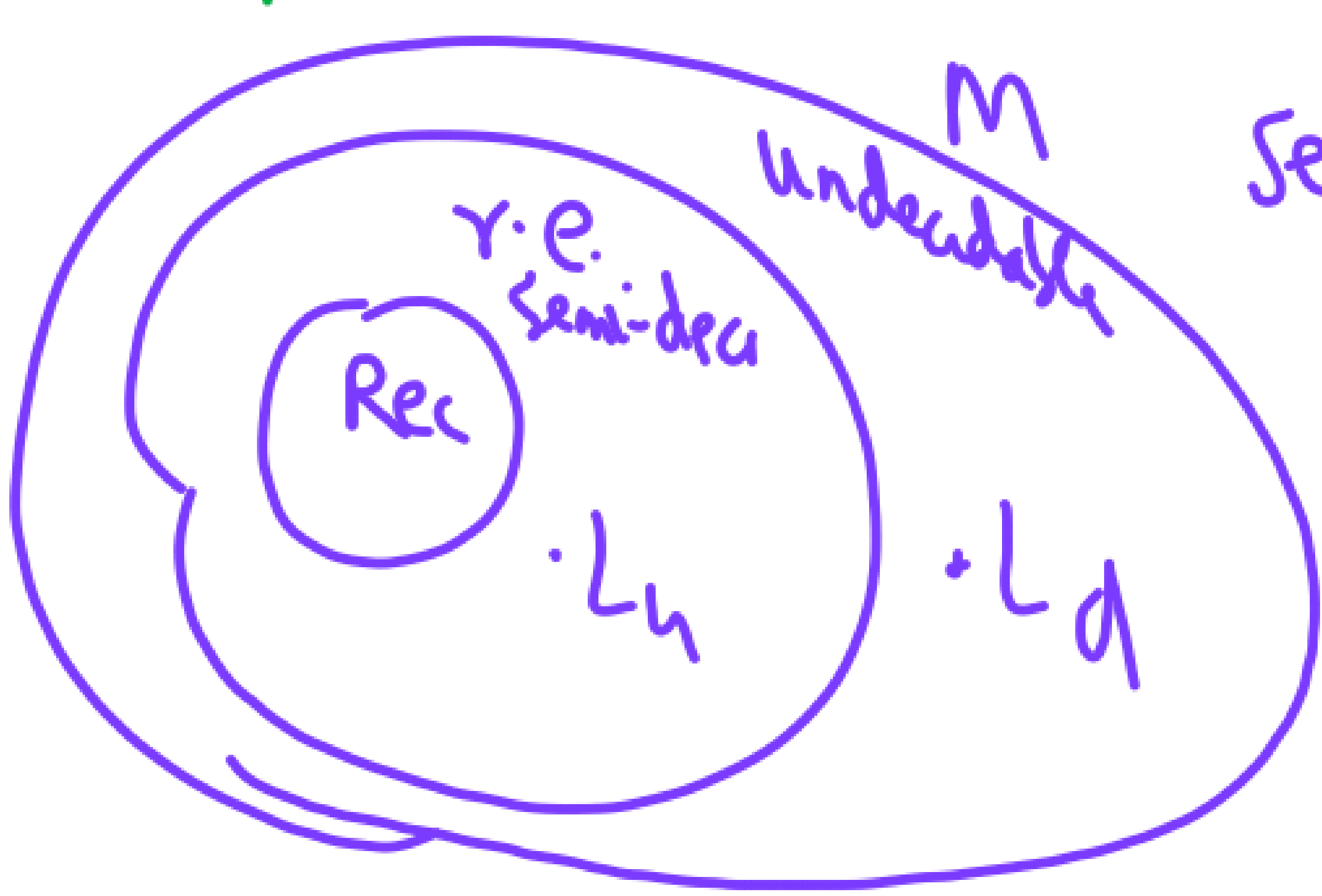
111 . . . . . 111# 0110 .

M w

Seperator

Simulation

Can behave like any other TM



8.10.2025

Language

- Closure Properties

TM

Models of Computation

a. λ-calculus

b. Term Rewriting

c. Rec. Fn. Theory

Church-Turing thesis

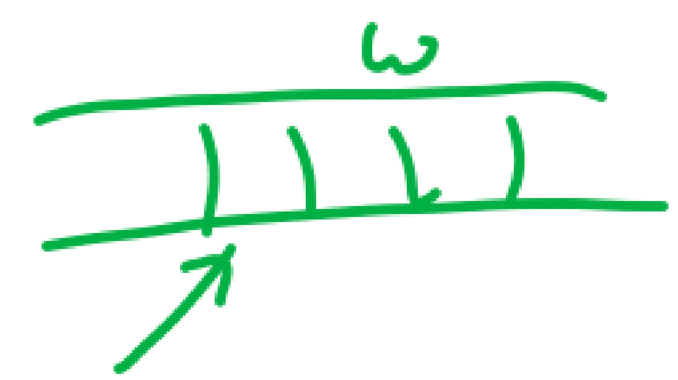
* Quiz 2 on 15/10/25

→ Assgn 4.

→ NP-completeness –

$$P \stackrel{?}{=} NP$$

r.e.: L is Rec En. $\overset{def}{\Longleftrightarrow}$ TM M accepts L



$L = \{ww \mid w \in (0+1)^*\}$

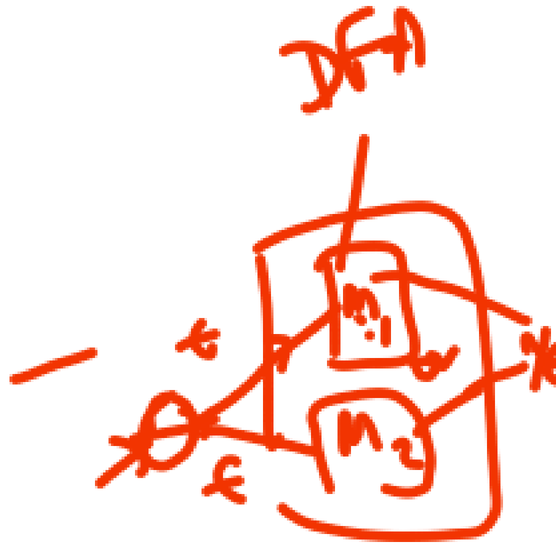
rec.: Recursive $\Longleftrightarrow$ $\vdash^{L}$ halts in reject state if $w \notin L$

DFA/NFA

L_1, L_2 are regular

$L_1 \cap L_2$

$L_1 \cup L_2$

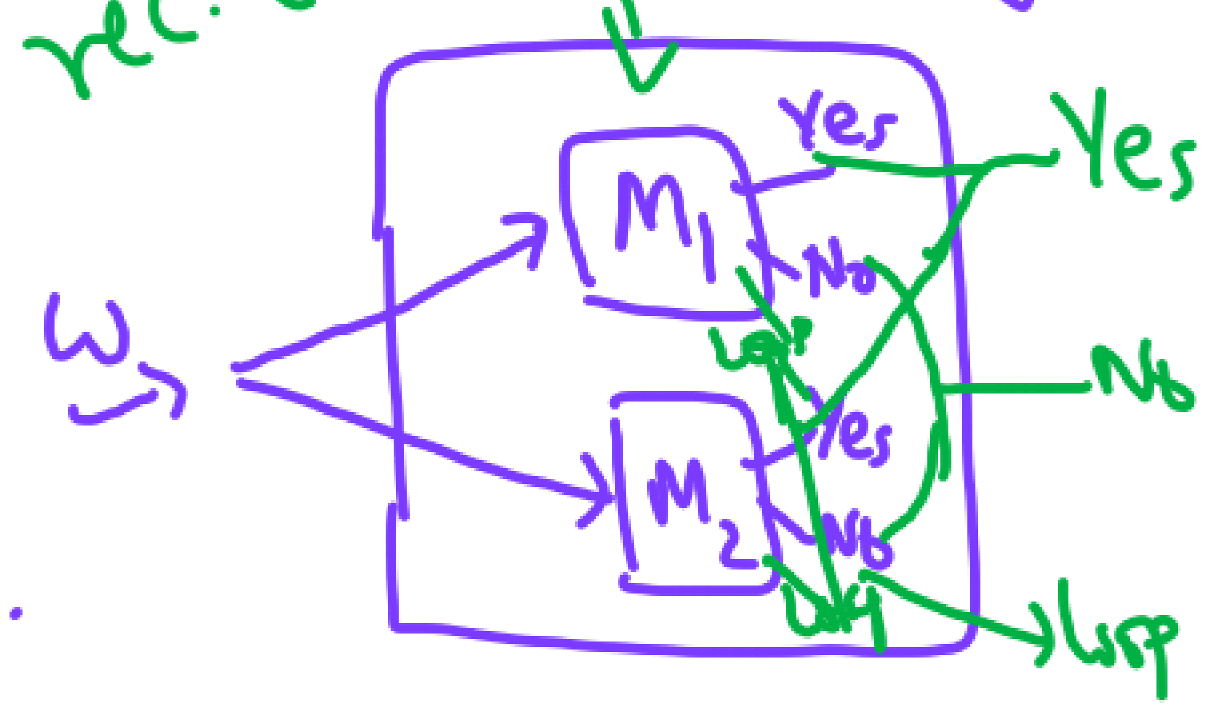


$L_1 L_2$

L_1^R

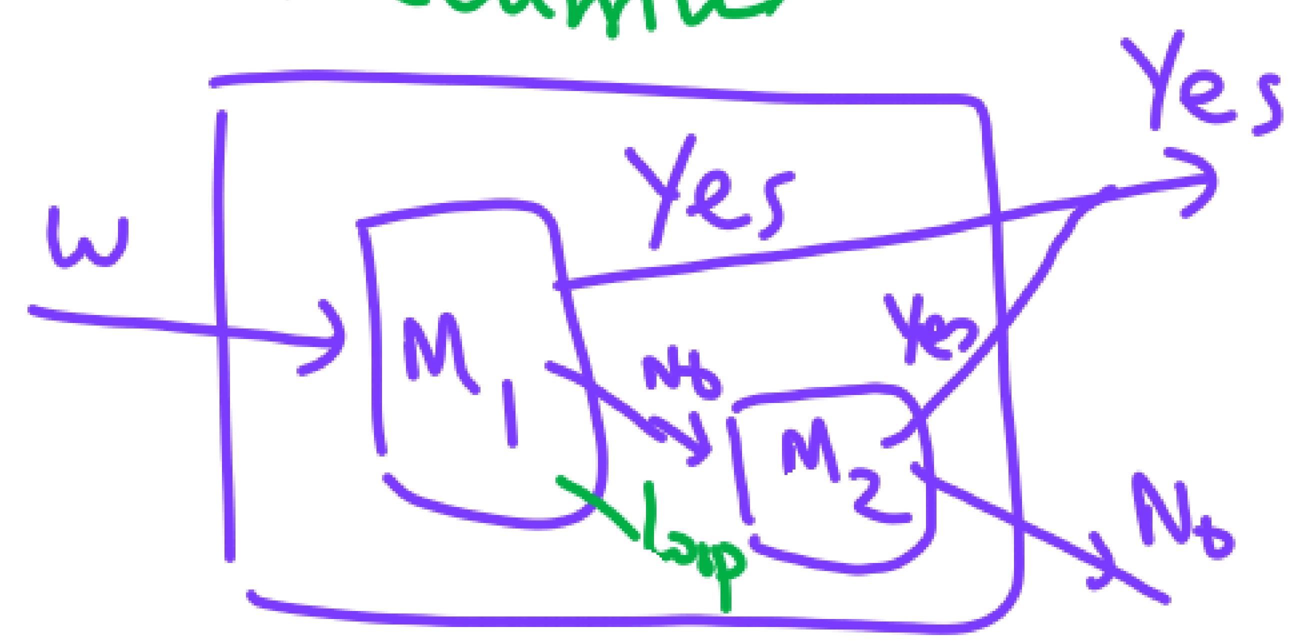
Union L_1, L_2 are r.e / rec what about $L_1 \cup L_2$?

rec. enumerable



allowed 3 behaviours

Recursive



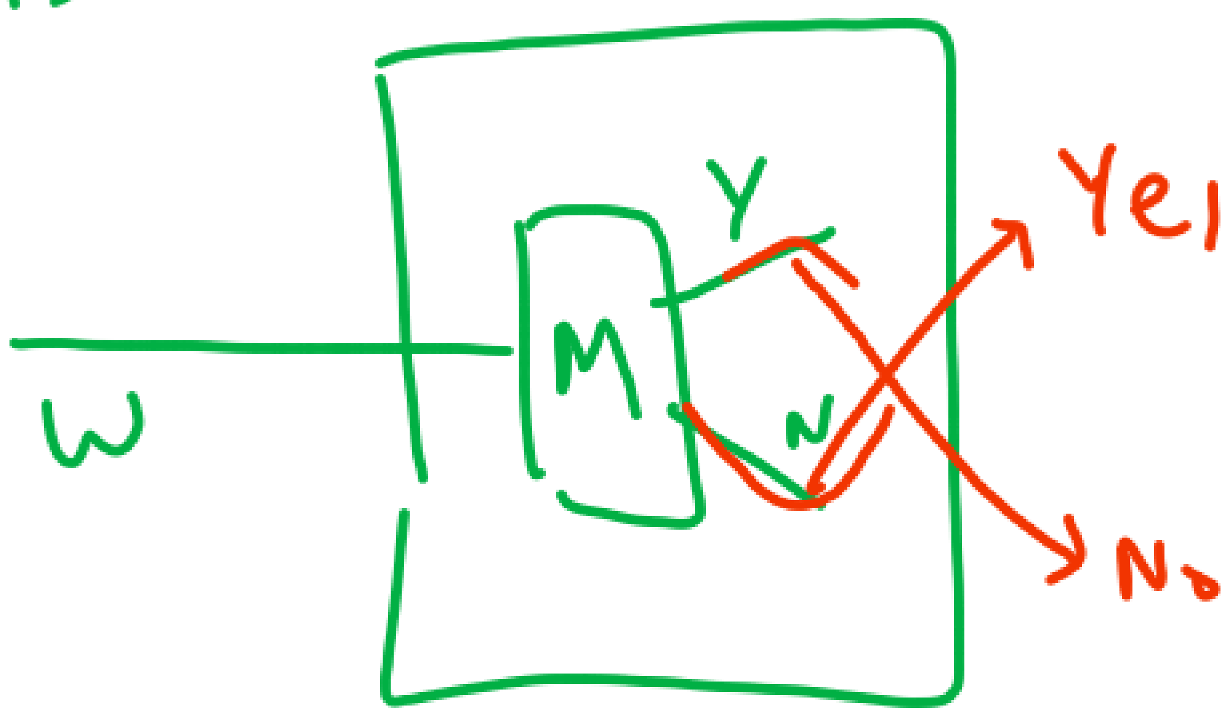
only 2 behaviours

Complement

r.e → h/w

Rec

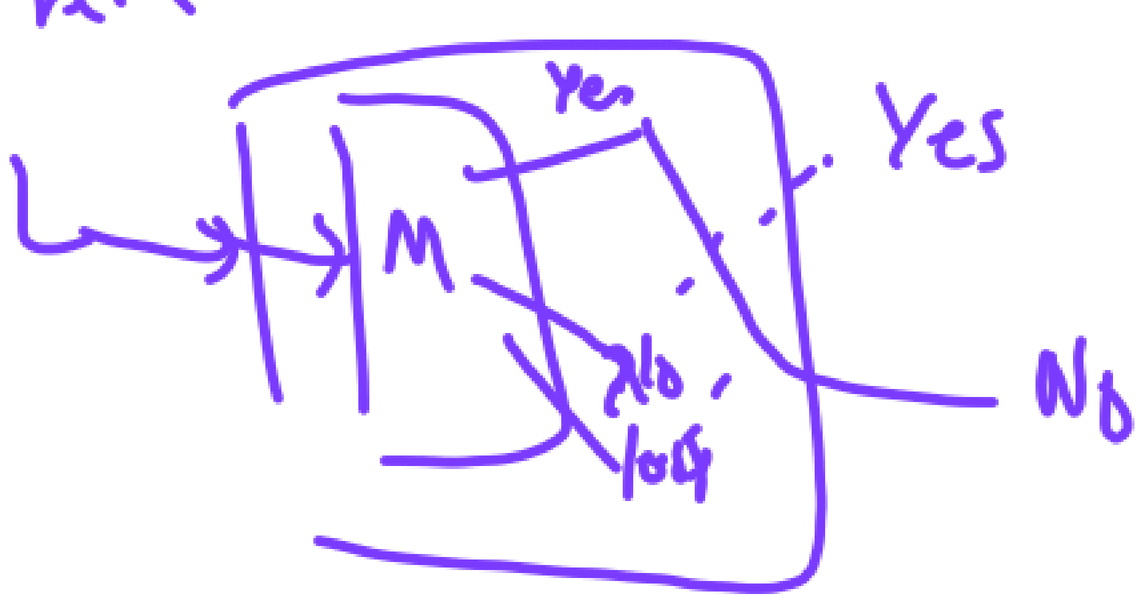
L is rec - L^c ?



flip answer

works for r.e.?

No!

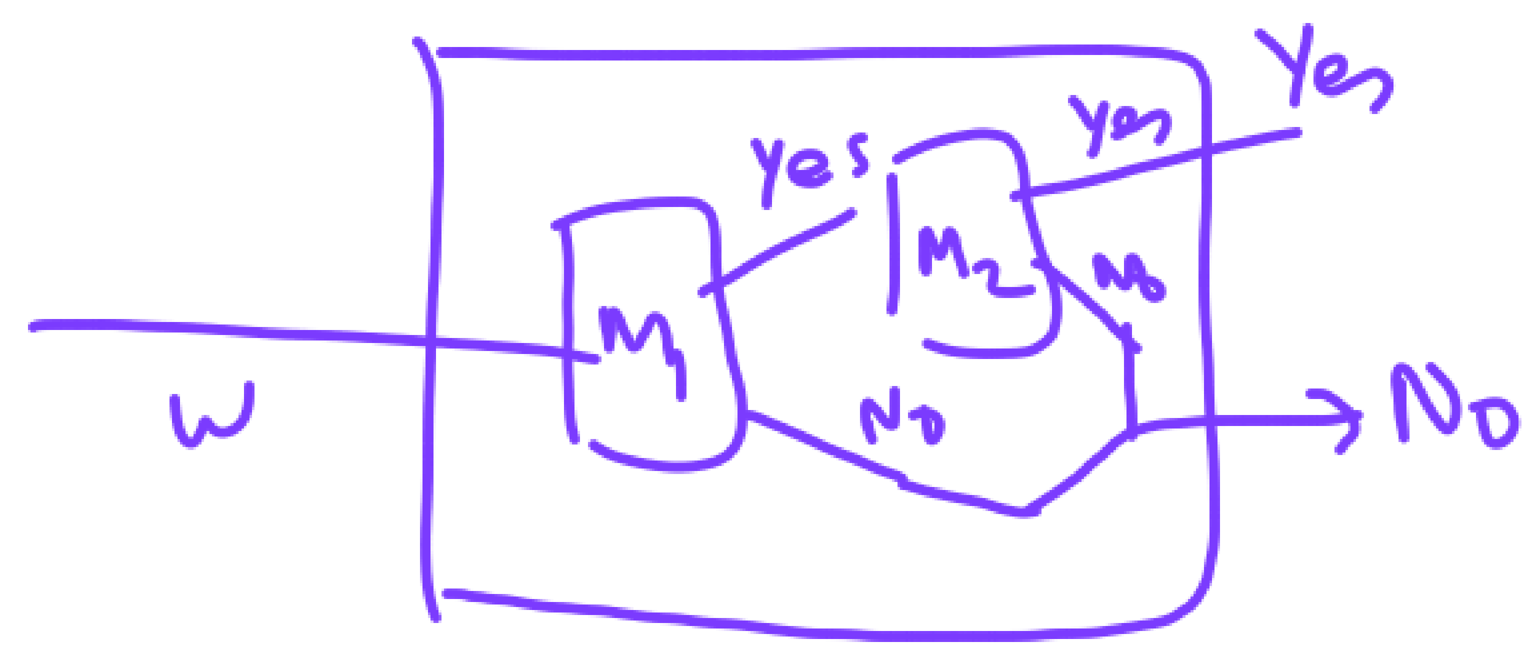


Intersection

r.e → h/w

L_1 is rec L_2 is rec

$L_1 \cap L_2$?



works for r.e.?

H/w → If L_1 is r.e. and $\overline{L_1}$ is r.e. then L_1 is recursive ⇐

Rec

$L_1 L_2$

L_1^*

L_1^{Rev}

Computations

Can be embedded as Language.

$a^i b^j c^{i*j}$

$a^i b^{2i^2+3i+9}$

$f(x_1, x_2, \ldots, x_k) \to N$

$N \to f \to N$

$N^k \to N$

$\langle k, \langle n_1, \ldots n_k \rangle \rangle$

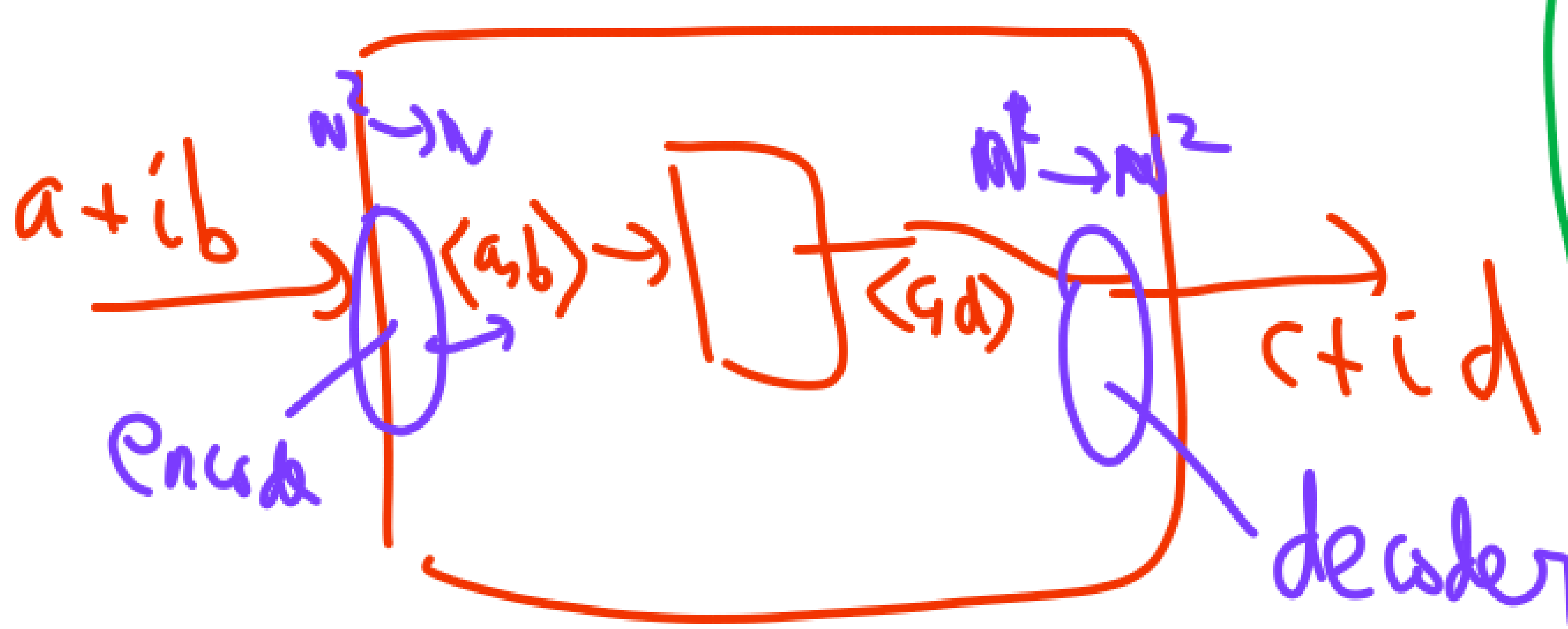
N

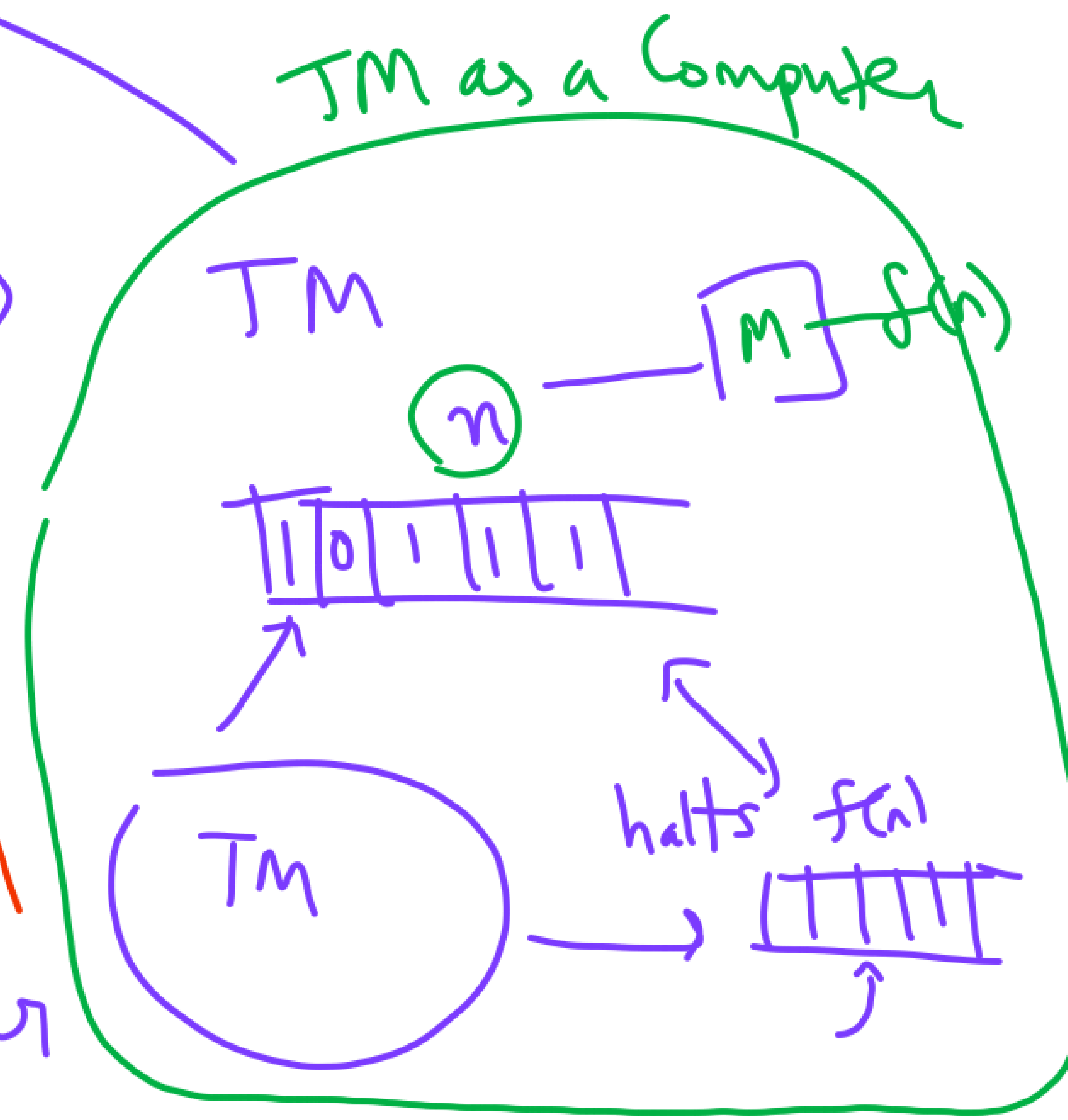
encoded

Real

Complex

?





Equivalent Computational models.

T

λ-calculus

Abstraction — Application

} β-reductions

λ-Expression

Base

Variable

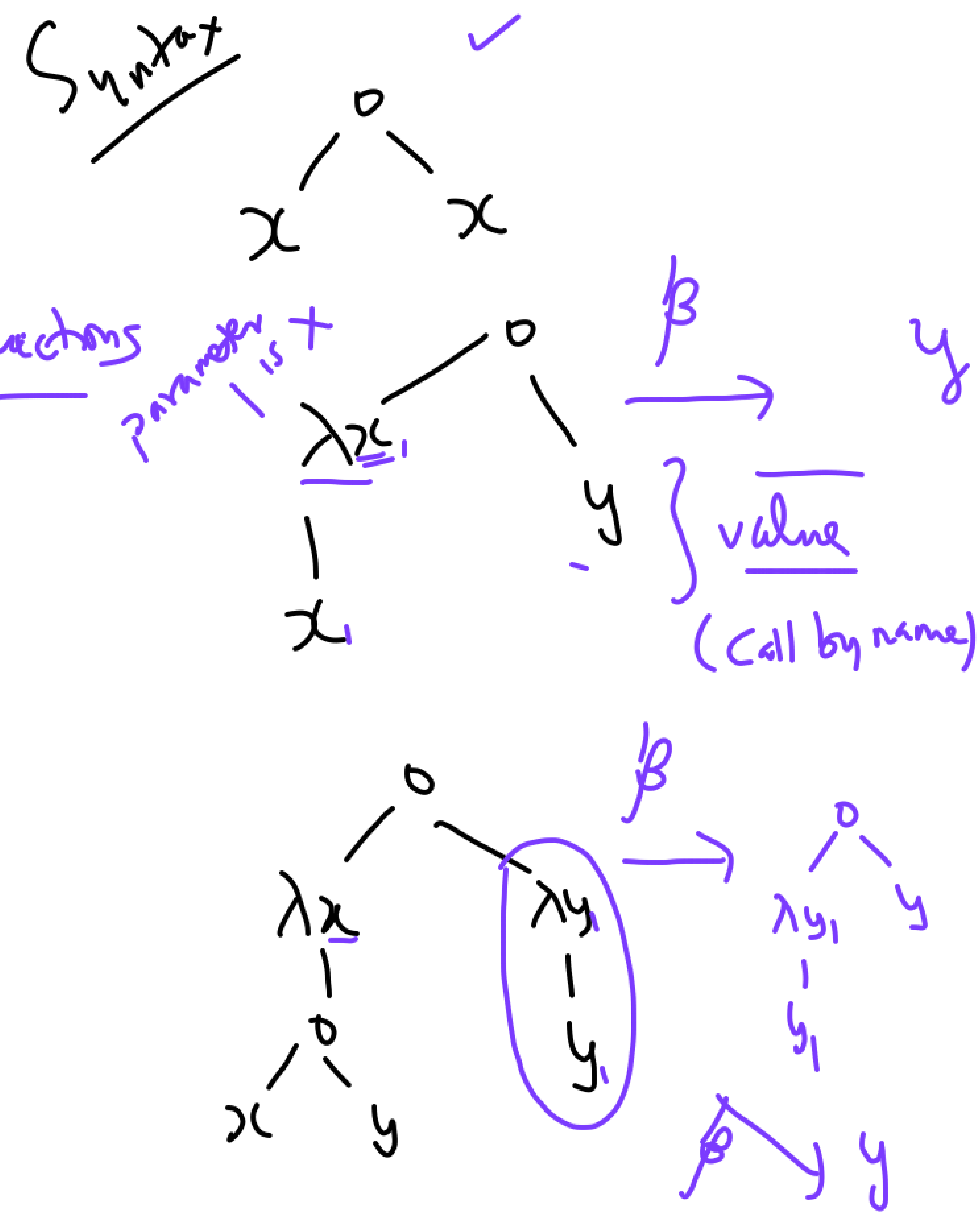
x, y, z

Inductive

λx
|
E

any λ-express

$\circ$
/ \
E_1 E_2



Goal

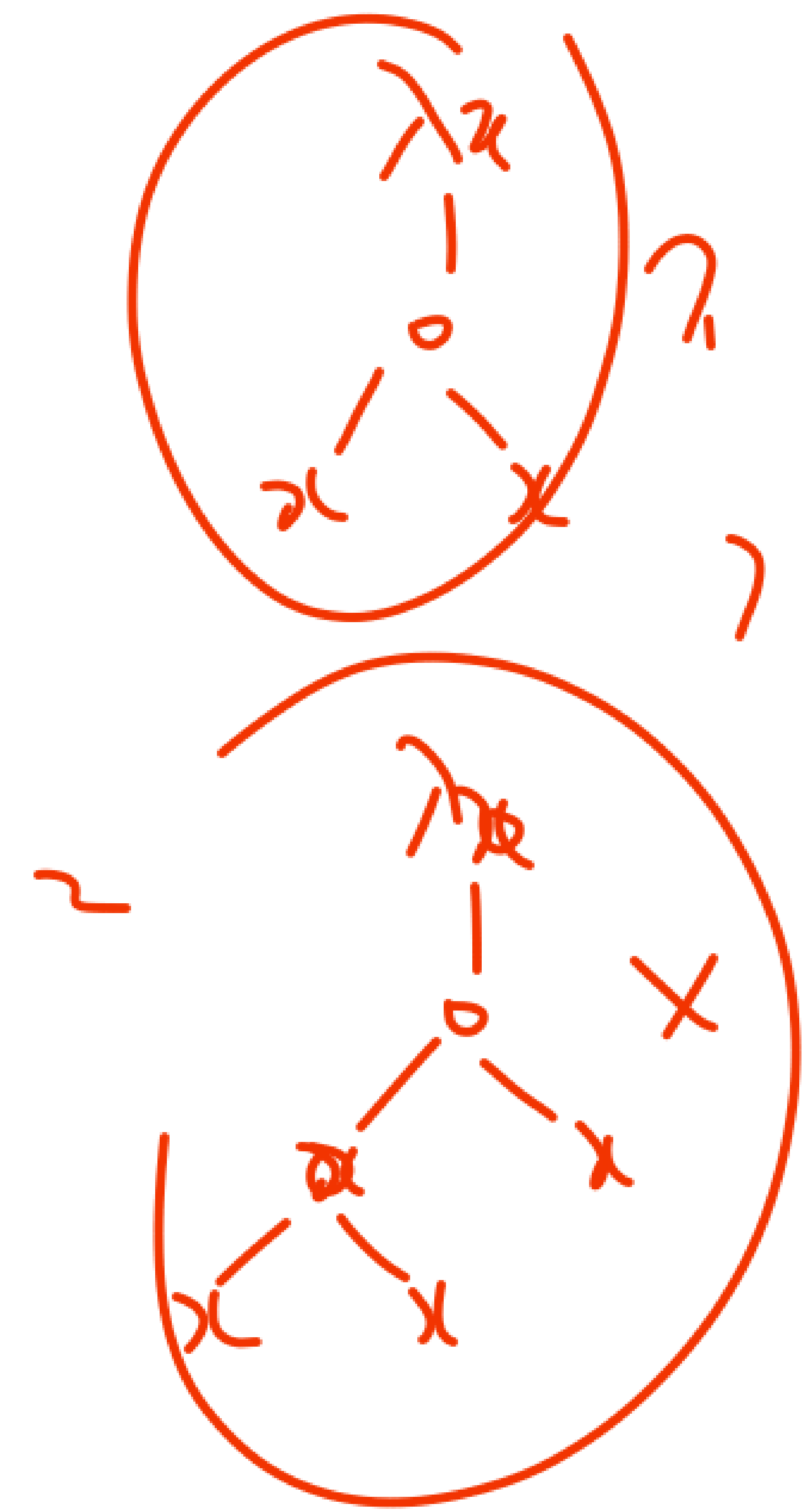
Numbers — Church numerals $\in 0, 1$

addition expr

mult expr

$\downarrow \beta *$

$n_1 + n_2$



Term Rewriting

H/w gcd square

fact(s(0)) → *(s(0), fact(0))

fact(s(x)) → *(s(x), fact(x))

/ Equality

=

"replace equals by equals"

$3 * 7 = 21$

*(s(s(s(0))), s(…(0))) → → → s(…(s(0)))   21

1. $+(0, x) \to x$
2. $+(s(x), y) \to s(+(x, y))$
3. $*(0, x) \to 0$
4. $*(s(x), y) \to +(*(x, y), y)$

Terms

fn. symbol

variable

0   s(0)   +(s(0), s(s(s(0))))   *(x, s(+(x, x)))

Constants

0   s(0)   s(s(0)) . . .   +(0)

+ ✓   * ✓

Constructors (domain)

defined by rules

10.10.2025

$\boxed{\text{Quiz 2} \\ 15/10 \;\; 11\text{–}12{:}30}$

1. λ-calculus
2. Term Rewriting
3. Recursive Function Theory

Not in Quiz 2
↓
Qn 3 &
I will post
pointers on moodle

$n \rightarrow \boxed{f} \rightarrow f(n)$

λ-Expression → Variables
x, y, z
f, m, n

λx
$|$
E
↓
Abstraction

formal ≈ Parameter

o
$E_1 \quad E_2$
↓
application

β-redn

o
$\lambda x \quad E_2$
$|$
E_1

$\xrightarrow{\beta} E_1[x \rightarrow E_2]$

Type D

TM practice

w

1	0	1	1

Copy
↓

1	0	1	1	1	0	1	1

$\underbrace{}_{w} \quad \underbrace{}_{w}$

& halt

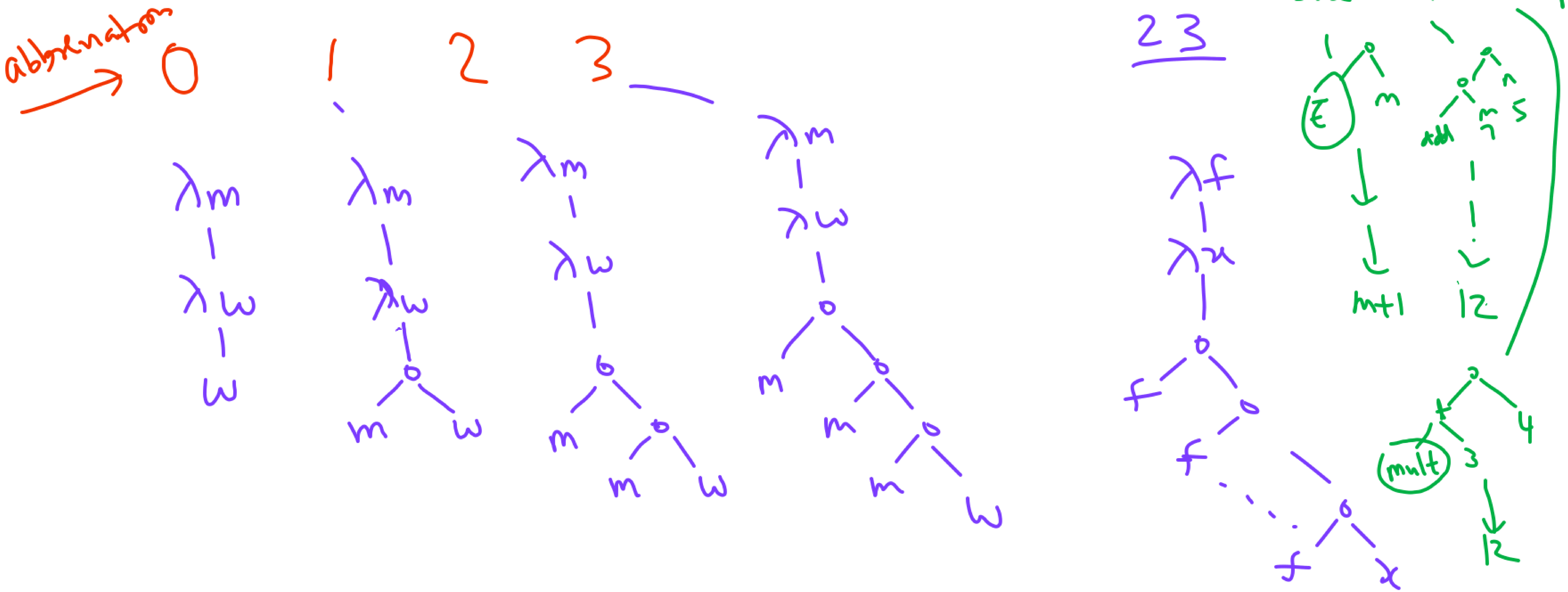
www

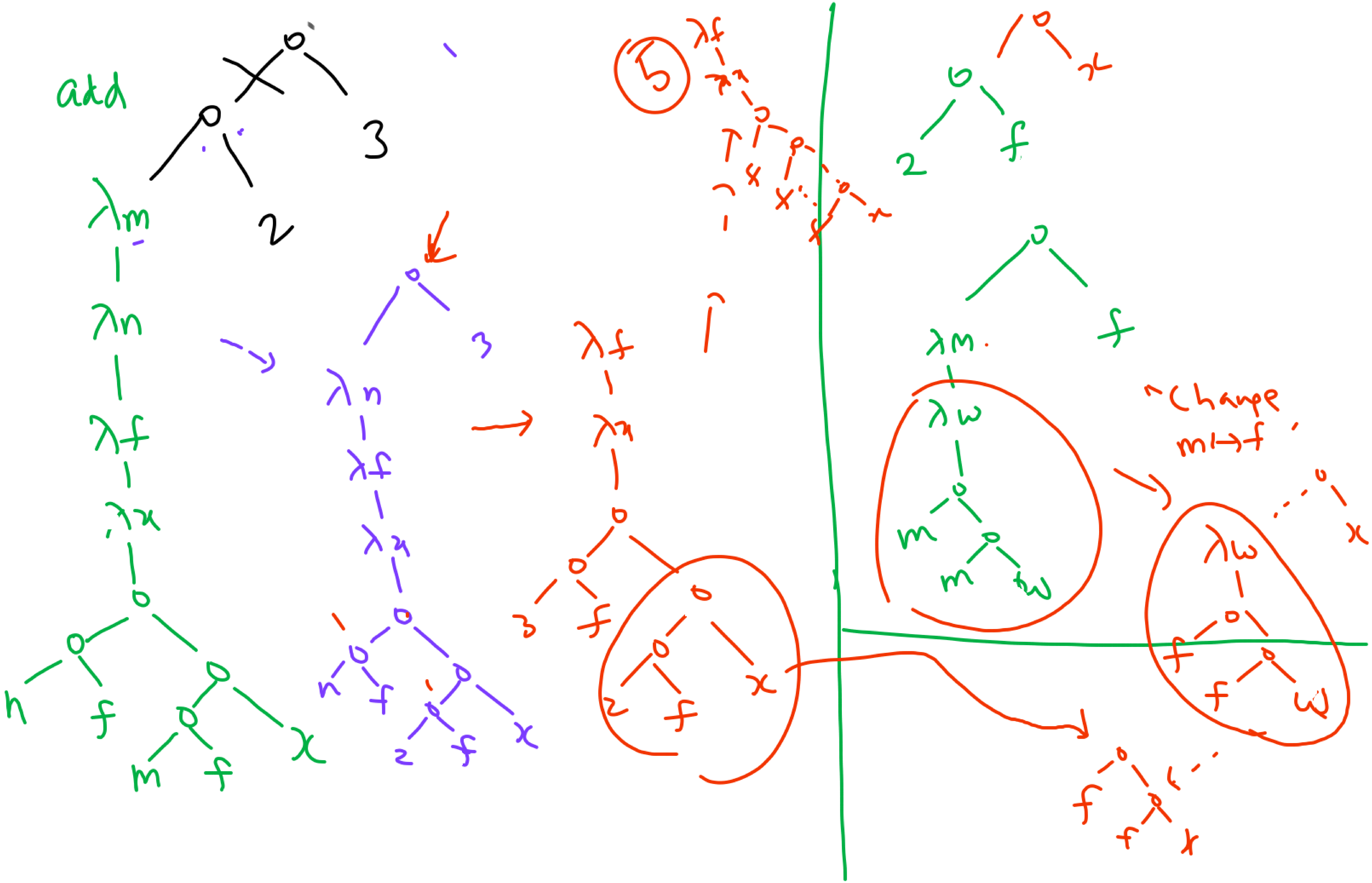
Church numerals

Numbers

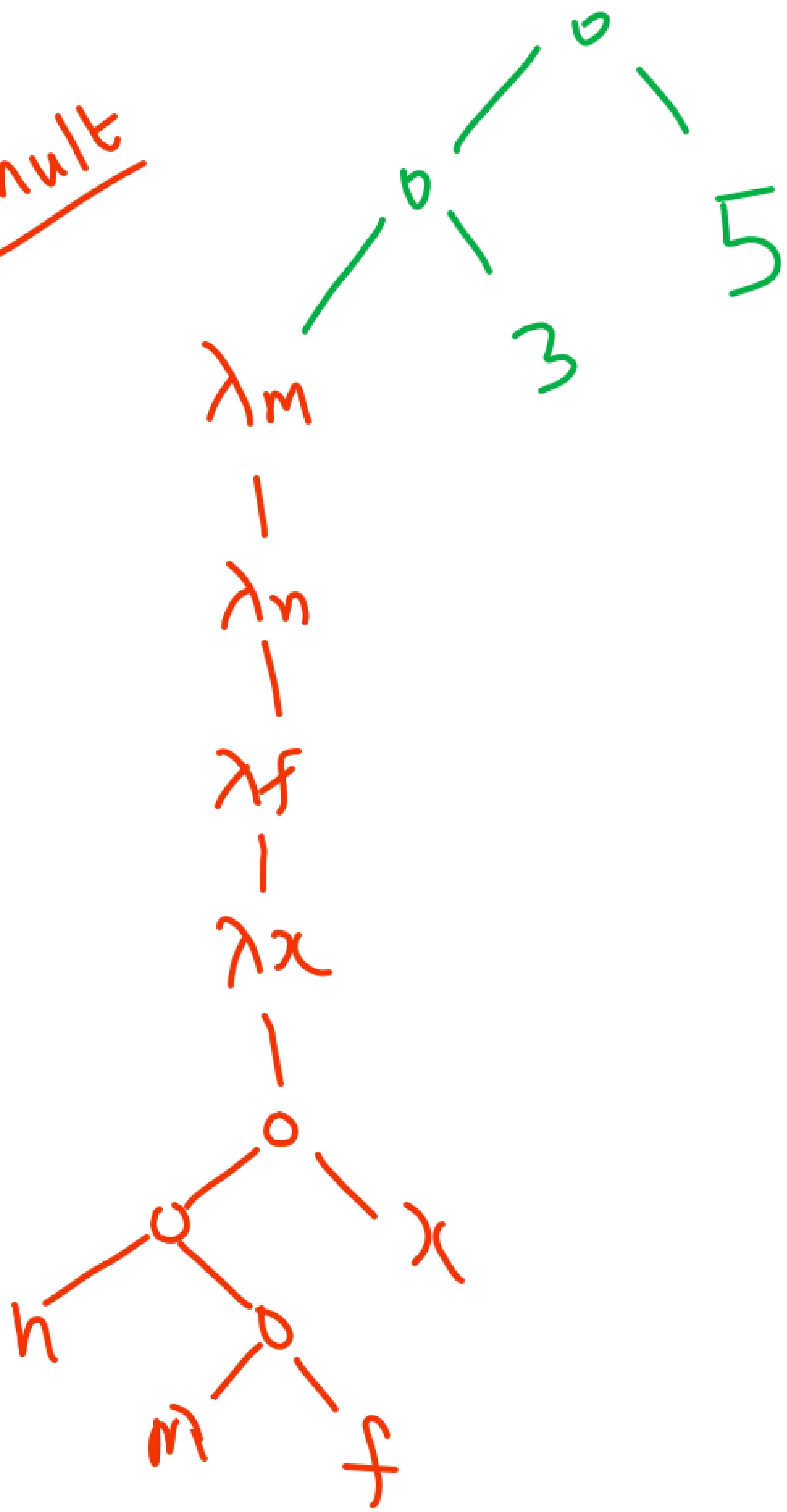
Goal

Succ add mult exp

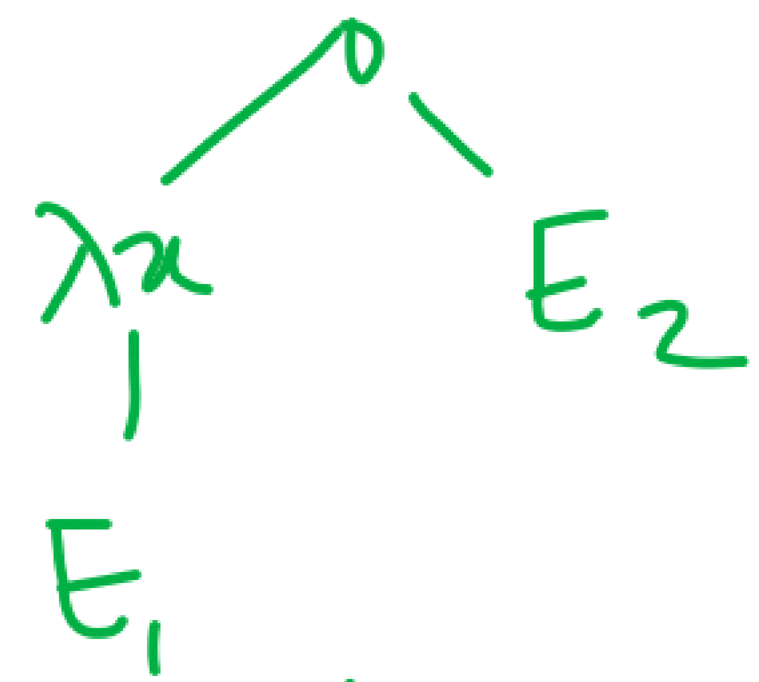



add
5
change
m↦f

mult



β-reduction



$$E_1[x \mapsto E_2]$$

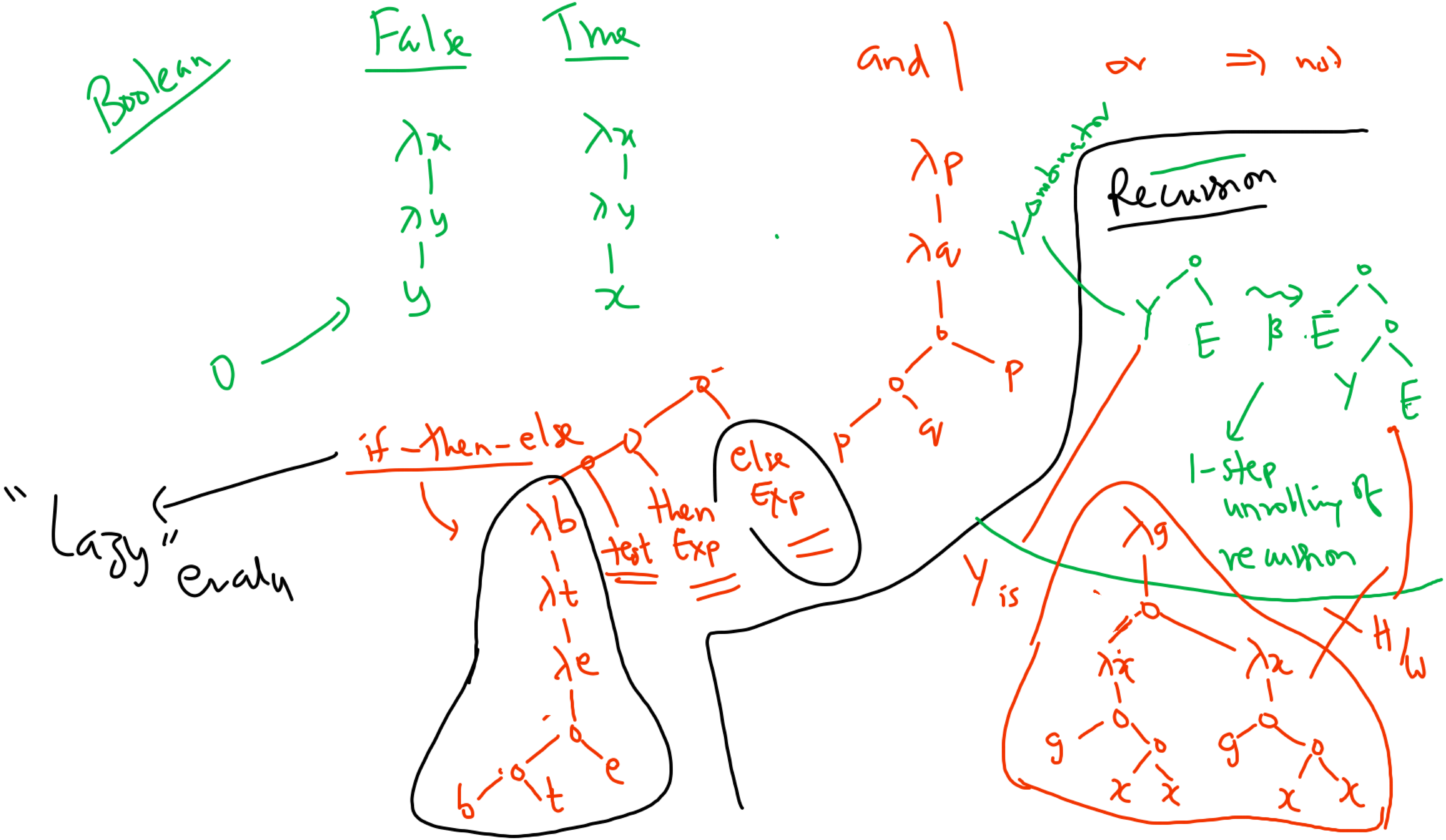
"replace x by E_2 in E_1"

Arithmetic

- exponentiation
- succ
- pred $0 \to 0$
- minus $\dot{-}$

recursion

- gcd
- fact


Boolean
False
True
λx
λy
y
λx
λy
x
0
and
or
⇒
not
λp
λq
p
p
q
Y-combinator
Recursion
Y
E
β
E
Y
E
1-step unrolling of recursion
if-then-else
"Lazy" evalu
λb
λt
λe
b
t
e
test
then Exp
else Exp
Y is
λg
λx
λx
g
x
x
g
x
x
H/W

Term Rewriting Systems → rules

replace equals by equals

$l \to r$

Vars x, y, z

Const 0, nil, . . .

list

Constructors s, cons

Functions + * f g

Numbers

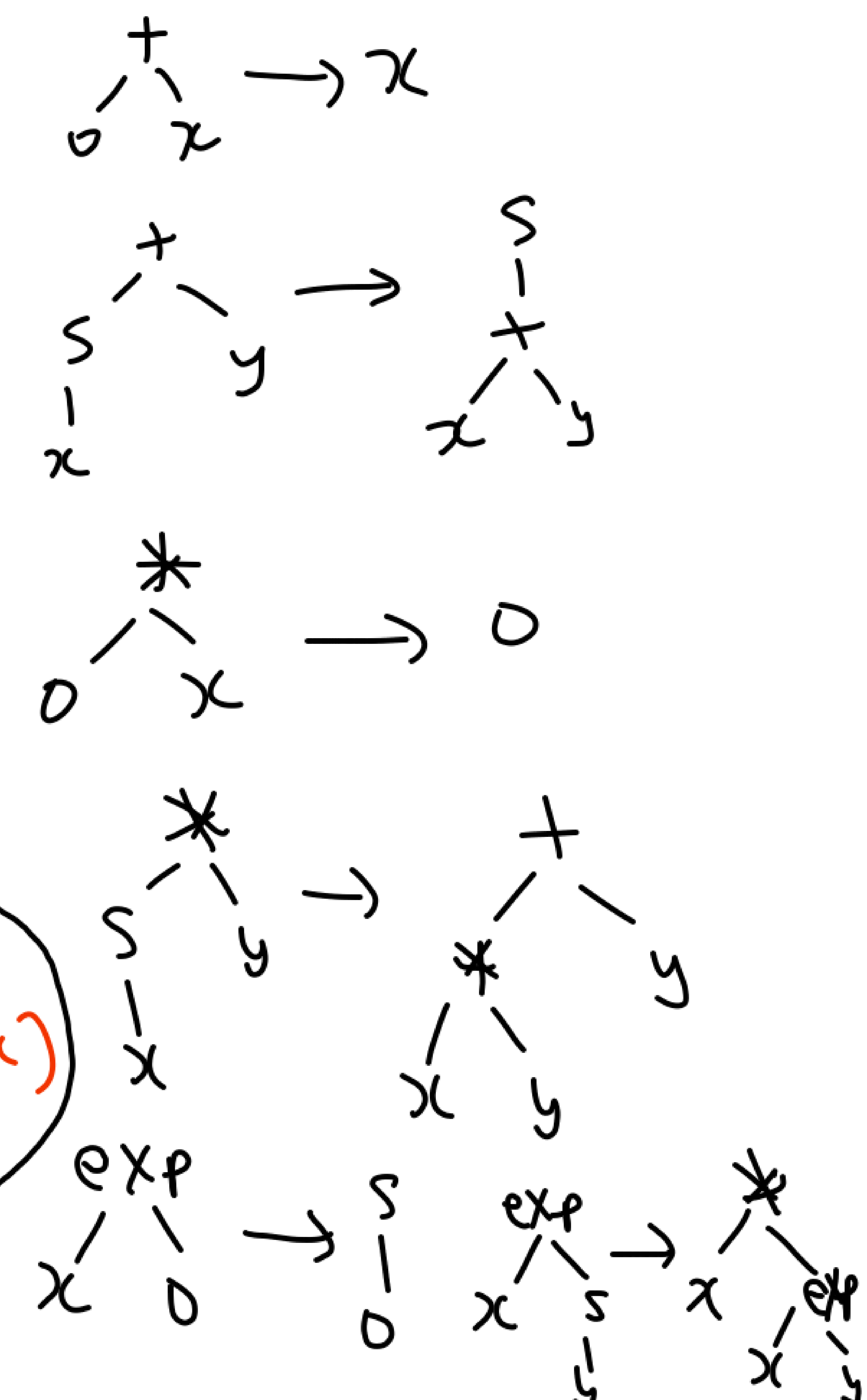
0 s s 0 s s s 0 s

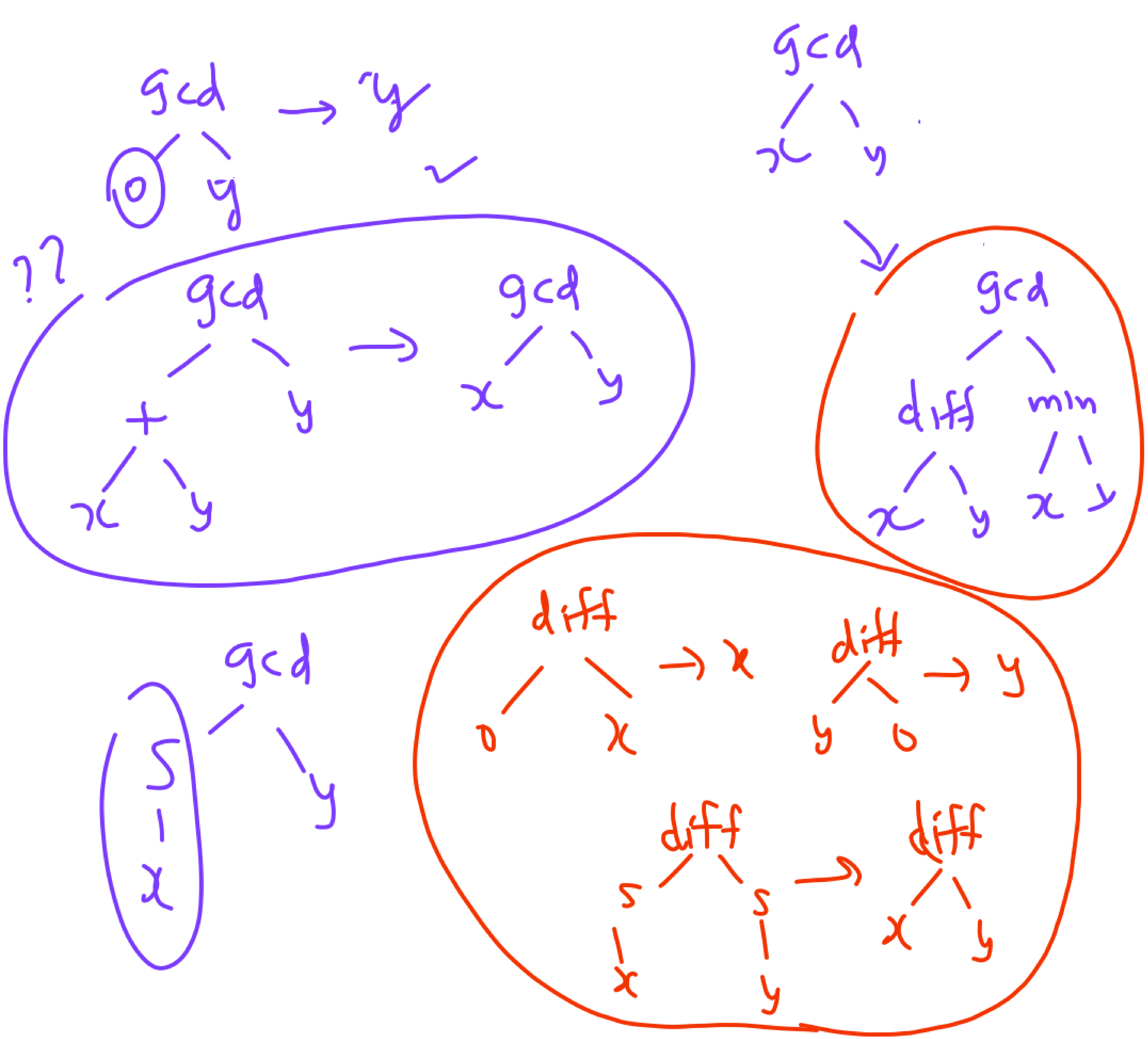
Lists

nil

cons

[a, b, c]





λ-calc ⟷ fnl prog (higher order)

first order

term rew

Logic Prog.

$+(3, 7) \to^* 10$

$+(x, y) \stackrel{?}{=} s^{10}(0)$

$x=0, y=10 \quad x=1, y=9 \quad \ldots \quad x=10, y=0$

22.10.25

Rec. Fn. Theory

N $\qquad N^k \longrightarrow N$

$n_1, \ldots, n_k \to f \to n_0$

(Building block)

how to define functions f

in a formal mathematical way

0

$S(n) \to n+1$

successor

Computation

Theory

B — (Logic)

Semantics

PL

A — (Algorithms

Discrete math)

Quiz 3

Wed 29, Oct

Functional — Haskell

1) λ-calculus

2) Term Rewriting — Prolog

(Equational Logic) — Logic Prog

3) Recursive Function Theory — Maths

Homework to be released soon

Quiz 2 marks released.

Basic Functions

1. Zero $\quad Z(n) = 0$

2. Succ $\quad S(n) = n + 1$

3. Projection $\quad P^{k}_{i}(n_1, \dots, n_k) = n_i$

$k \geq i$

P^{32}_{7}

32 → # of inputs

7 → projected output

4. Composition

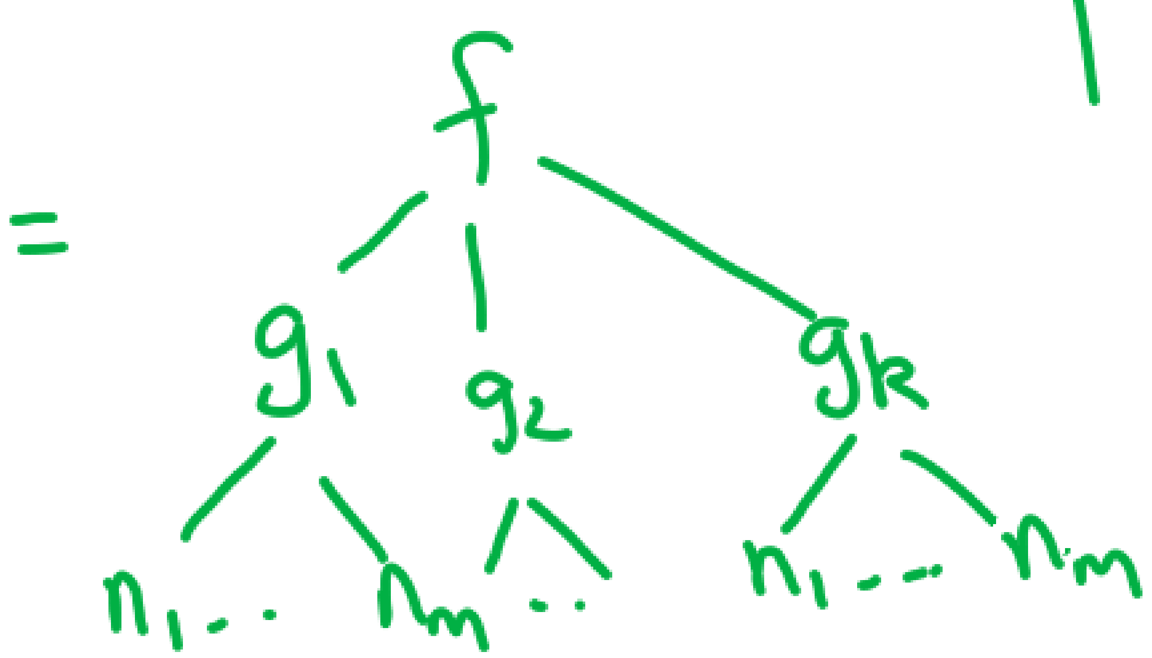
arity: $f \to k$

$g_i \to m$

$f : N^k \to N$

$g_1, \dots, g_k : N^m \to N$

$f \circ (g_1 \dots g_k) =$



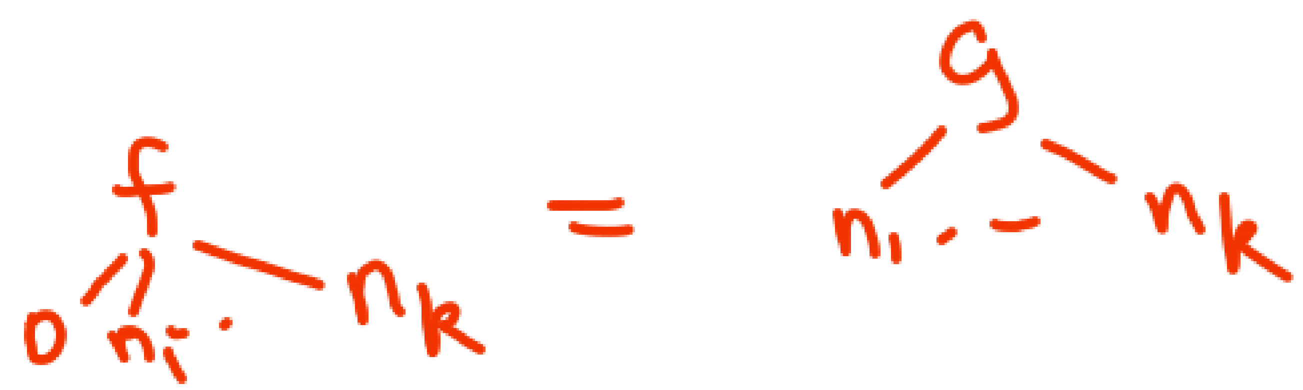
Zero, Succ, Proj

Primitive Recursive functions

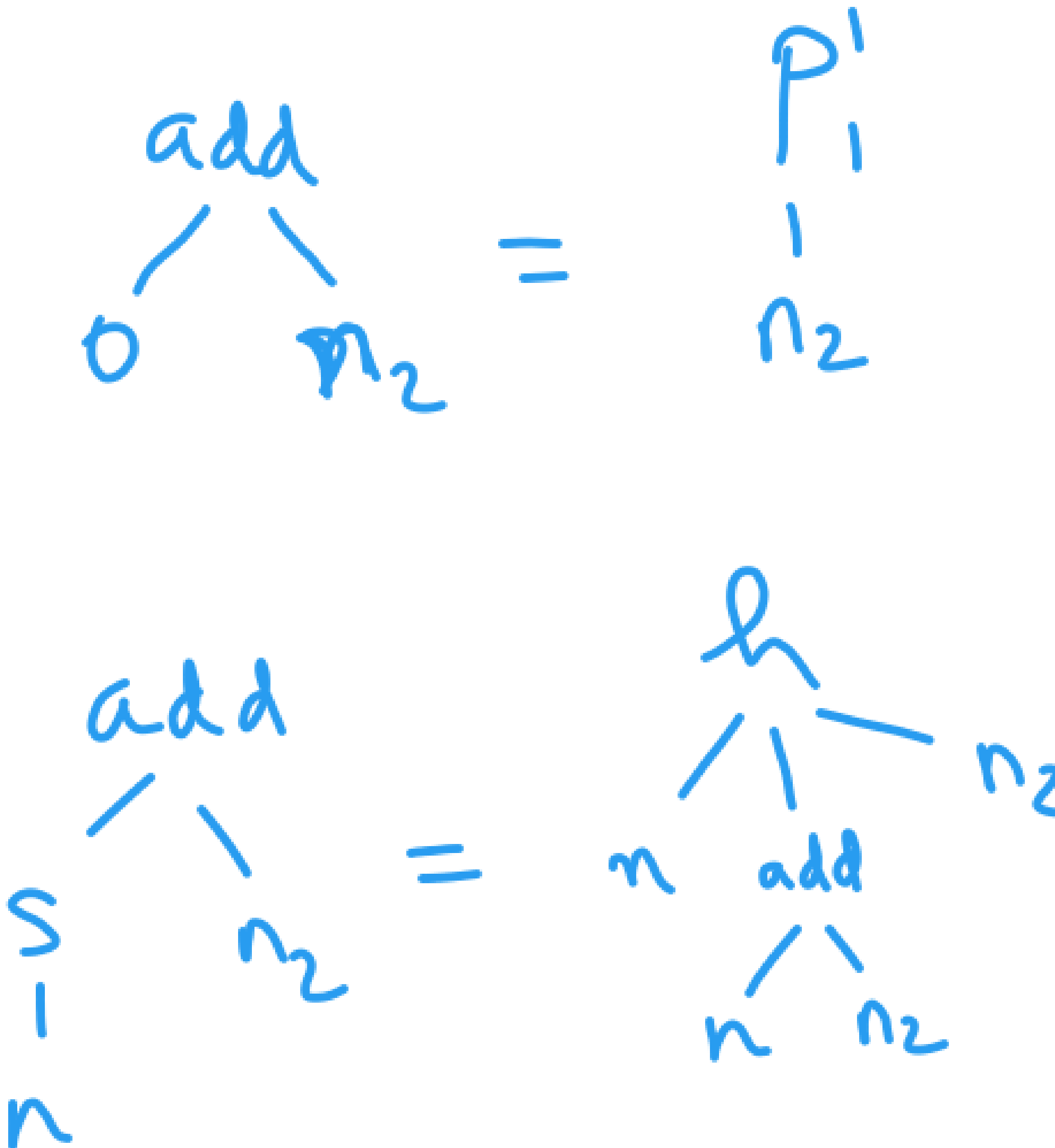
given $g : N^k \to N$ $h : N^{k+2} \to N$

define $f : N^{k+1} \to N$

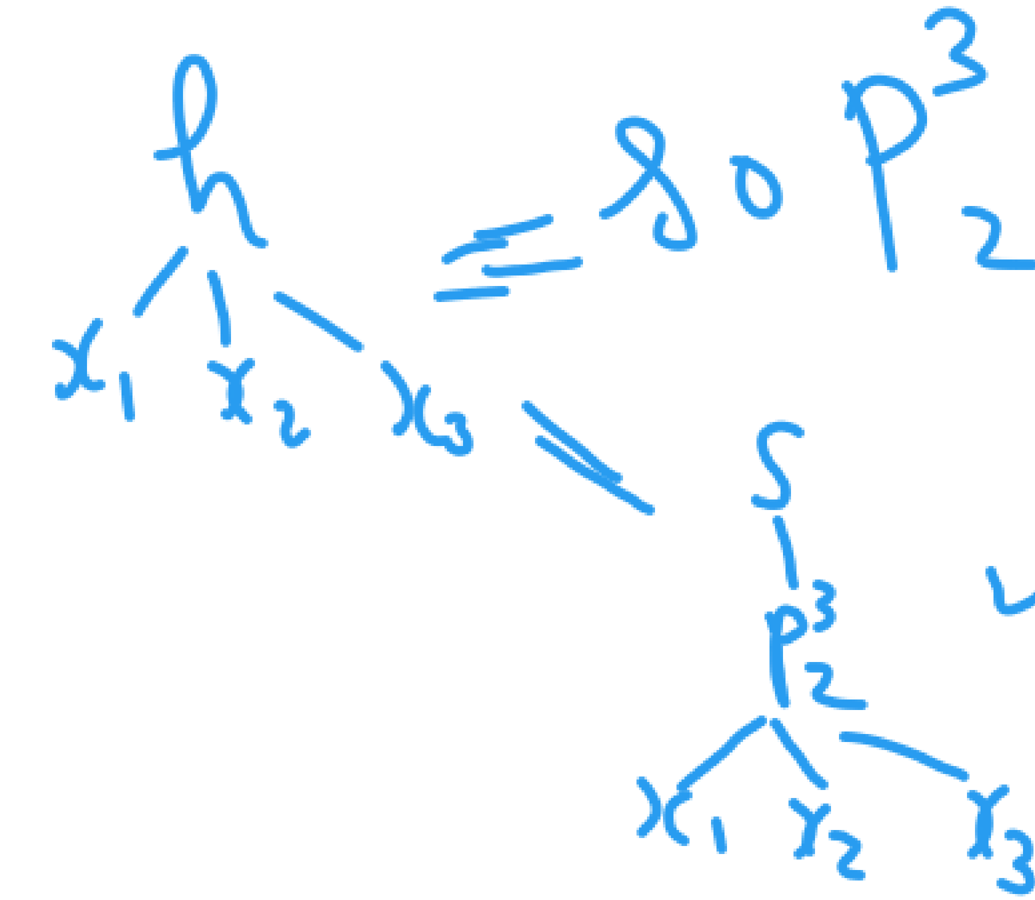
Base Case →

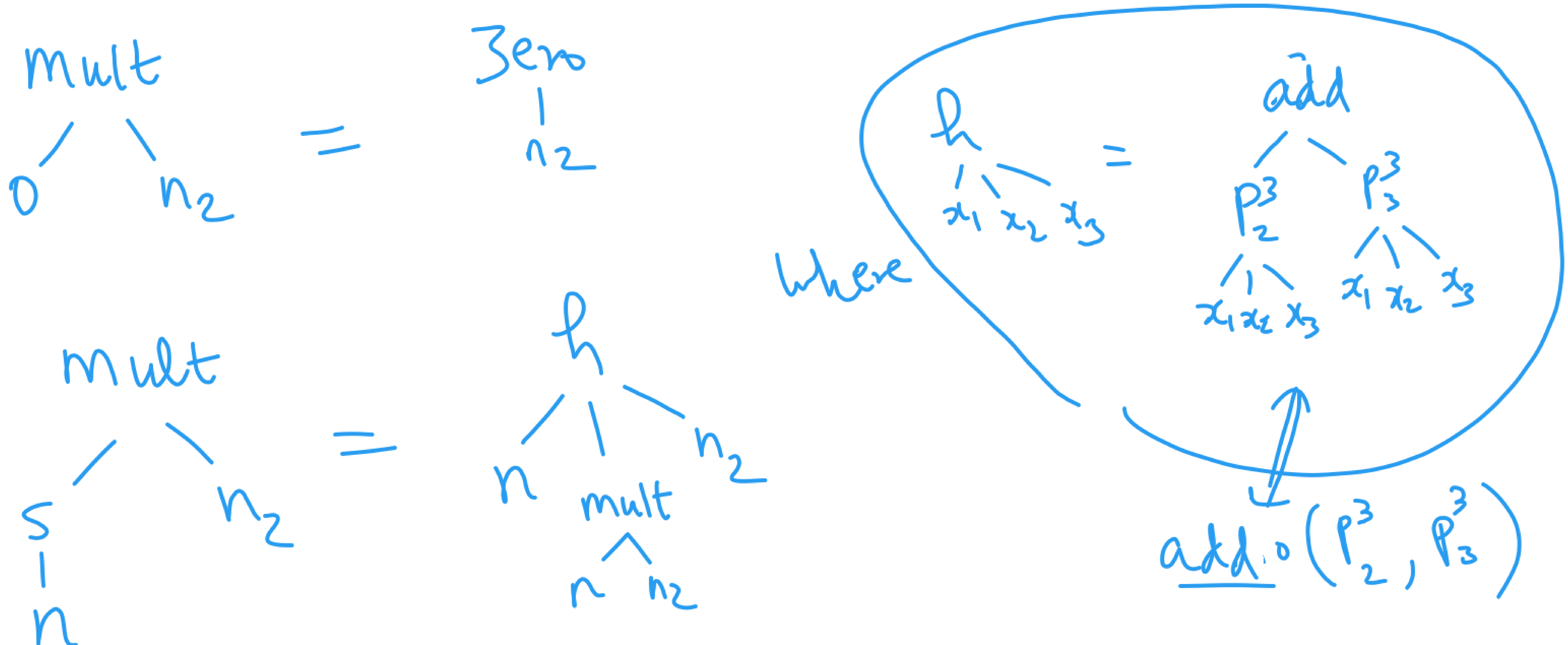


Rec Case



where





H/w → monus

$$x \dot{-} y = \begin{cases} 0 & \text{if } x \leq y \\ x - y & \text{otherwise} \end{cases}$$

→ exp → fact → fib

Ackermann Fn.

$a(3,3) \rightarrow 61$

$a(4,2) \rightarrow 2^{2^{16}} - 3$

$$a(0,n) = n+1$$

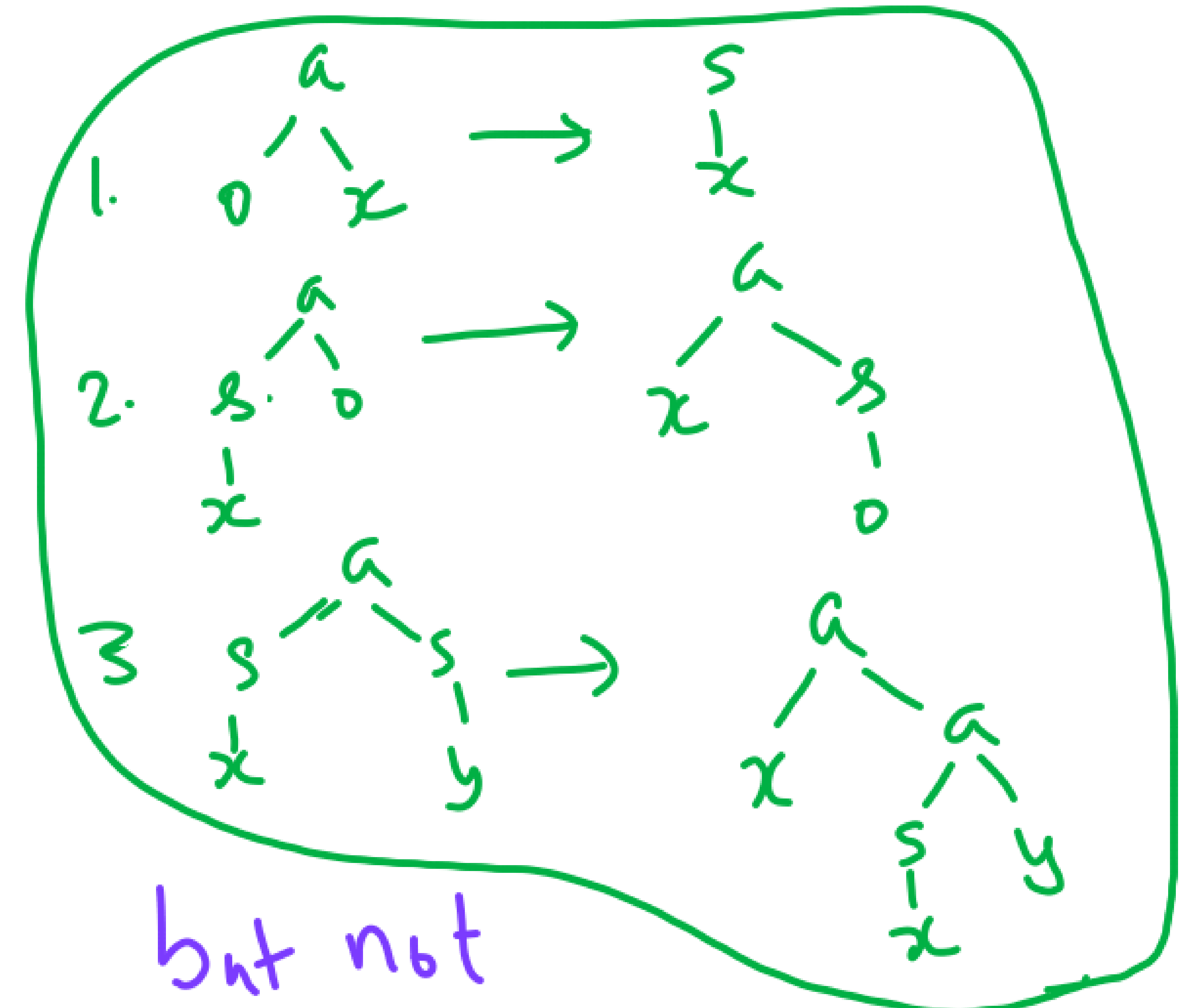
$$a(m,0) = a(m-1,1)$$

$$a(m,n) = a(m-1, a(m,n-1))$$

$a(2,3) \rightarrow a(1, \underline{a(2,2)}) \rightarrow a(1, \underline{a(1, a(2,1))})$

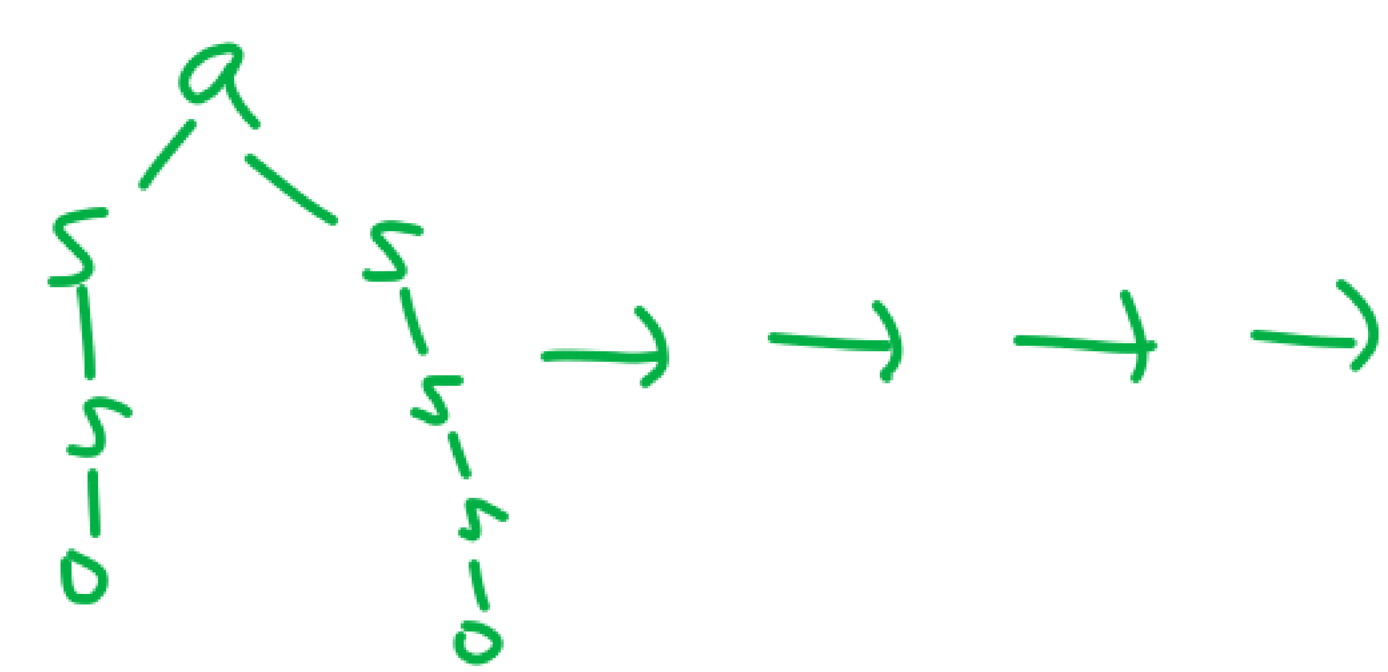
$\rightarrow \quad \rightarrow \quad 9$

Term Rewriting



but not

primitive recursion



Total

Partial functions

Minimization

Given $g : N^{k+1} \to N$

define $f : N^{k} \to N$

using minimization

$$f(n_1 \dots n_k) = \mu_n \, g(n, n_1 \dots n_k) = 0$$

Smallest value of n for which g becomes 0

$f(n_1 \dots n_k) \equiv n$

$f(13, 45)$

$= g(0, 13, 45) \to g(1, 13, 45) \to g(2, 13, 45) \to$

H/W $y^{38} \% 7 = 3$ remainder

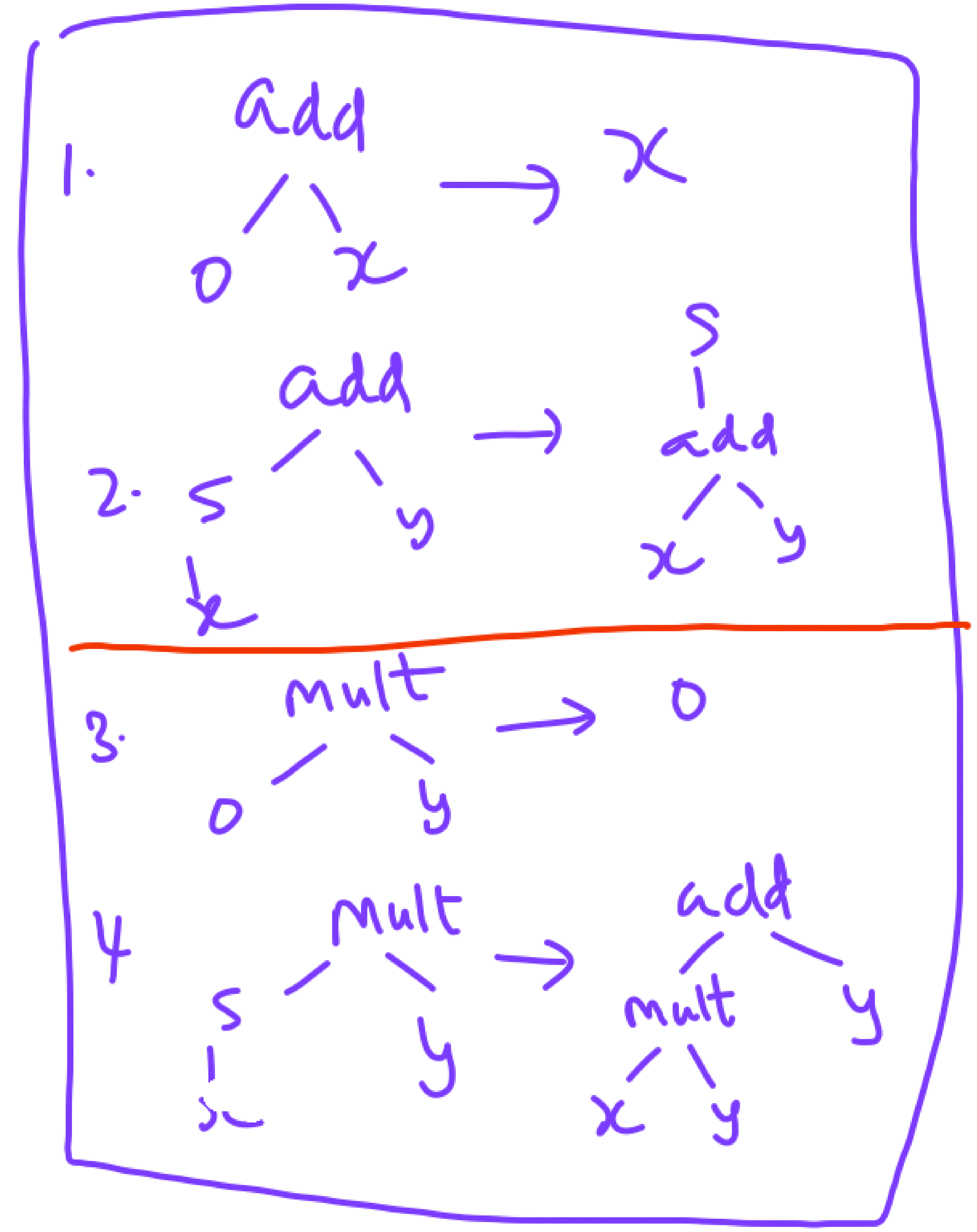
$38 / 7 = 5$ quotient

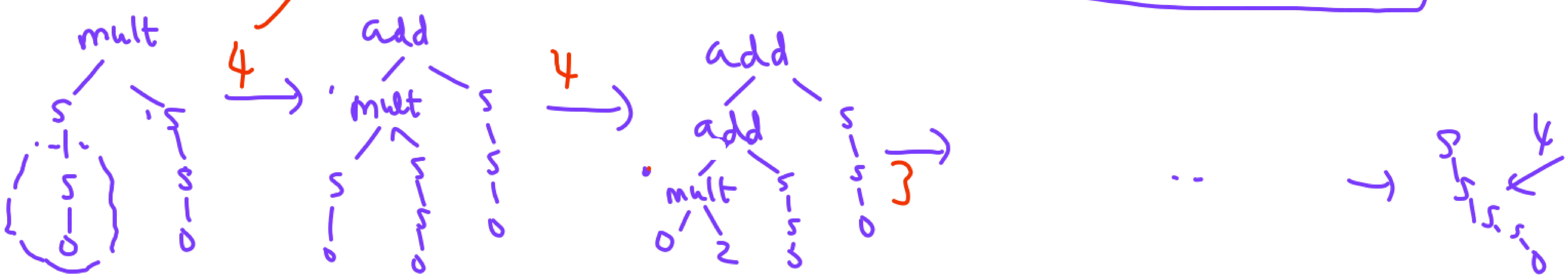
λ-calc   Rec. fn theory

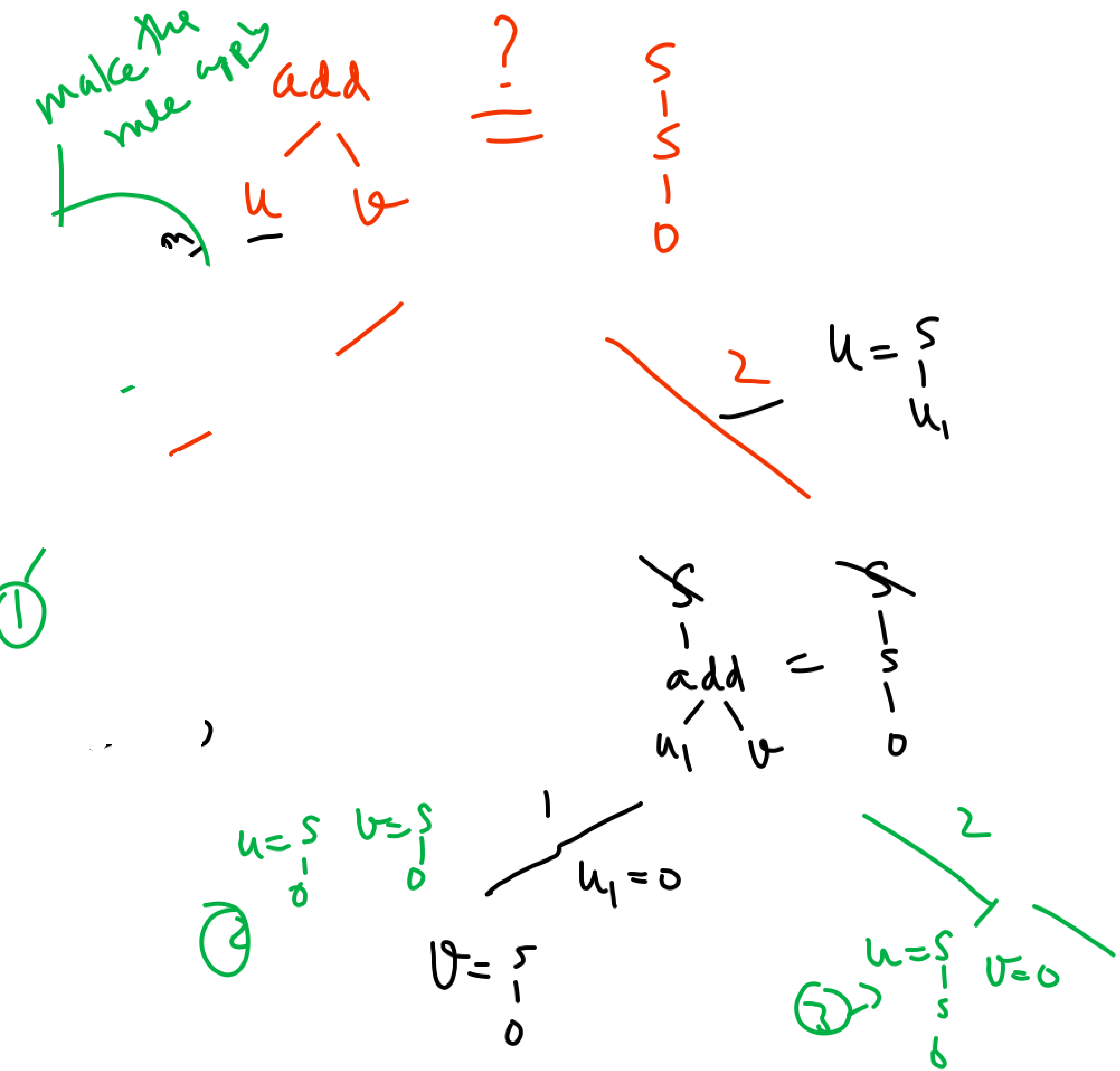
Term Rewriting

ackermann

matching & replacement







make the rule apply
add
u
v
?
=
S
S
0
2
u = S
u_1
1
S
add
u_1
v
=
S
0
1
u_1 = 0
v = S
0
2
u = S
0
v = S
0
u = S
S
0
v = 0
1
2
3

25.10.25

Lazy Evaluation → λ-expr $\xrightarrow{\beta}$ $\xrightarrow{\beta}$ - - - →

Quiz 3
Wed 29

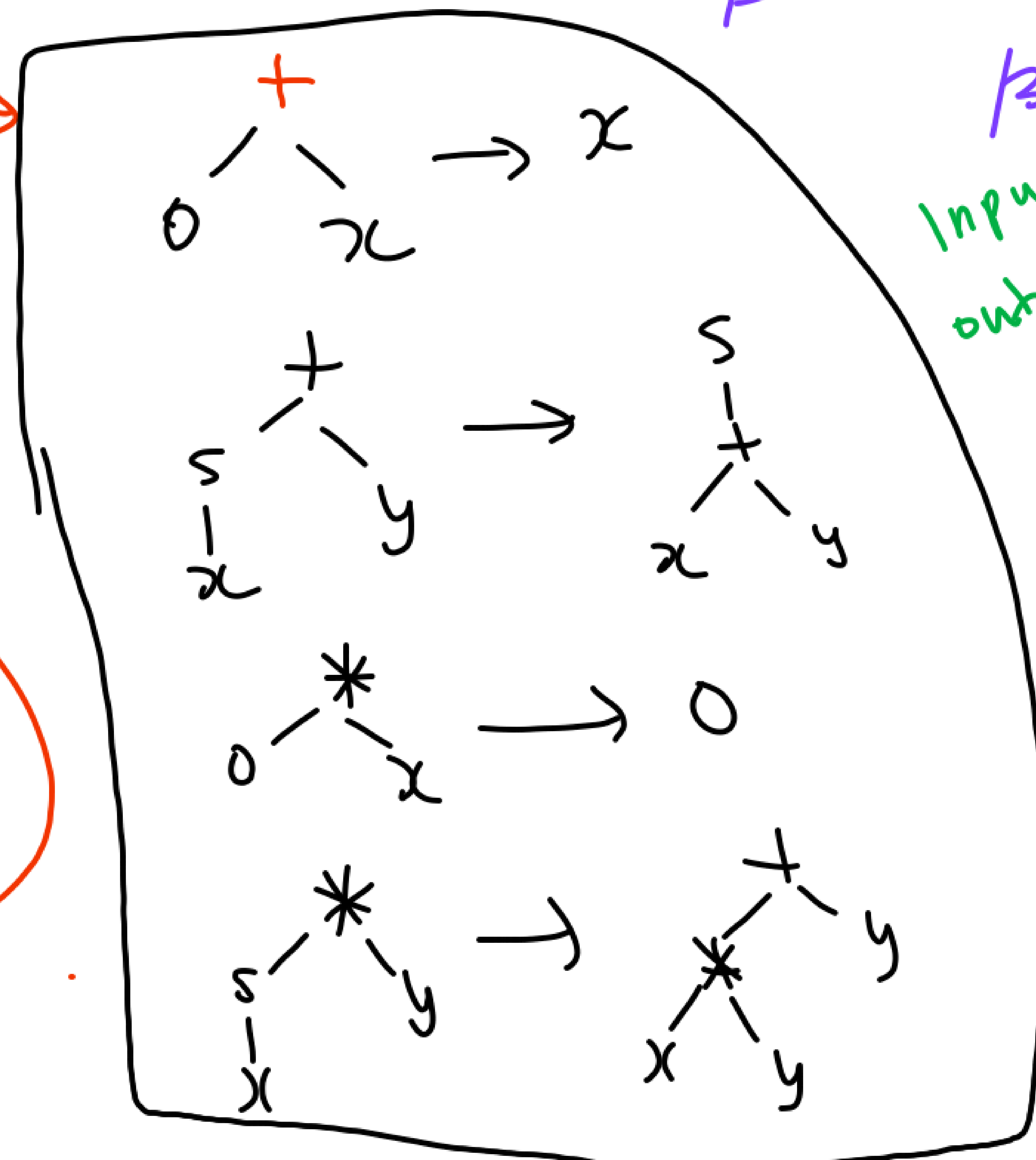
Prog Lang

Term

if-then-else
cond then else

```
def fact(x):
    if x=0 then 1
    else x * fact(x-1)
```

Spectre



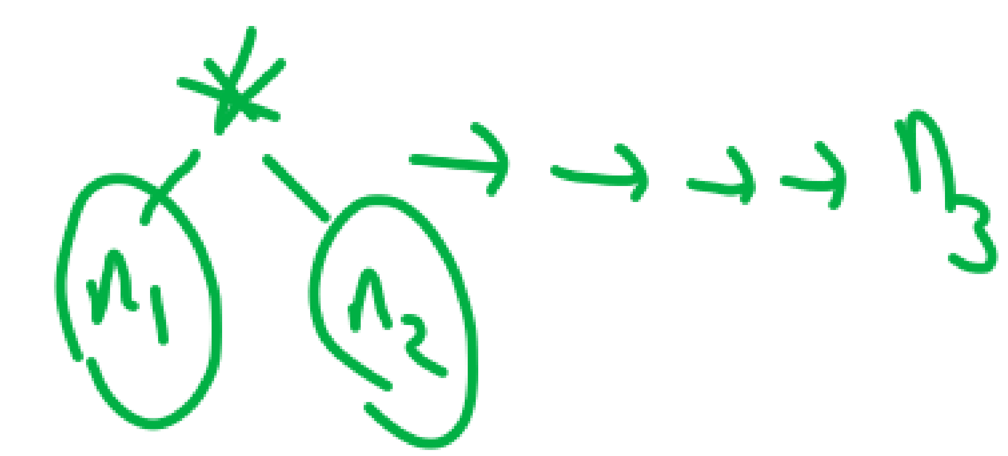
n_1, n_2

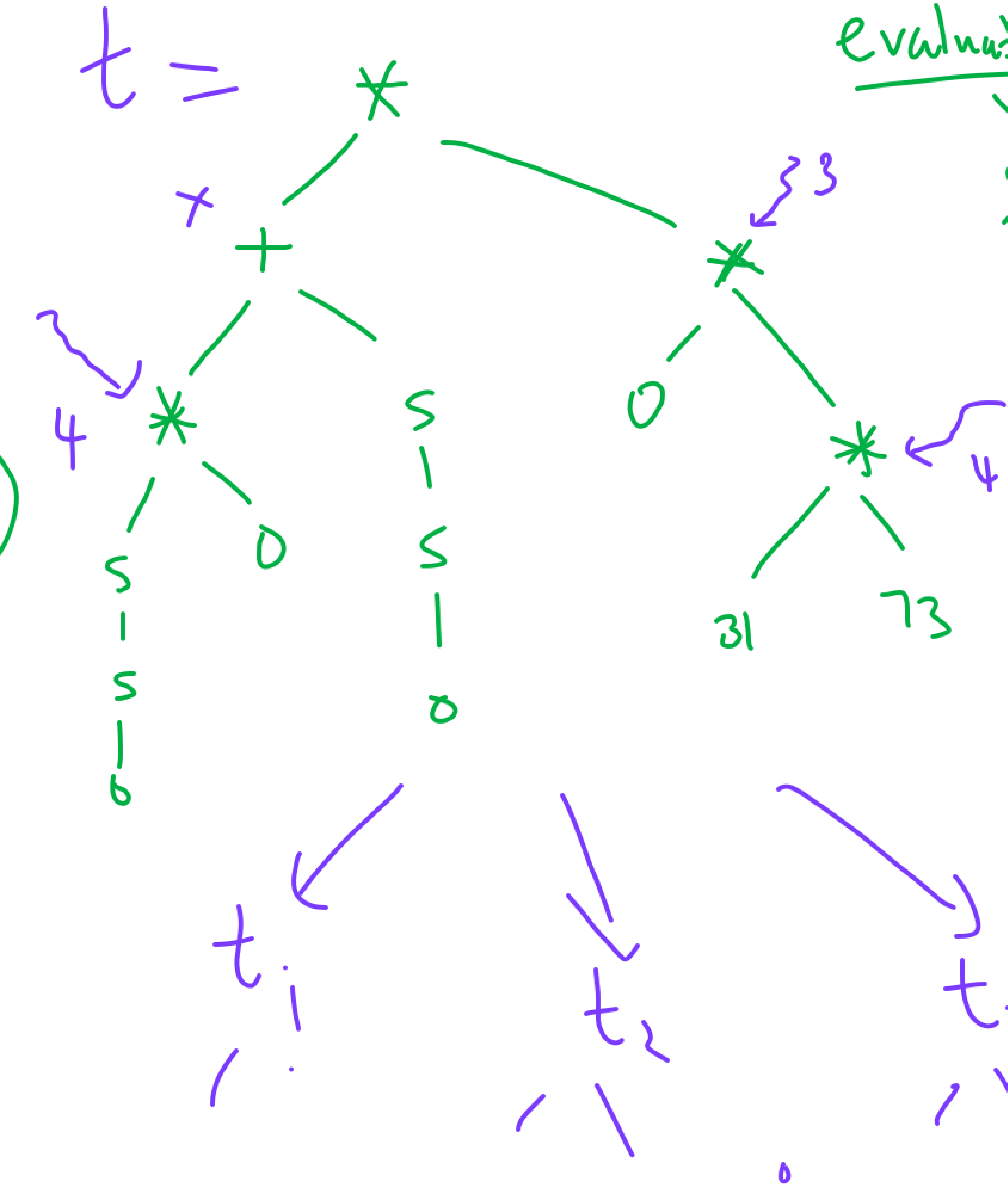
Input
output

0, 1, S
x
numbers

$+$ $*$ → defined fns

"Complete"





evaluate

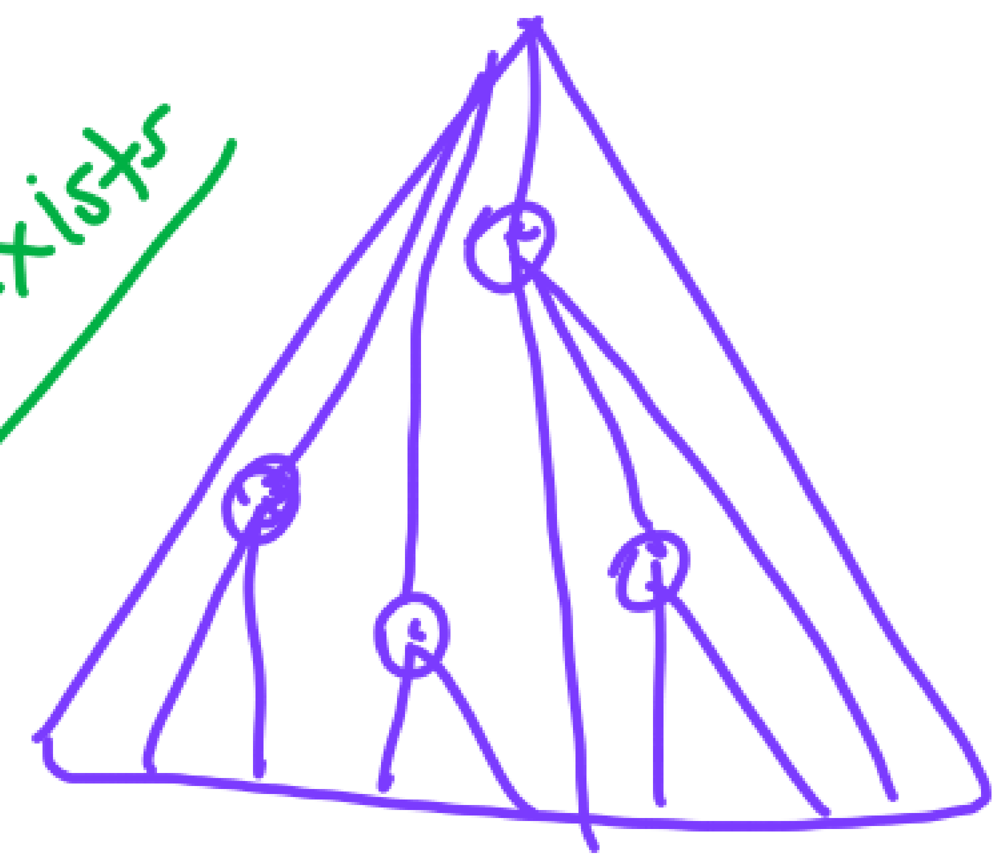
find its normal form

no rule applies

Redex

is a position where a rule can "apply"

Strategies for evaluating a term



guaranteed to find normal form iff it exists

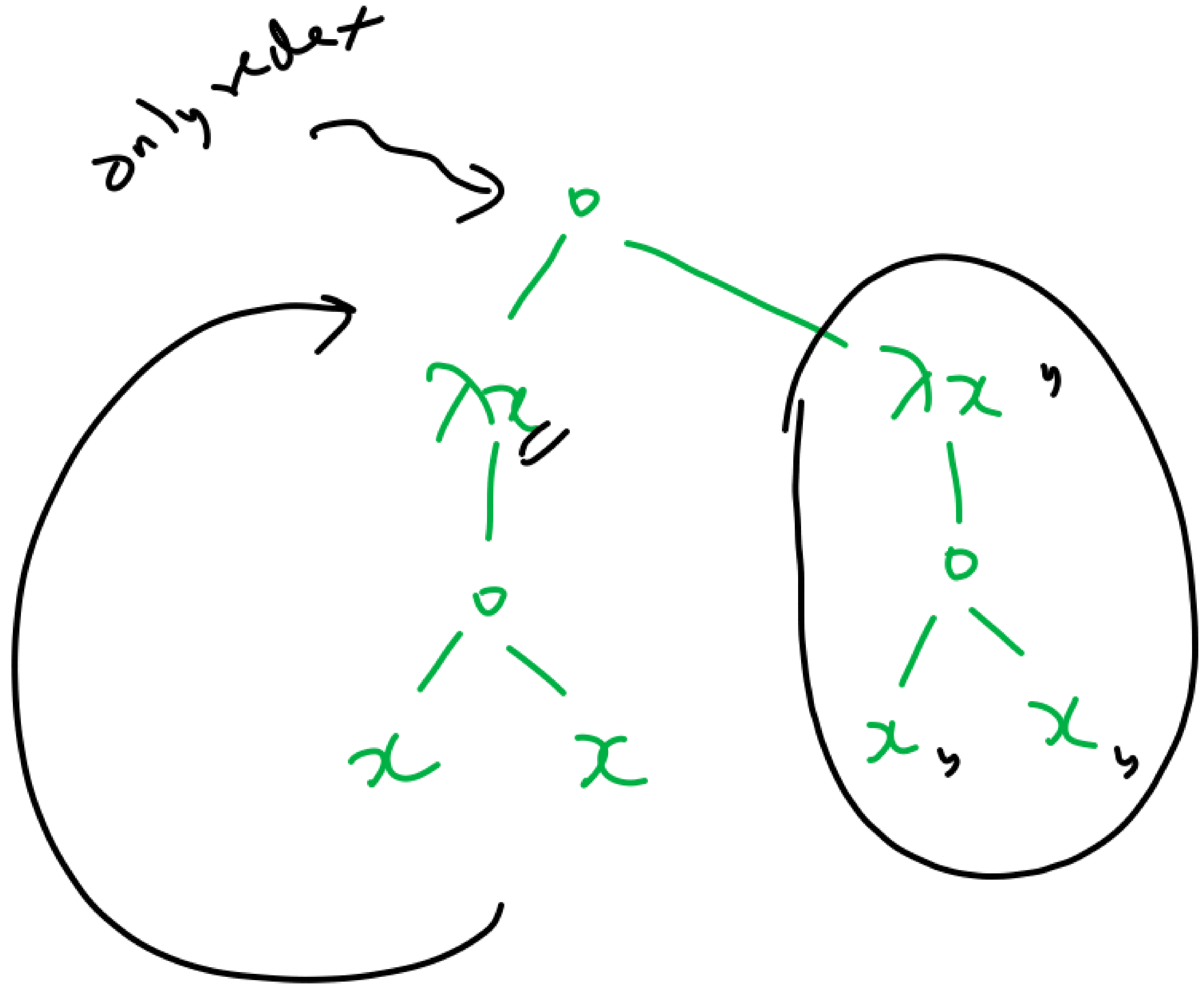
Outermost (Leftmost)

(Lazy)

Innermost (bottom-most)

Eager (strict)

Mixed (any)

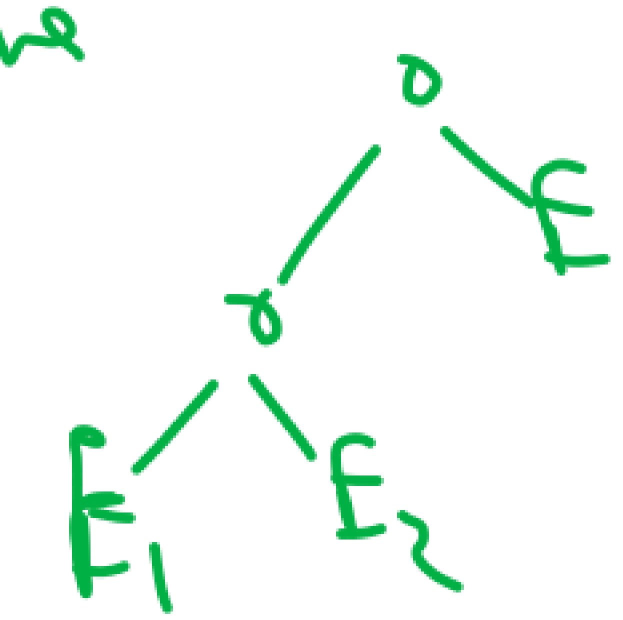


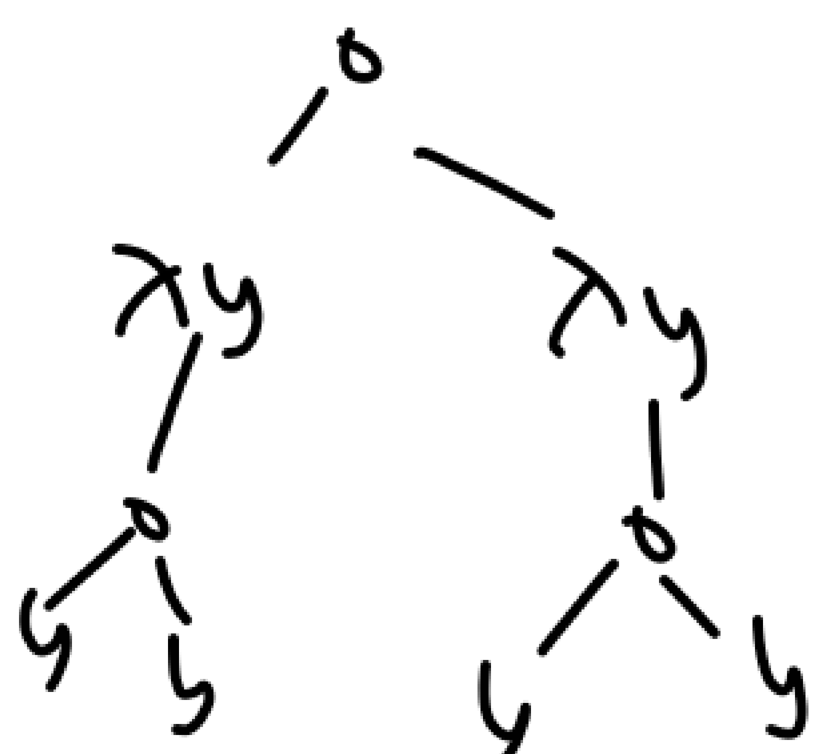
$$(\lambda x.\, x\, x) . (\lambda x\; x\; x)$$

$$E_1\ E_2\ E_3$$

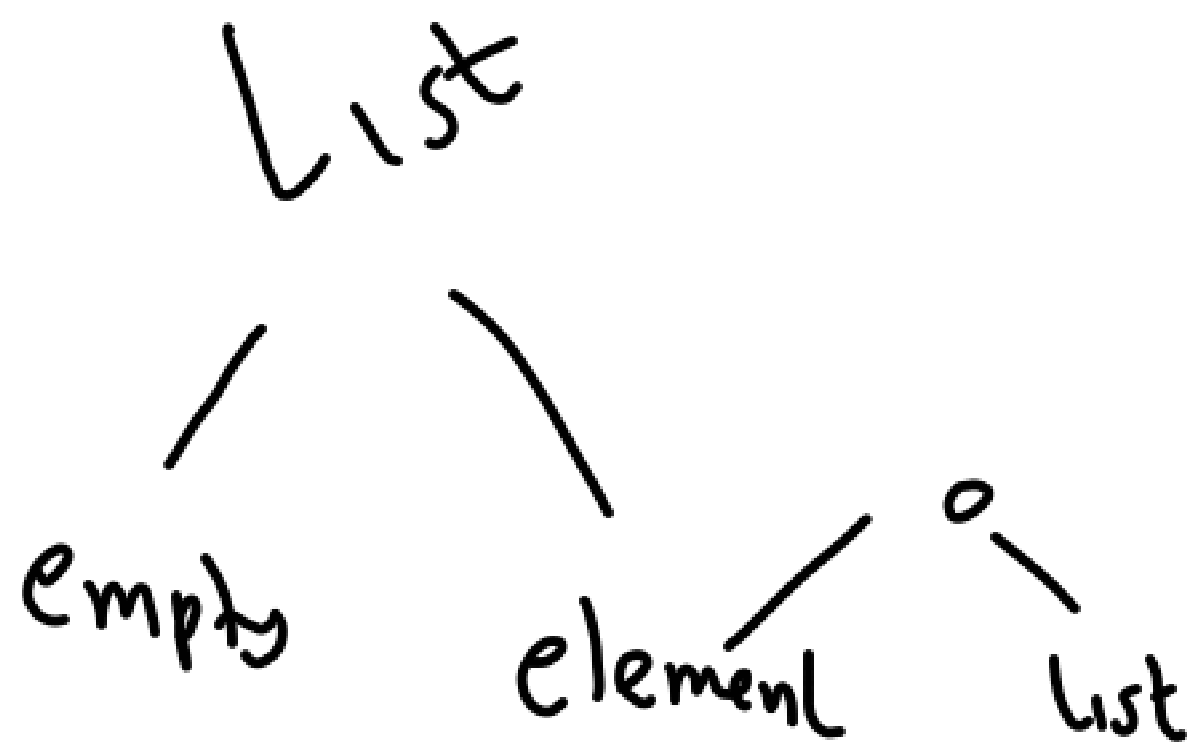
$$((E_1\ E_2)\ E_3)$$

left-associative





$$f(x) \rightarrow f(g(x))$$

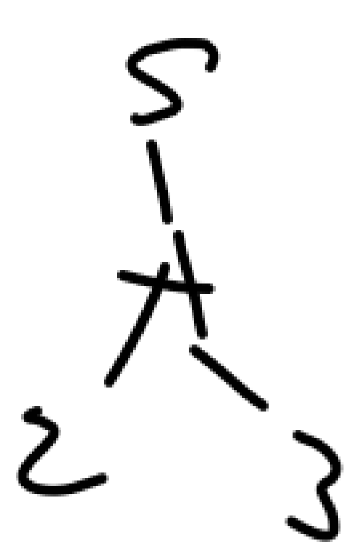


[ ] $\qquad$ [e | L]

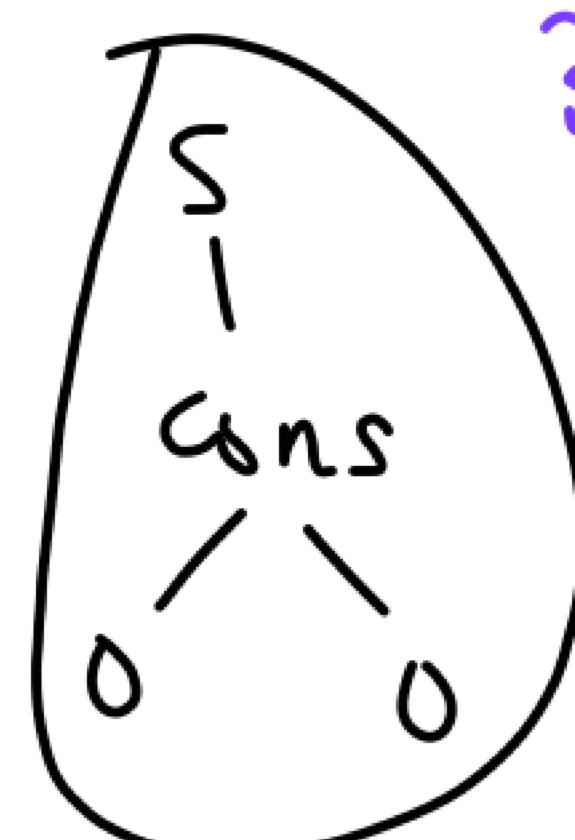
nil $\qquad$ cons(x, y)

L

[ ] $\qquad$ e . L



meaningless term



"x is a member of L" → 0 not there

→ s(0) if it is there

1. $\text{mem}(x, \underbrace{[\,]}_{\text{nil}}) \rightarrow 0$

2. $\text{mem}(x, \underbrace{[x \mid y]}_{\text{cons}(x,y)}) \rightarrow s(0)$

3. $\text{mem}(x, \underbrace{[y \mid z]}_{\text{cons}(y,z)}) \rightarrow \text{mem}(x, z)$

$0 \to$ false
$s(0) \to$ true

1. $mem(x, nil) \to 0$
2. $mem(x, cons(y, z)) \to ite(eq(x,y), 0, mem(x,z))$
3. $ite(0, x, y) \to y$
4. $ite(s(0), x, y) \to x$
5. $eq(0,0) \to s(0)$
6. $eq(0, s(x)) \to 0$
7. $eq(s(x), 0) \to 0$
8. $eq(s(x), s(y)) \to eq(x,y)$

Tail-Recursion

$f(0) \to s(0)$

$f(s(x)) \to *(s(x), f(x))$

1. $f(x) \to f1(x, s(0))$
2. $f1(0, y) \to y$
3. $f1(s(x), y) \to f1(x, *(s(x), y))$

1. $fib(0) \to s(0)$
2. $fib(s(0)) \to s(0)$
3. $fib(s(s(x))) \to +(fib(s(x)), fib(x))$

31.10.25

NP-Completeness

Polynomial # of steps

Turing Machine

Input $x_1 \mid x_2 \mid \cdot \mid \cdot \mid \ \mid x_n$ (n)

2^{nd} tape Guess → (factor) (m)

Verify in polynomial time

Non-deterministic — choice

s_3 —a→, —a→

Verifier "given by oracle" → "Guess"

→ Search \ don't know

any one correct choice ⇒ accept

vs find a solution

- Quiz 3
- Lectures
  1. 31/10
  2. ~~5/11~~
  3. 7/11

(Assignment - 6)

- Final Exam

(15/11)

Polynomial Time

Input Size

Knapsack

Stanford

Site

3 lectures

unary

numbers

log

binary

Deterministic

graph

Decision problems

vs

Computations or Optimizations

DTM

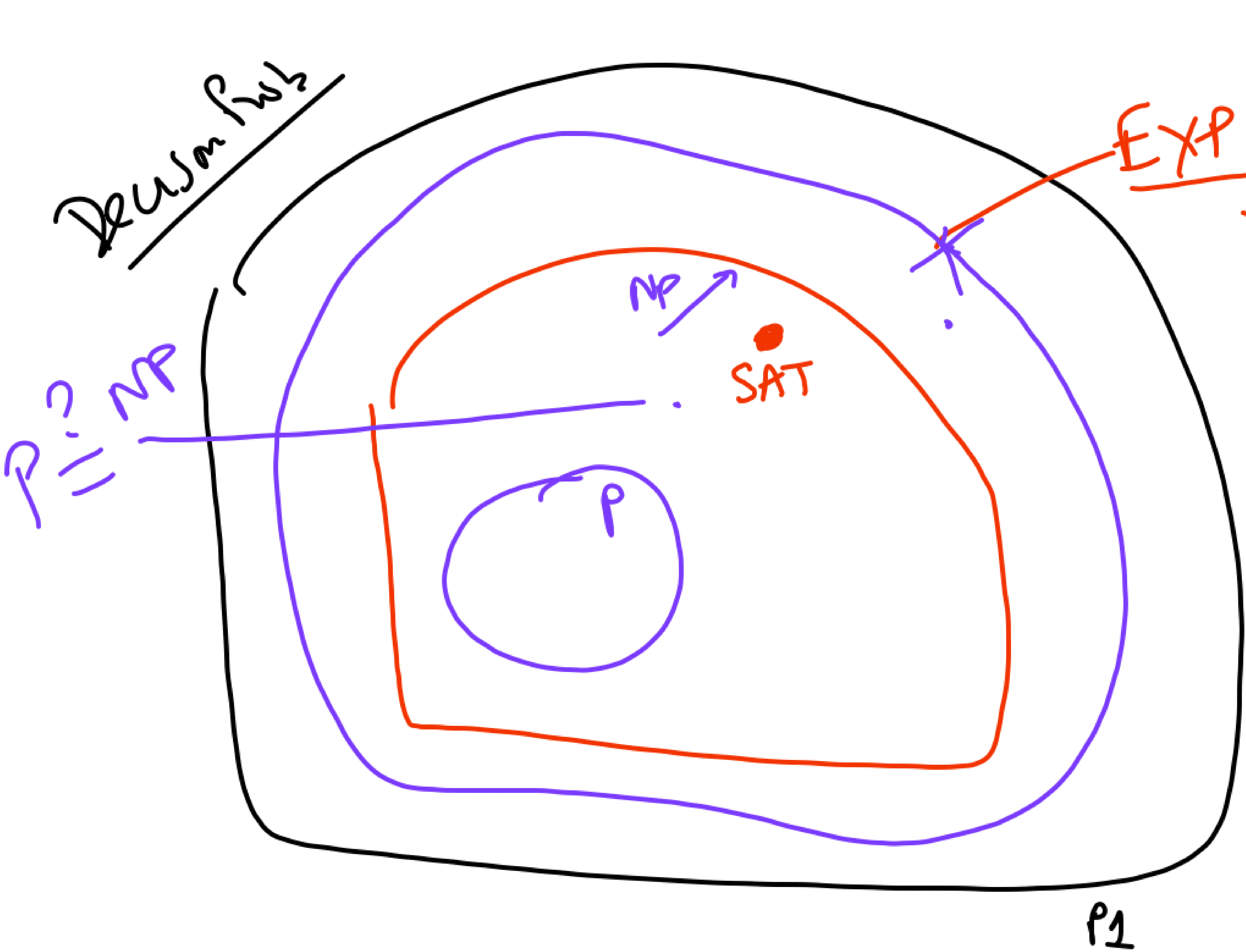
$\underbrace{n_1 \# n_2}_{\text{edge 1}} \# \underbrace{n_3 \# n_4}_{\text{edge 2}}$

Count # of steps

Polynomial time algorithms

$ID_0 \to ID_1 \to ID_2$

$\boxed{q_0 x_1 x_2 \ldots}$



requires exptime

P_2 is at least as difficult as P1

Polynomial time

a polynomial time soln for P_2 will give a poly soln for P1.

$P \subseteq NP$ ✓

$NP \subseteq P$?

1. Most-difficult problem in NP (SAT)
2. Show other problems are at least as difficult

Show that $\overline{\underline{3\text{-SAT}}}$ is atleast as difficult as $\overline{\underline{\text{SAT}}}$

Any prop logic formula is a $\underline{\text{CNF}}$ — formula

C_1	$x_1 \vee \overline{x_2} \vee x_4$
C_2	$\overline{x_3} \vee x_5 \vee x_1 \vee \overline{x_7} \vee \cdots$
$\vdots$	$\vdots$
C_k	

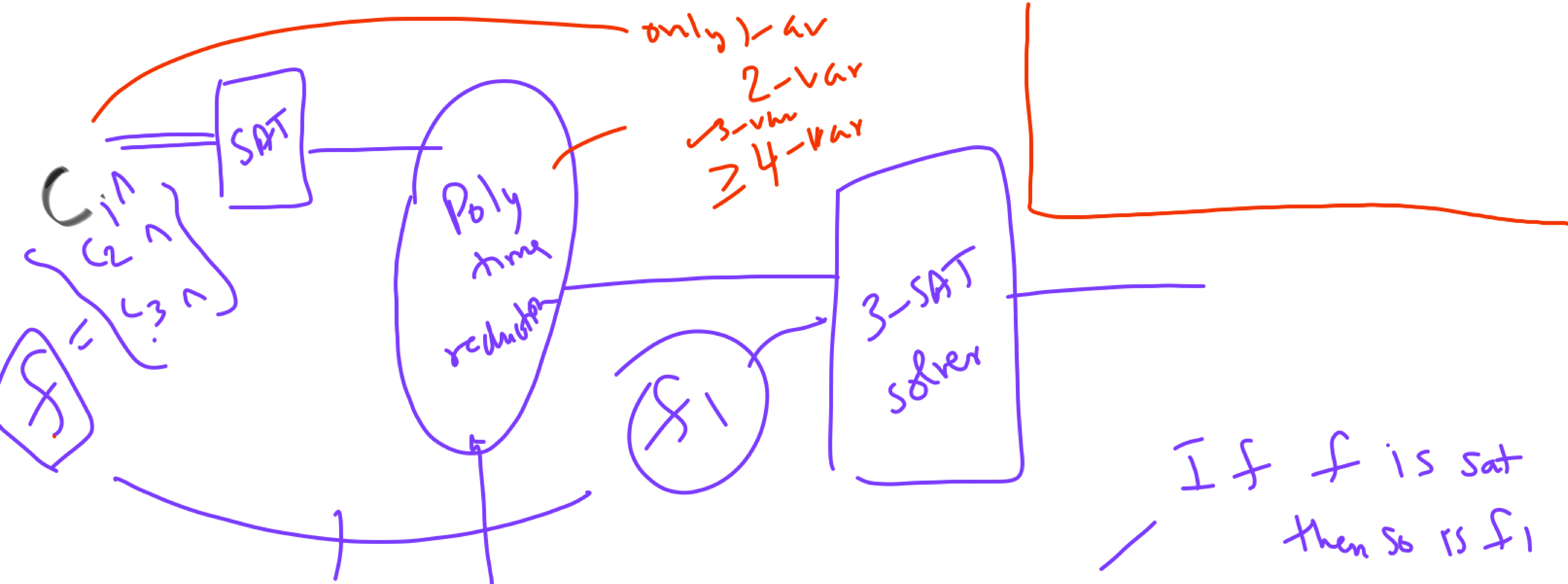
every clause exactly 3-literals

issatisfiable

Vars	truth
x_1	T
x_2	F
$\vdots$	
x_3	F
$\vdots$	

model

Literal	$P, \neg P$
Clause	$P \vee Q \vee \neg R \vee \neg S \cdots$
CNF	$\underbrace{C_1 \wedge C_2 \wedge \cdots \wedge C_k}$



If f is sat then so is f_1

a, b Convert f to f_1 → equiSatisfiable
(SAT) (3-SAT)

If f_1 is sat then so is f

a, b, c

f equivalent f_1 ~~not same~~

$a \vee b$ $(a \vee c) \wedge (\neg c \vee b)$ } $a, c = f$, $b = \text{true}$

SAT $\longrightarrow$ 3-SAT

x — l_1

$\bar{y}$ —

$l_1 \rightsquigarrow$ ~~2 new vars~~ $(l_1 \vee x_1 \vee x_2)$ $(l_1 \vee \bar{x}_1 \vee x_2)$ $(l_1 \vee x_1 \vee \bar{x}_2)$ $(l_1 \vee \bar{x}_1 \vee \bar{x}_2)$

$l_1 \vee l_2 \rightsquigarrow (l_1 \vee l_2 \vee x) \wedge (l_1 \vee l_2 \vee \bar{x})$

new variable

$\rightarrow \underbrace{l_1 \vee l_2 \vee \cdots \vee l_{100}}_{\text{97 new vars}}$

$x_1 \cdots x_{97}$

$(l_1 \vee l_2 \vee x_1)$

$(\bar{x}_1 \vee l_3 \vee x_2)$

$(\bar{x}_2 \vee l_4 \vee x_3)$

$\cdots$

$(\bar{x}_{97} \vee l_{99} \vee l_{100})$

98 clause

Recap.

Stanford lecture slides

any { NTM, polynomial verifier } $\leadsto$ SAT $\leadsto$ 3-SAT

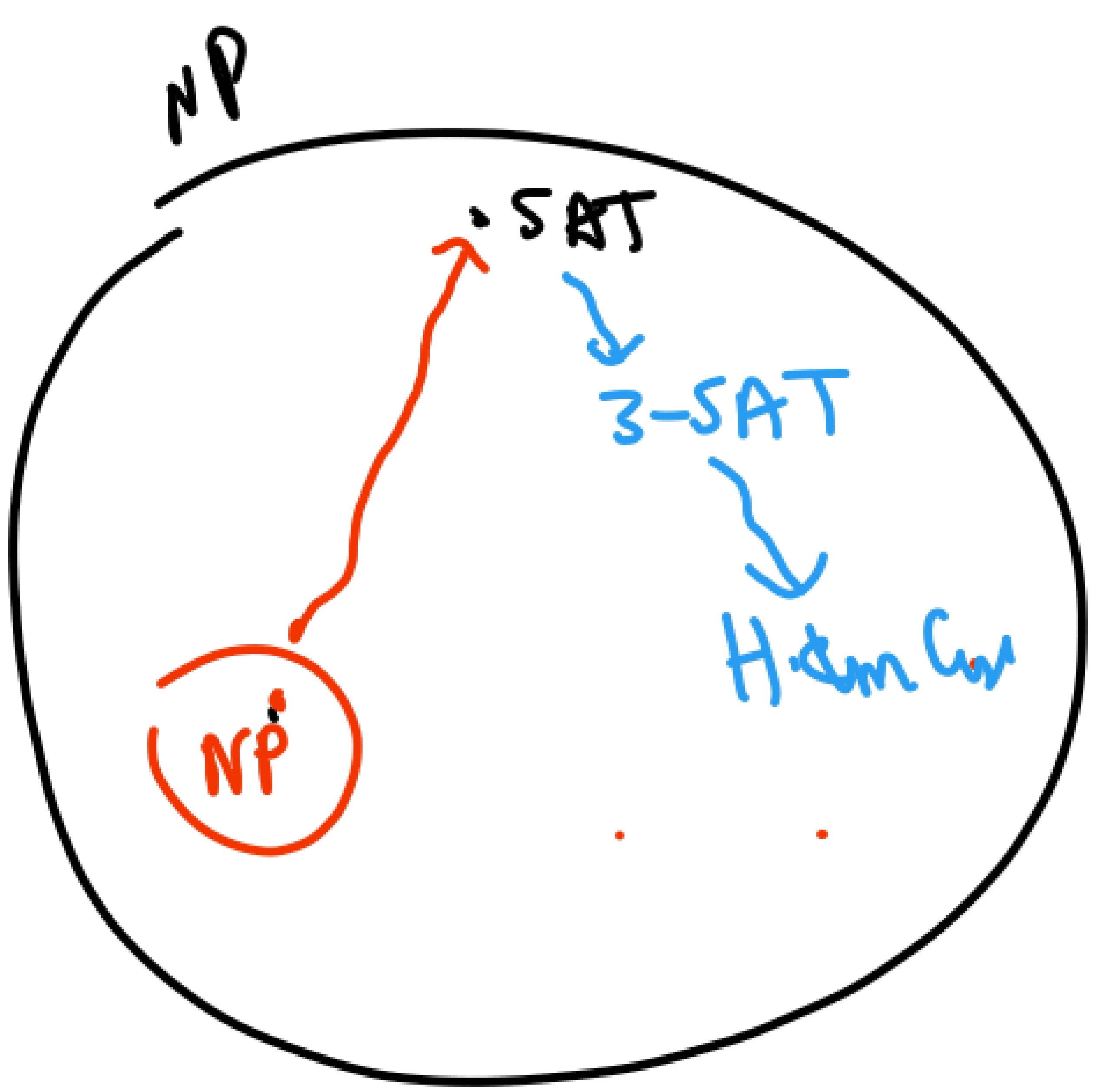
H/w 7/11

is the most difficult (NP-complete)

3-SAT

(a) Subset-Sum → Knapsack

(b) Hamiltonian Circuit (TSP) → clique → Vertex-cover



Yes ✓

3SAT → Poly (converges) → Ham Cir.

known NP-complete

C1. $x_1 \vee \overline{x_2} \vee x_4$

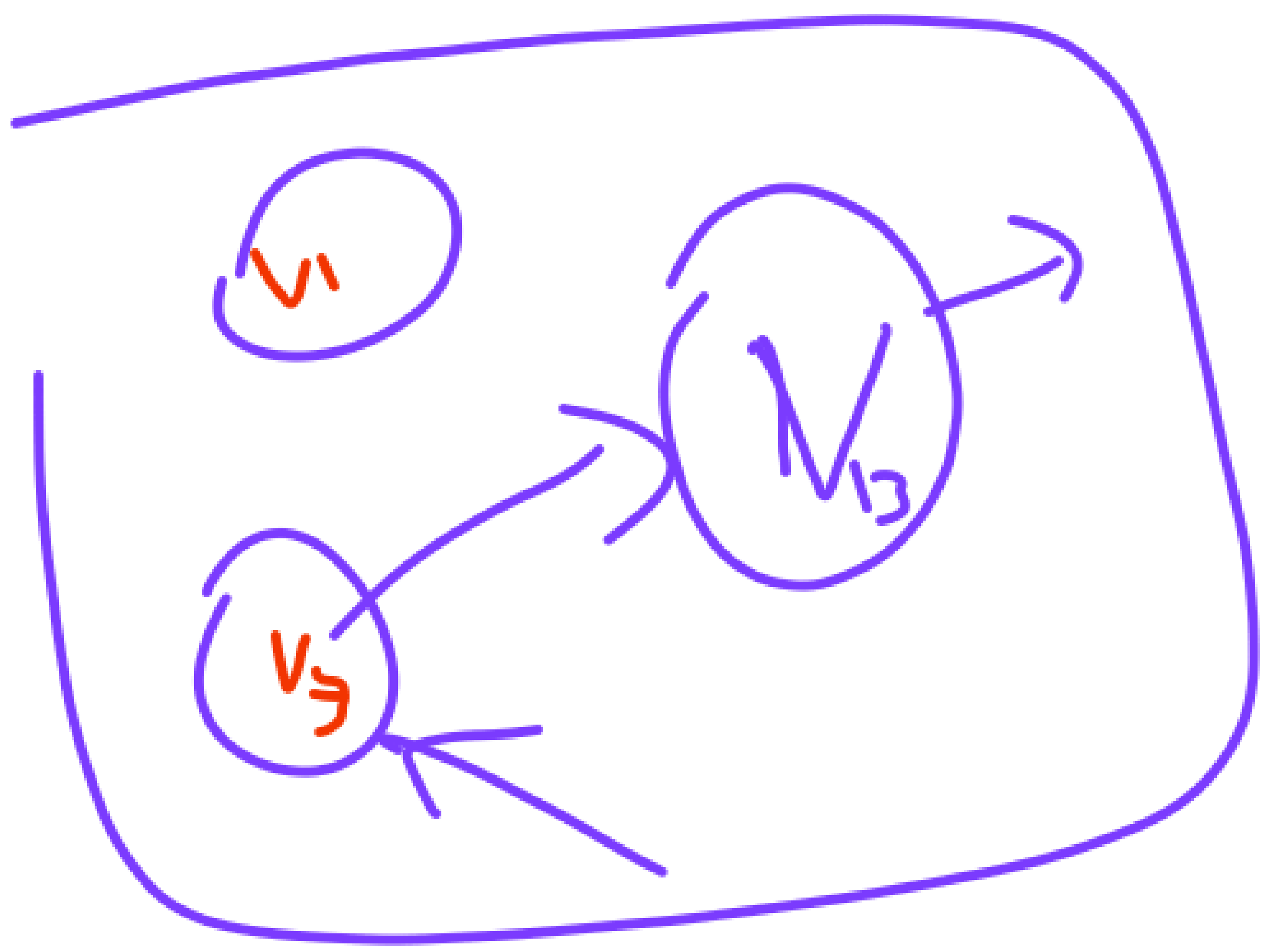
C2. $\overline{x_1} \vee x_3 \vee \overline{x_4}$

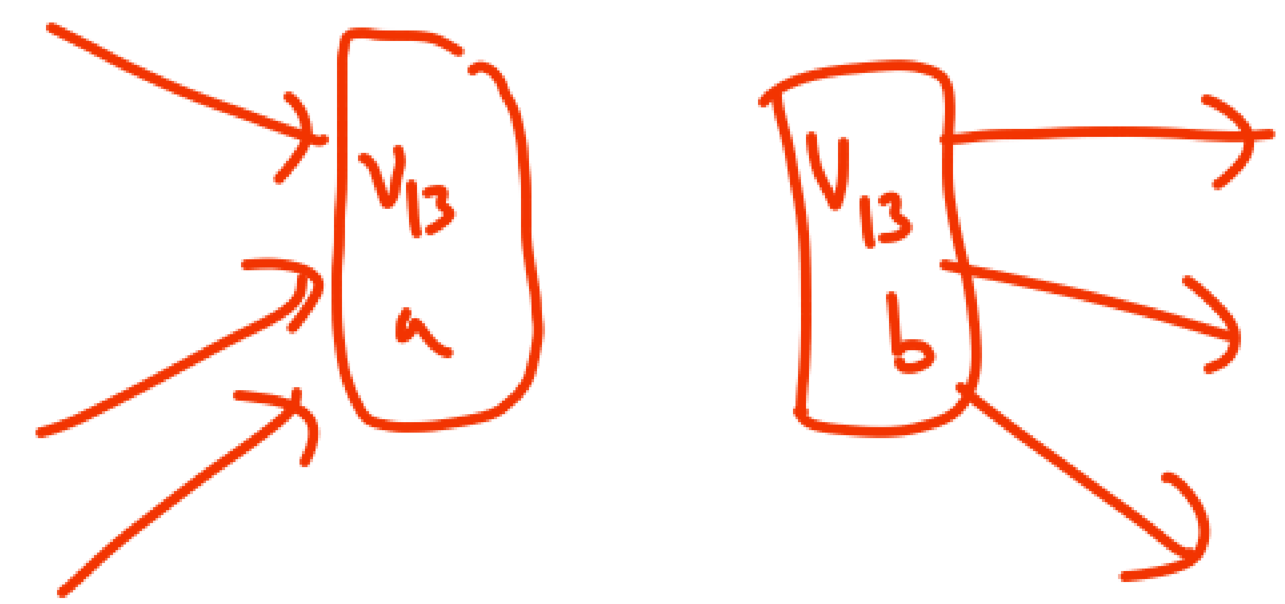
C3. $\overline{x_2} \vee \overline{x_3} \vee x_4$

Soln-preserving (equi satisfiable)

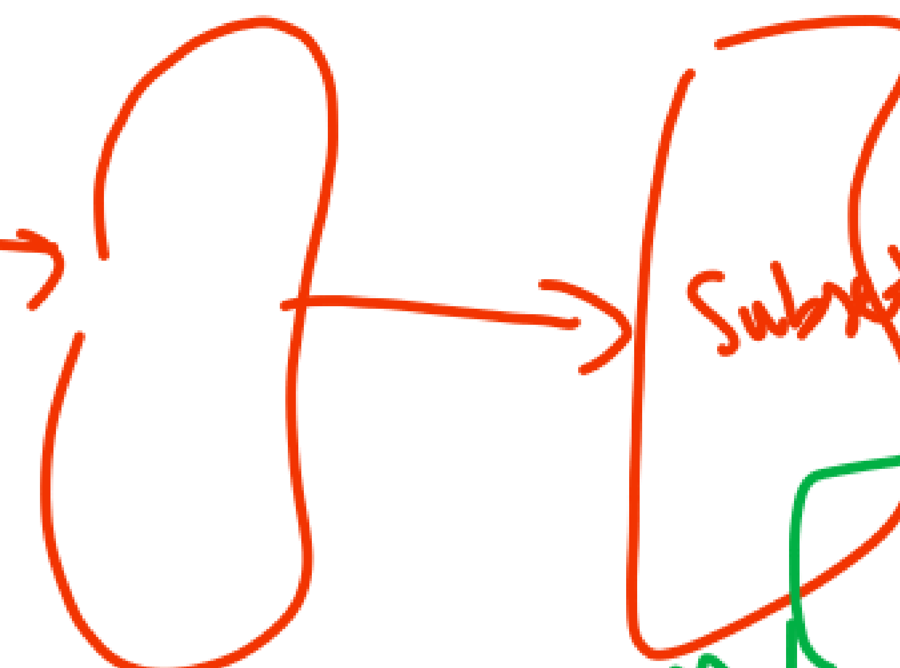
3 pairs of arrows | one for each literal

3SAT $\longrightarrow$ HC is NP-complete $\longrightarrow$ HP-path is NP complete





Subset-Sum

3-SAT →  → Subset Sum

C_1 $x_1 \vee \overline{x_2} \vee x_3$

C_2 $x_2 \vee \overline{x_3} \vee \overline{x_4}$

S has 12 numbers

$S = \{n_1 \dots n_k\}$, M

Subset $Q \subseteq S$ → sum of numbers in $Q = M$

slack

S ↓

	x_1	x_2	x_3	x_4	C_1	C_2
M	1	1	1	1	3	3
x_1 → p_1	1	0	0	0	1	0
x_1 → n_1	1	0	0	0	0	0
x_2 → p_2	0	1	0	0	0	1
x_2 → n_2	0	1	0	0	1	0
p_3	0	0	1	0		
n_3	0	0	1	0		
p_4	0	0	0	1		
n_4	0	0	0	1		
C_1 → s_1, t_1	0	0	0	0	1	0
C_2 → s_2, t_2	0	0	0	0	1	0

M

C_1

7.11.25

CS 310

Σ, alphabet

States & transitions

$q_i \xrightarrow{a,\ b \to \alpha} q_j$ (stack) — stack symbols; a,ϵ

$a \to b,\ L/R$ (Tape) — tape symbols

Final states F, q_f

FSA	PDA	(TM) L
ID $\langle q_0, w\rangle$	$\langle q_0, w, Z_0\rangle$	$\langle q_0, \vdash w\rangle$ (tape)
Accept (Run): $\langle q_0,w\rangle \vdash \langle\ \rangle \cdots \vdash \langle q_f, \epsilon\rangle$	Run: $\langle q_0,w,Z_0\rangle \vdash \cdots \vdash \langle q_f, \epsilon, \alpha\rangle$, $\alpha \in \Gamma^*$ (accept by final state)	Run: $\langle q_0,w\rangle \vdash \vdash \cdots \vdash \langle q_f, \alpha\rangle$ (anything on tape) — final state
move (1-step)	$\langle q_0,w,Z_0\rangle \vdash \cdots \vdash \langle q, \epsilon, \epsilon\rangle$ (accept by empty stack)	accept by Halt: $\langle q_0,w\rangle \vdash \cdots \vdash \langle q,\alpha\rangle$ — no move possible — halt/crash
Variants (choice): NFA, NFA+ϵ	DPDA $\to$ ~~no choice~~ at most 1-choice (no more $\to$ halt/crash)	DTM, NTM | 2-tape, multi-track; stay S
	NPDA $\to$	

TM: L $\to$ rec. enumerable; $\to$ recursive (always halts)

1. Construct
2. Convert NFA+ϵ $\to$ DFA
3. Minimize
4. Not possible

2 × pass

DFA $\to$ Reg Exp

Grammars $\langle \Sigma, V, S, P \rangle$ — rules $l \to r$

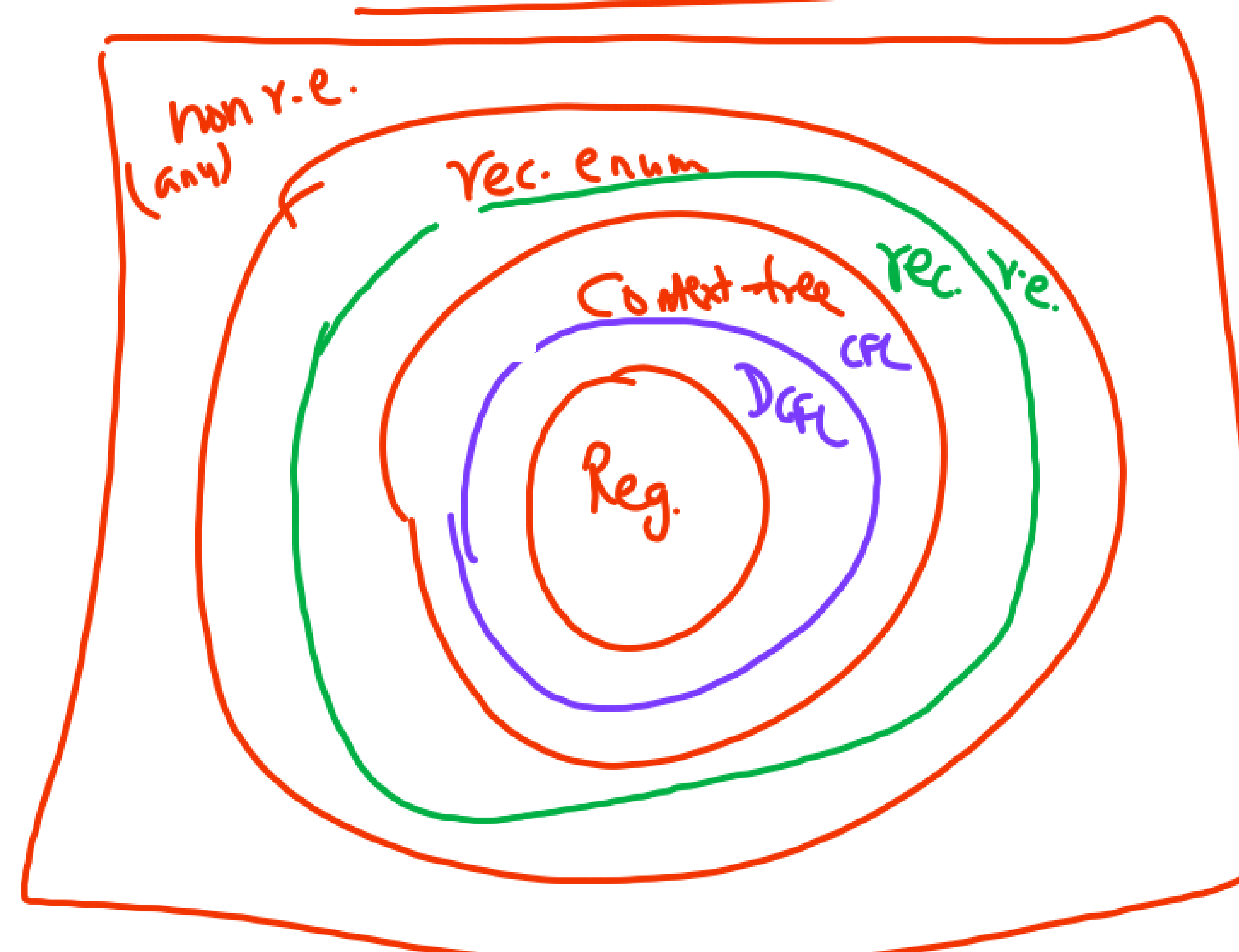
	l	r
FSA = Reg	$l \in V$ (simple var)	left-linear $l \to X\overline{abc}$; right-linear $l \to abcY$
D PDA ≠ PDA — Context free	$l \in V$ (simple var)	any string $(\Sigma \cup V)^*$
TM Type-0 r.e.	any	any

CSL

reg-exp

$(0+1)^* \, 1 \, 0^* (\ldots)$

CHOMSKY-Hierarchy



Extra.

Counting

Enumeration

not r.e.

Undecidability

Post Corr Problem

NP-completeness.

λ-calc

Equational Prog.

Recursive Function Theory

Topics

Closure Properties

$L_1 \cup L_2$, $L_1 \cap L_2$

$\vdots$

$L_1 - L_2$

Decision Problems

Is $w \in L$?

Normal Forms

Chomsky

Greibach

Time complexity of

Decision Problems

Recursive

TM always answers Yes/No

$P \stackrel{?}{=} NP$ ①

NP → Verify guessed soln in polynomial # of steps

DTM in polynomial number of steps

w $|w| = n$

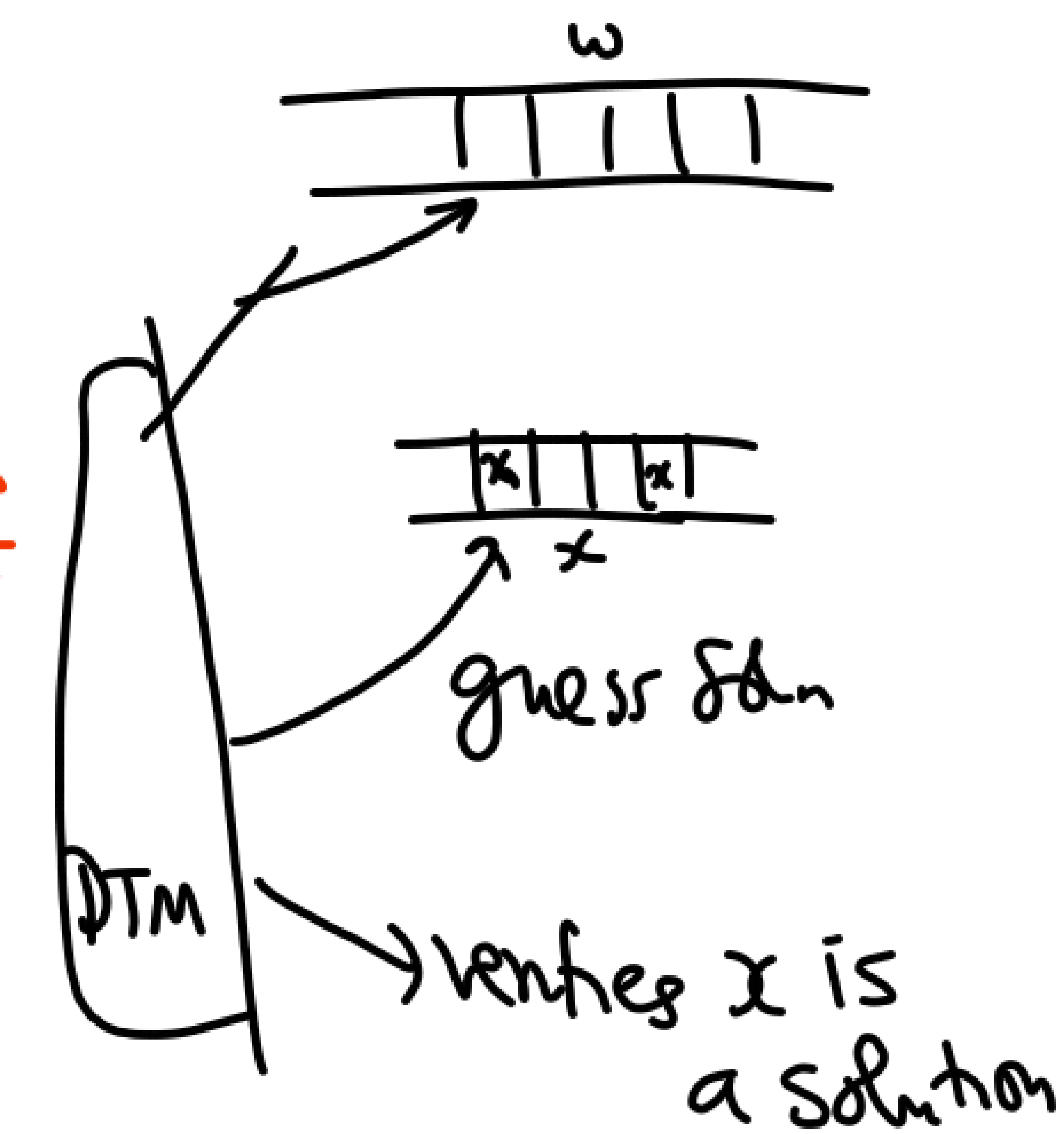
$\leftarrow n \rightarrow$

$\langle q_0, w \rangle \vdash \cdots \vdash$ Yes

$\vdash \vdash \vdash$ or No

n^3 n^8 $\underline{7n^3 + 9n^2 + \cdots}$

2nd tape



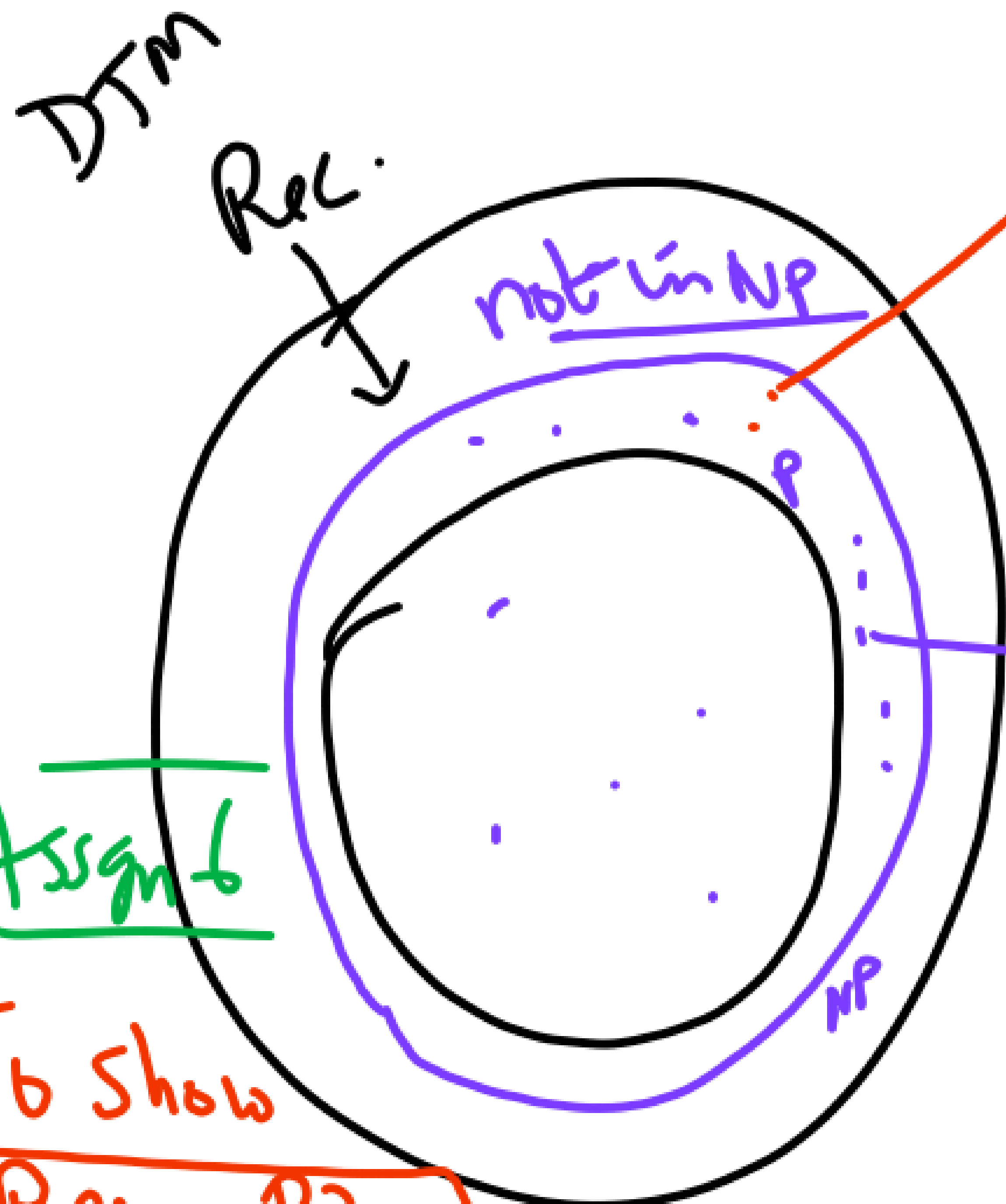
$P \overset{?}{=} NP$

NP-complete

SAT — most difficult problem in NP.

Polynomial time reduction from any other problem

"Stanford" lecture 2



Known to be in NP but not known to be in P

$P \leq NP$

strict inclusion

Assign 6

To show

Problem P3 is NP-complete

① Show $P3 \in NP$

② show known NP-comp. problem reduces in poly-time to P3.

Polynomial-reduction

$SAT \leq_p$ 3-SAT — Vertex-cover

knapsack

Solution preserving

Insem 70 — 30 Final.

15 25 15 15

Q1 M Q2 Q3

Assgn

-ve marks & tiebreakers